AOS-CX 10.13 Fundamentals Guide

4100i, 6000, 6100 Switch Series

Hewlett Packard Enterprise

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Notices

Acknowledgments

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Contents

Chapter 1
About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

- Aruba 4100i Switch Series (JL817A, JL818A)
- Aruba 6000 Switch Series (R8N85A, R8N86A, R8N87A, R8N88A, R8N89A, R9Y03A)
- Aruba 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in Support and Other Resources.

Command syntax notation conventions

Convention	Usage
`example-text`	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets (**[ ]**).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: ■ *`<example-text>`* ■ `<example-text>` ■ *`example-text`* ■ *example-text*	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: ■ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (**< >**). Substitute the text—including the enclosing angle brackets—with an actual value. ■ For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
`|`	Vertical bar. A logical **OR** that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.

Convention	Usage
`{ }`	Braces. Indicates that at least one of the enclosed items is required.
`[ ]`	Brackets. Indicates that the enclosed item or items are optional.
… or `. . .`	Ellipsis: ■ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. ■ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

`switch>`

The CLI prompt indicates the current command context. For example:

`switch>`

Indicates the operator command context.

`switch#`

Indicates the manager command context.

switch(*CONTEXT-NAME*)#

Indicates the configuration context for a feature. For example:

`switch(config-if)#`

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

`switch(config-vlan-100)#`

When referring to this context, this document uses the syntax:

`switch(config-vlan-<VLAN-ID>)#`

Where <*VLAN-ID*> is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

`member/slot/port`

On the 4100i Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.

On the 6000 and 6100 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.

Chapter 2
About AOS-CX

AOS-CX is a new, modern, fully programmable operating system built using a database-centric design that ensures higher availability and dynamic software process changes for reduced downtime. In addition to robust hardware reliability, the AOS-CX operating system includes additional software elements not available with traditional systems, including:

- Automated visibility to help IT organizations scale: The Aruba Network Analytics Engine allows IT to monitor and troubleshoot network, system, application, and security-related issues easily through simple scripts. This engine comes with a built-in time series database that enables customers and developers to create software modules that allow historical troubleshooting, as well as analysis of historical trends to predict and avoid future problems due to scale, security, and performance bottlenecks.
- Programmability simplified: A switch that is running the AOS-CX operating system is fully programmable with a built-in Python interpreter as well as REST-based APIs, allowing easy integration with other devices both on premise and in the cloud. This programmability accelerates IT organization understanding of and response to network issues. The database holds all aspects of the configuration, statistics, and status information in a highly structured and fully defined form.
- Faster resolution with network insights: With legacy switches, IT organizations must troubleshoot problems after the fact, using traditional tools like CLI and SNMP, augmented by separate, expensive monitoring, analytics, and troubleshooting solutions. These capabilities are built in to the AOS-CX operating system and are extensible.
- High availability: For switches that support active and standby management modules, the AOS-CX database can synchronize data between active and standby modules and maintain current configuration and state information during a failover to the standby management module.
- Ease of roll-back to previous configurations: The built-in database acts as a network record, enabling support for multiple configuration checkpoints and the ability to roll back to a previous configuration checkpoint.

AOS-CX system databases

The AOS-CX operating system is a modular, database-centric operating system. Every aspect of the switch configuration and state information is modeled in the AOS-CX switch configuration and state database, including the following:

- Configuration information
- Status of all features
- Statistics

The AOS-CX operating system also includes a time series database, which acts as a built-in network record. The time series database makes the data seamlessly available to Aruba Network Analytics Engine agents that use rules that evaluate network conditions over time. Time-series data about the resources monitored by agents are automatically collected and presented in graphs in the switch Web UI.

Aruba Network Analytics Engine introduction

The Aruba Network Analytics Engine is a first-of-its-kind built-in framework for network assurance and remediation. Combining the full automation and deep visibility capabilities of the AOS-CX operating system, this unique framework enables monitoring, collecting network data, evaluating conditions, and taking corrective actions through simple scripting agents.

This engine is integrated with the AOS-CX system configuration and time series databases, enabling you to examine historical trends and predict future problems due to scale, security, and performance bottlenecks. With that information, you can create software modules that automatically detect such issues and take appropriate actions.

With the faster network insights and automation provided by the Aruba Network Analytics Engine, you can reduce the time spent on manual tasks and address current and future demands driven by Mobility and IoT.

AOS-CX CLI

The AOS-CX CLI is an industry standard text-based command-line interface with hierarchical structure designed to reduce training time and increase productivity in multivendor installations.

The CLI gives you access to the full set of commands for the switch while providing the same password protection that is used in the Web UI. You can use the CLI to configure, manage, and monitor devices running the AOS-CX operating system.

Aruba NetEdit

Aruba NetEdit enables the automation of multidevice configuration change workflows without the overhead of programming.

The key capabilities of NetEdit include the following:

- Intelligent configuration with validation for consistency and compliance
- Time savings by simultaneously viewing and editing multiple configurations
- Customized validation tests for corporate compliance and network design
- Automated large-scale configuration deployment without programming
- Ability to track changes to hardware, software, and configurations (whether made through NetEdit or directly on the switch) with automated versioning

For more information about Aruba NetEdit, search for NetEdit at the following website:

www.hpe.com/support/hpesc

Ansible modules

Ansible is an open-source IT automation platform.

Aruba publishes a set of Ansible configuration management modules designed for switches running AOS-CX software. The modules are available from the following places:

- The `arubanetworks.aoscx_role` role in the Ansible Galaxy at: https://galaxy.ansible.com/arubanetworks/aoscx_role
- The `aoscx-ansible-role` at the following GitHub repository: https://github.com/aruba/aoscx-ansible-role

AOS-CX Web UI

The Web UI gives you quick and easy visibility into what is happening on your switch, providing faster problem detection, diagnosis, and resolution. The Web UI provides dashboards and views to monitor the status of the switch, including easy to read indicators for: power supply, temperature, fans, CPU use, memory use, log entries, system information, firmware, interfaces, VLANs, and LAGs. In addition, you use the Web UI to access the Network Analytics Engine, run certain diagnostics, and modify some aspects of the switch configuration.

AOS-CX REST API

Switches running the AOS-CX software are fully programmable with a REST (REpresentational State Transfer) API, allowing easy integration with other devices both on premises and in the cloud. This programmability—combined with the Aruba Network Analytics Engine—accelerates network administrator understanding of and response to network issues.

The AOS-CX REST API enables programmatic access to the AOS-CX configuration and state database at the heart of the switch. By using a structured model, changes to the content and formatting of the CLI output do not affect the programs you write. And because the configuration is stored in a structured database instead of a text file, rolling back changes is easier than ever, thus dramatically reducing a risk of downtime and performance issues.

The AOS-CX REST API is a web service that performs operations on switch resources using HTTPS `POST`, `GET`, `PUT`, and `DELETE` methods.

A switch resource is indicated by its Uniform Resource Identifier (URI). A URI can be made up of several components, including the host name or IP address, port number, the path, and an optional query string. The AOS-CX operating system includes the AOS-CX REST API Reference, which is a web interface based on the Swagger UI. The AOS-CX REST API Reference provides the reference documentation for the REST API, including resources URIs, models, methods, and errors. The AOS-CX REST API Reference shows most of the supported read and write methods for all switch resources.

In-band management

Management communications with a managed switch can be:

In band

In-band management communications occur through ports on the line modules of the switch, using common communications protocols such as SSH and SNMP.

When you use an in-band management connection, management traffic from that connection uses the same network infrastructure as user data. User data uses the `data plane`, which is responsible for moving data from source to destination. Management traffic that uses the data plane is more likely to be affected by traffic congestion and other issues affecting the user network.

SNMP-based management support

The AOS-CX operating system provides SNMP read access to the switch. SNMP support includes support of industry-standard MIB (Management Information Base) plus private extensions, including SNMP events, alarms, history, statistics groups, and a private alarm extension group. SNMP access is disabled by default.

User accounts

To view or change configuration settings on the switch, users must log in with a valid account. Authentication of user accounts can be performed locally on the switch, or by using the services of an external TACACS+ or RADIUS server.

Two types of user accounts are supported:

- Operators: Operators can view configuration settings, but cannot change them. No operator accounts are created by default.
- Administrators: Administrators can view and change configuration settings. A default locally stored administrator account is created with username set to **admin** and no password. You set the administrator account password as part of the initial configuration procedure for the switch.

Chapter 3
Initial Configuration

Perform the initial configuration of a factory default switch using one of the following methods:

- Load a switch configuration using zero-touch provisioning (ZTP). When ZTP is used, the configuration is loaded from a server automatically when the switch booted from the factory default configuration.
- Connect the management port on the switch to your network, and then use SSH client software to reach the switch from a computer connected to the same network. This requires that a DHCP server is installed on the network. Configure switch settings and features by executing CLI commands.
- Connect a computer running terminal emulation software to the console port on the switch. Configure switch settings and features by executing CLI commands.

Reserved ports or ports used by other applications/services with in the system are not recommended to be used for other services. When two services use the same port there is chance of unexpected behaviors from these services. Best practices is to use unique port for each service across system.

Initial configuration using ZTP

Zero Touch Provisioning (ZTP) configures a switch automatically from a remote server.

Prerequisites

- The switch must be in the factory default configuration.

Do not change the configuration of the switch from its factory default configuration in any way, including by setting the administrator password.

- Your network administrator or installation site coordinator must provide a Category 6 (Cat6) cable connected to the network that provides access to the servers used for Zero Touch Provisioning (ZTP) operations.

Procedure

1. Connect the network to a data port.

   See the *Installation Guide* for switch to determine the location of the switch ports.
2. If the switch is powered on, power off the switch.
3. Power on the switch. During the ZTP operation, the switch might reboot if a new firmware image is being installed. ZTP goes to "Failed" state if the switch receives DHCP IP for vlan1 and does not receive any ZTP options within 60 seconds.

Initial configuration using the CLI

This procedure describes how to connect to the switch for the first time and configure basic operational settings using the CLI. In this procedure, you use a computer to connect to the switch using the either the console port or management port.

Procedure

1. Connect to the console port or the management port.
2. Log into the switch for the first time.
3. Configure switch time using the NTP client.

Connecting to the console port

Prerequisites

- A switch installed as described in its hardware installation guide.
- A computer with terminal emulation software.

Procedure

1. Start the terminal emulation software on the computer and configure a new serial session with the following settings:
   - Speed: 115200 bps
   - Data bits: 8
   - Stop bits: 1
   - Parity: None
   - Flow control: None
2. Start the terminal emulation session.
3. Press **Enter** once. If the connection is successful, you are prompted to login.

Optional console port speed setting

If desired, the console port speed can be set with the `console baud-rate` command. For example, setting the console port speed to 9600 bps:

```
switch(config)# console baud-rate 9600
This command will configure the baud rate immediately for the active serial
console session. After the command is executed the user will be prompted to
re-login. The serial console will be inaccessible until the terminal client
settings are updated to match the baud rate of the switch.
Continue (y/n)? y
```

Showing the console port current speed:

```
switch# show console
Baud Rate: 9600
```

For details on the `console baud-rate` and `show console` commands, see Switch system and hardware commands.

Connecting to the in-band management port

Prerequisites

- Two Ethernet cables
- SSH client software

Procedure

1. By default, the in-band management interface is set to automatically obtain an IP address from a DHCP server, and SSH support is enabled. If there is no DHCP server on your network, you must configure a static address on the in-band management interface:
   a. Connect to the console port
   b. Configure using DHCP or static IP.
   c. Configure the in-band management interface and interface VLAN1.
2. Use an Ethernet cable to connect the management port to your network.
3. Use an Ethernet cable to connect your computer to the same network.
4. Start your SSH client software and configure a new session using the address assigned to the in-band management interface. (If the in-band management interface is set to operate as a DHCP client, retrieve the IP address assigned to the in-band management interface from your DHCP server.)
5. Start the session. If the connection is successful, you are prompted to log in.

Configure using DHCP or static IP

Users can use any data ports for in-band management purposes. IP DHCP is supported on interface VLAN 1 only. All switch ports are part of access VLAN 1 by default. Static IP address and IP DHCP configuration can co-exist on VLAN 1, however static addresses take precedence whenever configured.

DHCP Configuration

```
switch#: config
switch(config)#: vlan 1
Switch(config-vlan-1)#: description Management VLAN
Switch(config-vlan-1)#: end
Switch#
!
Switch(config)#: interface 1/1/1
Switch(config-if)#: description IN-BAND Management Port
Switch(config-if)#: vlan access 1
Switch(config-if)#: no shutdown
Switch(config-if)#: end
Switch#
!
Switch(config)#: interface vlan 1
Switch(config-if-vlan)#: description IN-BAND Management Interface
Switch(config-if-vlan)#: ip dhcp
Switch(config-if-vlan)#: no shutdown
Switch(config-if-vlan)#: end
Switch#
!
```

Without DHCP Configuration

```
switch#: config
switch(config)#: vlan 1
Switch(config-vlan-1)#: description Management VLAN
Switch(config-vlan-1)#: end
Switch#
!
Switch(config)#: interface 1/1/1
```

```
Switch(config-if)#: description IN-BAND Management Port
Switch(config-if)#: vlan access 1
Switch(config-if)#: no shutdown
Switch(config-if)#: end
Switch#
!
Switch(config)#: interface vlan 1
Switch(config-if-vlan)#: description IN-BAND Management Interface
Switch(config-if-vlan)#: no ip dhcp
Switch(config-if-vlan)#: ip address 192.168.10.1/24
Switch(config-if-vlan)#: no shutdown
Switch(config-if-vlan)#: end
Switch#
```

Logging into the switch for the first time

The first time you log in to the switch you must use the default administrator account. This account has no password, so you will be prompted on login to define one to safeguard the switch.

Procedure

1. When prompted to log in, specify **admin**. When prompted for the password, press **ENTER**. (By default, no password is defined.)

   For example:

   ```
   switch login: admin
   password:
   ```

2. Define a password for the **admin** account. The password can contain up to 32 alphanumeric characters in the range ASCII 32 to 127, which includes special characters such as asterisk (*), ampersand (&), exclamation point (!), dash (-), underscore (_), and question mark (?).

   For example:

   ```
   Please configure the 'admin' user account password.
   Enter new password: *******
   Confirm new password: *******
   switch#
   ```

3. You are placed into the manager command context, which is identified by the prompt: `switch#`, where `switch` is the model number of the switch. Enter the command `config` to change to the global configuration context `config`.

   For example:

   ```
   switch# config
   switch(config)#
   ```

Setting switch time using the NTP client

Prerequisites

- The IP address or domain name of an NTP server.
- If the NTP server uses authentication, obtain the password required to communicate with the NTP server.

Procedure

1. If the NTP server requires authentication, define the authentication key for the NTP client with the command `ntp authentication`.
2. Configure an NTP server with the command `ntp server`.
3. By default, NTP traffic is sent on the default VRF.
4. Review your NTP configuration settings with the commands `show ntp servers` and `show ntp status`.
5. See the current switch time, date, and time zone with the command `show clock`.

Example

This example creates the following configuration:

- Defines the authentication key **1** with the password **myPassword**.
- Defines the NTP server **my-ntp.mydomain.com** and makes it the preferred server.

```
switch(config)# ntp authentication-key 1 md5 myPassword
switch(config)# ntp server my-ntp.mydomain.com key 10 prefer
switch(config)# ntp vrf default
```

Configuring banners

1. Configure the banner that is displayed when a user connects to a device using a console port or in-band management interface. Use the command `banner motd`. For example:

```
switch(config)# banner motd ^
Enter a new banner. Terminate the banner with the delimiter you have chosen.
>> This is an example of a banner text which a connecting user
>> will see before they are prompted for their password.
>>
>> As you can see it may span multiple lines and the input
>> will be terminated when the delimiter character is
>> encountered.^
Banner updated successfully!
```

2. Configure the banner that is displayed after a user is authenticated. Use the command `banner exec`. For example:

```
switch(config)# banner exec &
Enter a new banner. Terminate the banner with the delimiter you have chosen.
>> This is an example of a different banner text. This time
>> the banner entered will be displayed after a user has
>> authenticated.
>>
```

```
>> & This text will not be included because it comes after the '&'
Banner updated successfully!
```

Using the Web UI

Prerequisites

- A connection to the switch CLI.

On the 6000 and 6100, the HTTPS server can only be enabled by the default VRF.

On the 4100i, the HTTPS server can only be enabled by the default VRF.

Procedure

1. Log in to the CLI.
2. Switch to `config` context.

   For example:

```
switch# config
switch(config)# https-server vrf default
```

3. The Web UI starts and you are prompted to log in.

Configuring the in-band management interface

Prerequisites

A connection to the console port.

Procedure

1. Switch to the in-band management interface context with the command `interface vlan 1`.
2. By default, the in-band management interface is enabled. If it was disabled, re-enable it with the command `no shutdown`.
3. Use the command `ip dhcp` to configure the in-band management interface to automatically obtain an address from a DHCP server on the network (factory default setting). Or, assign a static IPv4 or IPv6 address with the commands `ip address` or ipv6 address.
4. SSH is enabled by default on the default VRF. If disabled, enable SSH with the command `ssh server vrf default`.

Examples

This example enables the in-band management interface with static addressing:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip dhcp
switch(config-if-vlan)# ip address 192.168.100.200/24
switch(config-if-vlan)# no shutdown
```

Restoring the switch to factory default settings

Prerequisites

You are connected to the switch through its Console port.

This procedure erases all user information and configuration settings Consider backing up your running configuration first.

1. Optionally, back up the running configuration with either `copy running-config <REMOTE-URL>` or `copy running-config <STORAGE-URL>`. The `json` storage format is required for later configuration restoration.
2. Switch to the configuration context with the command `config`.
3. Erase all user information and configuration, restoring the switch to its factory default state with the command `erase all zeroize`. Enter `Y` when prompted to continue. The switch automatically restarts.
4. Optionally restore your saved configuration (it must be in `json` format) with either `copy <REMOTE-URL> running-config` or `copy <STORAGE-URL> running-config` followed by `copy running-config startup-config`.

Example

Backing up the running configuration to a file on a remote server (using TFTP), resetting the switch to its factory default state, and then restoring the saved configuration.

```
switch#  copy running-config tftp://10.100.1.12/backup_cfg json vrf default

  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 10340    0     0  100 10340      0  1329k --:--:-- --:--:-- --:--:-- 1329k
100 10340    0     0  100 10340      0  1313k --:--:-- --:--:-- --:--:-- 1313k
switch#
switch#
switch# erase all zeroize
This will securely erase all customer data and reset the switch
to factory defaults. This will initiate a reboot and render the
switch unavailable until the zeroization is complete.
This should take several minutes to one hour to complete.
Continue (y/n)? y
The system is going down for zeroization.
[  OK  ] Stopped PSPO Module Daemon.
[  OK  ] Stopped AOS-CX Switch Daemon for BCM.
...
[  OK  ] Stopped Remount Root and Kernel File Systems.
[  OK  ] Reached target Shutdown.
reboot: Restarting system
Press Esc for boot options
ServiceOS Information:
    Version:          GT.01.03.0006
```

```
    Build Date:       2018-10-30 14:20:44 PDT
    Build ID:         ServiceOS:GT.01.03.0006:8ee0faaa52da:201810301420
    SHA:              xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
...

################ Preparing for zeroization #################

################ Storage zeroization ######################
################ WARNING: DO NOT POWER OFF UNTIL  ##########
################          ZEROIZATION IS COMPLETE ##########
################ This should take several minutes ##########
################ to one hour to complete          ##########

################ Restoring files ###########################

Boot Profiles:

0. Service OS Console
1. Primary Software Image [XL.10.02.0010]
2. Secondary Software Image [XL.10.02.0010]

Select profile(primary):

Booting primary software image...
Verifying Image...

Image Info:
        Name: AOS-CX
     Version: XL.10.02.0010
    Build Id: AOS-CX:XL.10.02.0010:feaf5b9b7f09:201901292014
  Build Date: 2019-01-29 12:43:50 PST

Extracting Image...
Loading Image...
Done.
kexec_core: Starting new kernel
System is initializing
fips_post_check[5473]: FIPS_POST: Cryptographic selftest started...SUCCESS
[  OK  ] Started Login banner readiness check.
...
8400X login: admin
Password:

switch#
switch#
switch# copy tftp://192.168.1.10/backup_cfg running-config json vrf default
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 10340  100 10340    0     0  2858k      0 --:--:-- --:--:-- --:--:-- 2858k
100 10340  100 10340    0     0  2804k      0 --:--:-- --:--:-- --:--:-- 2804k
Large configuration changes will take time to process, please be patient.
switch#
switch#
switch# copy running-config startup-config
Large configuration changes will take time to process, please be patient.
switch#
```

NTP commands

ntp authentication

`ntp authentication`

```
no ntp authentication
```

Description

Enables support for authentication when communicating with an NTP server.

The **no** form of this command disables authentication support.

Examples

Enabling authentication support:

```
switch(config)# ntp authentication
```

Disabling authentication support:

```
switch(config)# no ntp authentication
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

ntp authentication-key

```
ntp authentication-key <KEY-ID> {md5 | sha1}
    [{ <PLAINTXT-KEY> [trusted] | ciphertext <ENCRYPTED-KEY> }]
no ntp authentication-key <KEY-ID>  {md5 | sha1}
    [{ <PLAINTXT-KEY> [trusted] | ciphertext <ENCRYPTED-KEY> }]
```

Description

Defines an authentication key that is used to secure the exchange with an NTP time server. This command provides protection against accidentally synchronizing to a time source that is not trusted.

The **no** form of this command removes the authentication key.

Parameter	Description
<KEY-ID>	Specifies the authentication key ID. Range: 1 to 65534.
`md5`	Selects MD5 key encryption.
`sha1`	Specifies SHA1 key encryption.
<PLAINTXT-KEY>	Specifies the plaintext authentication key. Range: 8 to 40 characters. The key may contain printable ASCII characters

Parameter	Description
	excluding "#" or be entered in hex. Keys longer than 20 characters are assumed to be hex. To use an ASCII key longer than 20 characters, convert it to hex.
`trusted`	Specifies that this is a trusted key. When NTP authentication is enabled, the switch only synchronizes with time servers that transmit packets containing a trusted key.
`ciphertext` *<ENCRYPTED-KEY>*	Specifies the ciphertext authentication key in Base64 format. This is used to restore the NTP authentication key when copying configuration files between switches or when uploading a previously saved configuration. **NOTE:** When the key is not provided on the command line, plaintext key prompting occurs upon pressing Enter, followed by prompting as to whether the key is to be trusted. The entered key characters are masked with asterisks.

Examples

Defining key 10 with MD5 encryption and a provided plaintext trusted key:

```
switch(config)# ntp authentication-key 10 md5 F82#450b trusted
```

Defining key 5 with SHA1 encryption and a prompted plaintext trusted key:

```
switch(config)# ntp authentication-key 5 sha1
Enter the NTP authentication key: *********
Re-Enter the NTP authentication key: *********

Configure the key as trusted (y/n)? y
```

Removing key 10:

```
switch(config)# no ntp authentication-key 10
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

ntp disable

`ntp disable`

Description

Disables the NTP client on the switch. The NTP client is disabled by default.

Examples

Disabling the NTP client.

```
switch(config)# ntp disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

ntp enable

```
ntp enable
no ntp enable
```

Description

Enables the NTP client on the switch to automatically adjust the local time and date on the switch. The NTP client is disabled by default.

The **no** form of this command disables the NTP client.

Examples

Enabling the NTP client.

```
switch(config)# ntp enable
```

Disabling the NTP client.

```
switch(config)# no ntp enable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ntp server

```
ntp server <IP-ADDR> [key <KEY-NUM>] [minpoll <MIN-NUM>] [maxpoll <MAX-NUM>][burst |
iburst][prefer] [version <VER-NUM>]
no ntp server <IP-ADDR> <IP-ADDR> [key <KEY-NUM>] [minpoll <MIN-NUM>] [maxpoll <MAX-NUM>]
[burst | iburst] [prefer] [version <VER-NUM>]
```

Description

Defines an NTP server to use for time synchronization, or updates the settings of an existing server with new values. Up to eight servers can be defined.

The **no** form of this command removes a configured NTP server.

The default NTP version is 4; it is backwards compatible with version 3.

Parameter	Description
server *<IP-ADDR>*	Specifies the address of an NTP server as a DNS name, an IPv4 address (**x.x.x.x**), where **x** is a decimal number from 0 to 255, or an IPv6 address (**xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F. When specifying an IPv4 address, you can remove leading zeros. For example, the address **192.169.005.100** becomes **192.168.5.100**. When specifying an IPv6 address, you can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address **2222:0000:3333:0000:0000:0000:4444:0055** becomes **2222:0:3333::4444:55** .
key *<KEY-NUM>*	Specifies the key to use when communicating with the server. A trusted key must be defined with the command **ntp authentication-key** and authentication must be enabled with the command **ntp authentication**. Range: 1 to 65534.
minpoll *<MIN-NUM>*	Specifies the minimum polling interval in seconds, as a power of 2. Range: 4 to 17. Default: 6 (64 seconds).
maxpoll *<MAX-NUM>*	Specifies the maximum polling interval in seconds, as a power of 2. Range: 4 to 17. Default: 10 (1024 seconds).
burst	Send a burst of packets instead of just one when connected to the server. Useful for reducing phase noise when the polling interval is long.
iburst	Send a burst of six packets when not connected to the server. Useful for reducing synchronization time at startup.
prefer	Make this the preferred server.

Parameter	Description
`version <VER-NUM>`	Specifies the version number to use for all outgoing NTP packets. Range: 3 or 4. Default: 4. **NOTE:** NTP is backwards compatible.

Usage

For features such as Activate and ZTP, a switch that has a factory default configuration will automatically be configured with pool.ntp.org. NTP server configurations via DHCP options are supported. The DHCP server can be configured with maximum of two NTP server addresses which will be supported on the switch. Only IPV4 addresses are supported.

NTP uses a stratum to describe the distance between a network device and an authoritative time source:

- A stratum 1 time server is directly attached to an authoritative time source (such as a radio or atomic clock or a GPS time source).
- A stratum 2 NTP server receives its time through NTP from a stratum 1 time server.

When using multiple servers with same stratum setting, the best practice to configure a preferred server, so NTP will attempt to use the preferred server as the primary NTP connection. If a preferred server is not manually set when NTP is enabled, the configured server with the lowest stratum will automatically be set as the preferred server. If there are servers with the same stratum, this auto prefer status will prevent AOS-CX from toggling between different servers as the primary server. Auto prefer selection of servers with same stratum (if not manually selected) may change after reconfiguring the switch, or after executing the **reboot** command.

Examples

Defining the ntp server pool.ntp.org, using iburst, and NTP version 4.

```
switch(config)# ntp server pool.ntp.org iburst version 4
```

Removing the ntp server pool.ntp.org.

```
switch(config)# no ntp server pool.ntp.org
```

Defining the ntp server my-ntp.mydomain.com and makes it the preferred server.

```
switch(config)# ntp server my-ntp.mydomain.com prefer
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

ntp trusted-key

```
ntp trusted-key <KEY-ID>
no ntp trusted-key <KEY-ID>
```

Description

Sets a key as trusted. When NTP authentication is enabled, the switch only synchronizes with time servers that transmit packets containing a trusted key.

The **no** form of this command removes the trusted designation from a key.

Parameter	Description
<KEY-ID>	Specifies the identification number of the key to set as trusted. Range: 1 to 65534.

Examples

Defining key 10 as a trusted key.

```
switch(config)# ntp trusted-key 10
```

Removing trusted designation from key 10:

```
switch(config)# no ntp trusted-key 10
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

ntp vrf

```
ntp vrf <VRF-NAME>
no ntp vrf <VRF-NAME>
```

Description

6000 and 6100 only support default VRF.

4100i only supports default VRF.

Specifies the VRF on which the NTP client communicates with an NTP server.
The **no** form of the command returns to default VRF.

Parameter	Description
<VRF-NAME>	Specifies the name of a VRF.

Example

Setting the switch to use the default VRF for NTP client traffic.

```
switch(config)# ntp vrf default
```

Returning the switch to use the default VRF for NTP client traffic.

```
switch(config)# no ntp vrf
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

show ntp associations

`show ntp associations`

Description

Shows the status of the connection to each NTP server. The following information is displayed for each server:

- Tally code : The first character is the Tally code:
  - (blank): No state information available (e.g. non-responding server)
  - x : Out of tolerance (discarded by intersection algorithm)
  - . : Discarded by table overflow (not used)
  - - : Out of tolerance (discarded by the cluster algorithm)
  - + : Good and a preferred remote peer or server (included by the combine algorithm)
  - # : Good remote peer or server, but not utilized (ready as a backup source)

- * : Remote peer or server presently used as a primary reference
- o : PPS peer (when the prefer peer is valid)

- ID: Server number.
- NAME: NTP server FQDN/IP address (Only the first 24 characters of the name are displayed).
- REMOTE: Remote server IP address.
- REF_ID: Reference ID for the remote server (Can be an IP address).
- ST: (Stratum) Number of hops between the NTP client and the reference clock.
- LAST: Time since the last packet was received in seconds unless another unit is indicated.
- POLL: Interval (in seconds) between NTP poll packets. Maximum (1024) reached as server and client sync.
- REACH: 8-bit octal number that displays status of the last eight NTP messages (377 = all messages received).

Example

```
switch# show ntp associations
----------------------------------------------------------------------
 ID             NAME           REMOTE          REF-ID ST LAST POLL REACH
----------------------------------------------------------------------
  1       192.0.1.1       192.0.1.1          .INIT. 16    -   64     0
* 2   time.apple.com    17.253.2.253         .GPSs.  2   70  128   377
----------------------------------------------------------------------
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ntp authentication-keys

`show ntp authentication-keys`

Description

Shows the currently defined authentication keys.

Examples

```
switch# show ntp authentication-keys
--------------------------------
Auth key  Trusted   MD5 password
--------------------------------
```

```
10          No      **********
20          Yes     **********
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show ntp servers

`show ntp servers`

Description

Shows all configured NTP servers, including any DHCP servers, default pool servers or any server with the status **auto prefer**.

Example

```
switch# show ntp servers
------------------------------------------------
     NTP SERVER KEYID MINPOLL MAXPOLL OPTION VER
------------------------------------------------
     192.0.1.18     -       5      10 iburst   3
     192.0.1.19     -       6      10   none   4
     192.0.1.20     -       6       8  burst   3 prefer
------------------------------------------------
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ntp statistics

`show ntp statistics`

Description

Shows global NTP statistics. The following information is displayed:

- Rx-pkts: Total NTP packets received.
- Current Version Rx-pkts: Number of NTP packets that match the current NTP version.
- Old Version Rx-pkts: Number of NTP packets that match the previous NTP version.
- Error pkts: Packets dropped due to all other error reasons.
- Auth-failed pkts: Packets dropped due to authentication failure.
- Declined pkts: Packets denied access for any reason.
- Restricted pkts: Packets dropped due to NTP access control.
- Rate-limited pkts: Number of packets discarded due to rate limitation.
- KOD pkts: Number of Kiss of Death packets sent.

Examples

```
switch(config)# show ntp statistics
                Rx-pkts 100
Current Version Rx-pkts 80
    Old Version Rx-pkts 20
               Err-pkts 2
       Auth-failed-pkts 1
          Declined-pkts 0
        Restricted-pkts 0
      Rate-limited-pkts 0
               KoD-pkts 0
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ntp status

`show ntp status`

Description

Shows the status of NTP on the switch.

Examples

Displaying the status information when the switch is not synced to an NTP server:

```
switch# show ntp status
NTP is enabled.
NTP authentication is enabled.
NTP is using the default VRF for NTP server connections.

Wed Nov 23 23:29:10 PDT 2016
NTP uptime: 187 days, 1 hours, 37 minutes, 48 seconds

Not synchronized with an NTP server.
```

Displaying the status information when the switch is synced to an NTP server:

```
switch# show ntp status
NTP is enabled.
NTP authentication is enabled.
NTP is using the default VRF for NTP server connections.

Wed Nov 23 23:29:10 PDT 2016
NTP uptime: 187 days, 1 hours, 37 minutes, 48 seconds

Synchronized to NTP Server 17.253.2.253 at stratum 2.
Poll interval = 1024 seconds.
Time accuracy is within 0.994 seconds
Reference time: Thu Jan 28 2016 0:57:06.647 (UTC)
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Chapter 4
Telnet access

Telnet server enables switches to accept Telnet connections from clients to manage the switch. The user authentication is password based authentication (RADIUS, TACACS+ or locally stored password). The Telnet server can be enabled on a desired VRF using the `telnet server` command. The maximum number of Telnet sessions per VRF is five.

In the default configuration, Telnet access is disabled.

Telnet commands

show telnet server

`show telnet server`

Description

Shows the Telnet server configuration.

Examples

Showing the Telnet server configuration:

```
switch(config)# show telnet server
TELNET Server Configuration:

    IP Version          : IPv4
    TCP Port            : 23
    Enabled VRFs        : default, vrf1, vrf2,
                          red, green
```

Command History

Release	Modification
10.11	Command introduced on all other platforms.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

show telnet server sessions

`show telnet server sessions [vrf <VRF-NAME> | all-vrfs]`

Description

Shows all active Telnet sessions for the specified VRF or all VRFs. If no VRF is provided, the Telnet sessions on the **default** VRF are shown.

Parameter	Description
`vrf <VRF-NAME>`	Specifies the Telnet sessions for a specific VRF.
`all-vrfs`	Specifies the Telnet sessions for all VRFs

Examples

Showing the Telnet session on the **default** VRF:

```
switch(config)# show telnet server sessions
TELNET sessions on VRF default:

    IPv4 TELNET Sessions:
        Server IP         : 10.1.1.1
        Client IP         : 10.1.1.2
        Client Port       : 58835
```

Showing the Telnet sessions on all VRFs:

```
switch(config)# show telnet server sessions all-vrfs
TELNET sessions on VRF mgmt:

    IPv4 TELNET Sessions:
        Server IP         : 10.1.1.1
        Client IP         : 10.1.1.2
        Client Port       : 58835

TELNET sessions on VRF default:

    IPv4 TELNET Sessions:
        Server IP         : 20.1.1.1
        Client IP         : 20.1.1.2
        Client Port       : 58837
```

Command History

Release	Modification
10.11	Command introduced on all other platforms.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

telnet server

```
telnet server vrf <VRF-NAME>
no telnet server vrf <VRF-NAME>
```

Description

Enables the Telnet server on the desired VRF. Telnet server is disabled by default.

The **no** form of this command disables the Telnet server.

Parameter	Description
`vrf <VRF-NAME>`	Specifies the VRF on which the Telnet server will be enabled or disabled.

Examples

Configuring the Telnet server on the **mgmt** VRF:

```
switch(config)# telnet server vrf mgmt
```

Command History

Release	Modification
10.11	Command introduced on all other platforms.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

Configuring a layer 2 interface

Procedure

1. Change to the interface configuration context for the interface with the command `interface`.
2. Set the interface MTU (maximum transmission unit) with the command `mtu`.
3. Review interface configuration settings with the command `show interface`.

Single source IP address

Certain IP-based protocols used by the switch (such as RADIUS, sFlow, TACACS, and TFTP), use a client-server model in which the client's source IP address uniquely identifies the client in packets sent to the server. By default, the source IP address is defined as the IP address of the outgoing switch interface on which the client is communicating with the server. Since the switch can have multiple routing interfaces, outgoing packets can potentially be sent on different paths at different times. This can result in different source IP addresses being used for a client, which can create a client identification problem on the server. For example, it can be difficult to interpret system logs and accounting data on the server when the same client is associated with multiple IP addresses.

To resolve this issue, you can use the commands `ip source-interface` and `ipv6 source-interface` to define a single source IP address that applies to all supported protocols (RADIUS, sFlow, TACACS, and TFTP), or an individual address for each protocol. This ensures that all traffic sent by a client to a server uses the same IP address.

Unsupported transceiver support

Transceiver products (optical, DAC, AOCs) that are listed as supported by a switch model are detailed in the *Transceiver Guide*. Transceiver products that are not listed, are considered unsupported; this would include transceivers that are:

- Non-Aruba branded products
- HPE branded products that were designed for non-AOS-CX switch models (e.g. Comware)
- HPE branded products designated for use in HPE Compute Servers or Storage
- Transceivers originally designated for use in Aruba WLAN controllers or former Mobility Access Switch (MAS) products
- End-of-life Aruba Transceivers

The unsupported transceiver mode (UT-mode) is designed to allow the possible use of these unsupported products. Not all unsupported products can be recognized and enabled; they may be unable to be identified (do not follow the proper MSA standards for identification). These unsupported transceiver products are enabled only on a best-effort basis and there are no guarantees implied for their continued operation.

This feature is enabled by default. A periodic system log will be generated by default at an interval of 24 hours listing the ports on which unsupported transceivers are present. The log interval is configurable and can be disabled by setting the log-interval to `none`.

Configuring an interface persona

A persona is a template interface. It is a mechanism intended to eases the process of configuring one or multiple interfaces that behave in the same manner. For example, a persona could be an uplink interface with a well-known collection of VLANs to which it belongs. Instead of manually configuring each interface one-by-one, the persona collects the common configuration settings and allows them to be applied on the interface or a collection of interfaces.

Persona configuration is similar to configuring other interfaces. Most of the commands in the interface context are available. Because of the nature of the commands, several of them do not apply to the persona (for example, applying the same IP address configuration to several ports) and have been removed from the available list.

The following commands are not supported within the interface context when configuring a persona:

```
aaa authentication port-access dot1x authenticator macsec
arp ipv4 <IPV4_ADDR> mac <MAC_ADDR>
downshift-enable
energy-efficient-ethernet
error-control { auto | none | base-r-fec | rs-fec }
ip bootp-gateway <IPV4-ADDR>
ip forward-protocol udp <IPV4-ADDR> {<PORT-NUM> | <PROTOCOL>}
ip helper-address <IPV4-ADDR> [vrf <VRF-NAME>]
ip igmp router-alert-check [enable | disable]
ip igmp snooping [auto vlan <VLAN-LIST>]
ip igmp snooping [blocked vlan <VLAN-LIST>]
ip igmp snooping [fastleave vlan <VLAN-LIST>]
ip igmp snooping [forced-fastleave vlan <VLAN-LIST>]
ip igmp snooping [forward vlan <VLAN-LIST>]
ip ospf <PROCESS-ID> area <AREA-ID>
ip ospf passive
ip rip <PROCESS-ID> {all-ip | ip-address}
ip rip all-ip disable
ip rip all-ip enable
ip rip all-ip receive disable
ip rip all-ip send disable
ipv6 helper-address multicast {all-dhcp-servers | <MULTICAST-IPV6-ADDR>} egress <PORT-NUM>
ipv6 mld snooping [auto vlan <VLAN-LIST>]
ipv6 mld snooping [blocked vlan <VLAN-LIST>]
ipv6 mld snooping [fastleave vlan <VLAN-LIST>]
ipv6 mld snooping [forced-fastleave vlan <VLAN-LIST>]
ipv6 mld snooping [forward vlan <VLAN-LIST>]
ipv6 neighbor <IPV6-ADDR> mac <MAC-ADDR>
ipv6 ospfv3 <PROCESS-ID> area <AREA-ID>
ipv6 ospfv3 passive
ipv6 ripng <PROCESS-ID>
lacp port-id <PORT-ID>
lacp port-priority <PORT-PRIORITY>
lag <ID>
link-poe
lldp dot3 eee
lldp dot3 poe
lldp med poe
lldp med poe [priority-override]
persona {access | uplink | custom <PERSONA-NAME>} [copy | attach]
```

```
port-access device-profile <DEVICE-PROFILE-NAME>
power-over-ethernet
power-over-ethernet allocate-by {usage | class}
power-over-ethernet assigned-class {3 | 4 | 6 | 8}
power-over-ethernet pd-class-override
power-over-ethernet power-pairs {alt-a|alt-a-and-alt-b}
power-over-ethernet pre-std-detect
power-over-ethernet priority {critical|high|low}
ptp lag-role {primary | secondary}
spanning-tree cost <PORT-COST>
spanning-tree instance <INSTANCE-ID> cost <PORT-COST>
spanning-tree instance <INSTANCE-ID> port-priority <PRIORITY-MULTIPLIER>
spanning-tree port-priority <PRIORITY-MULTIPLIER>
spanning-tree vlan <A:1-4094> cost <0-200000000>
spanning-tree vlan <A:1-4094> port-priority <0-15>
split [confirm]
track by <OBJECT-ID>
vsx-sync {access-lists | qos | rate-limits | vlans | policies | irdp}
```

Modes

There are two supported modes:

1. Copy—The `copy` mode is a one-step configuration that copies the persona configuration to an interface. Further changes to the persona will not be applied to the interfaces using that mode.

2. Attach—Unlike the `copy` mode, besides applying the configuration to the interface immediately, the `attach` mode also follows the persona configuration. It means that the subsequent changes to the persona will be applied to the interfaces attached to it.

Predefined and custom persona names

There are two predefined interface persona names:

- uplink
- access

These names have no predefined configuration. To use them, they must be manually configured as needed. You can also create personas with a custom name. These custom personas can be created and configured in the same manner as the predefined ones. The only difference is the command used to apply them to the interface. The procedure below provides the details.

Creating and configuring an interface persona

Follow these steps to create and configure an interface persona:

1. Create the interface **persona** with the command `interface persona <PERSONA-NAME>`.
2. Set the interface configuration as any other physical interface.
3. Review the configuration with the command `show running-configuration current-context`.
4. Switch to an interface context with the command `interface <PORT>`.
5. Apply the persona configuration to the interface and set the mode with the command `persona custom <PERSONA-NAME> <mode>`. Note that `<custom>` is an optional argument, required only if the persona is not one of the predefined names (neither **uplink** nor **access**).

For information on this feature, see the related video on the Aruba AirHeads Broadcasting Channel.

Examples

To copy a predefined persona name configuration to an interface:

1. Configure the interface persona:

```
switch(config)# interface persona uplink
switch(config-if)# no shutdown
switch(config-if)# no routing
switch(config-if)# vlan access 100
switch(config-if)# exit
```

2. Apply the configuration with **copy** mode:

```
switch(config)# interface 1/1/1
switch(config-if)# persona custom mypersona copy
switch(config-if)# exit
```

To attach a custom persona name to several interfaces simultaneously:

1. Configure the interface persona:

```
switch(config)# interface persona mytemplate
switch(config-if)# no shutdown
switch(config-if)# vrf attach upstream
switch(config-if)# exit
```

2. Apply the configuration with **attach** mode:

```
switch(config)# interface 1/1/1-1/1/24
switch(config-if)# persona custom mypersona attach
switch(config-if)# exit
```

Monitor mode

Monitor mode displays interface statistics in real time, updating frequently. The update interval varies by product, number of visible ports, and display options in use. View the current update interval in the help menu.

Formatting options can be supplied either at command launch or while monitor mode is active. Press `?` to see the available formatting options available in the current command context. Exit the help menu with `q`.

Monitor mode detects terminal size to adjust the number of interfaces and statistics viewable on screen. Additional interfaces or statistics columns can be viewed with the navigation keys: Arrow keys, PageUp, PageDown, Home, and End. Serial console does not automatically detect terminal setting changes, resulting in unexpected output. Recommended recovery steps are to exit and restart.

Monitor mode refreshes data automatically until exited with `q`.

Interface commands

allow-unsupported-transceiver

```
allow-unsupported-transceiver [confirm | log-interval {none | <INTERVAL>}]
no allow-unsupported-transceiver
```

Description

Allows unsupported transceivers to be enabled or establish connections. Transceivers with speeds up to 100G are enabled by this command.

This command is enabled by default, allowing the use of third party transceiver products without adding the command in the configuration. Disabling this command with the **no** form will now disable the command in the running and stored configurations.

The **no** form of this command disallows using unsupported transceivers.

Parameter	Description
`confirm`	Specifies that unsupported transceiver warnings are to be automatically confirmed.
`log-interval none`	Disables unsupported transceiver logging.
`log-interval` *`<INTERVAL>`*	Sets the unsupported transceiver logging interval in minutes. Default: 1440 minutes. Range: 1440 to 10080 minutes.

Usage

When none of the parameters are specified it will display a warning message to accept the warranty terms. With **confirm** option the warning message is displayed but the user is not prompted to **(y/n)** answering. Warranty terms must be agreed to as part of enablement and the support is on best effort basis.

Examples

Allowing unsupported transceivers with follow-up confirmation:

```
switch(config)# allow-unsupported-transceiver
Warning: The use of unsupported transceivers, DACs, and AOCs is at your
own risk and may void support and warranty. Please see HPE Warranty terms
and conditions.

Do you agree and do you want to continue (y/n)? y
```

Allowing unsupported transceivers with confirmation in command syntax:

```
switch(config)# allow-unsupported-transceiver confirm
Warning: The use of unsupported transceivers, DACs, and AOCs is at your
own risk and may void support and warranty. Please see HPE Warranty terms
and conditions.
```

Configuring unsupported transceiver logging with an interval of every 48 hours:

```
switch(config)# allow-unsupported-transceiver log-interval 2880
```

Disabling unsupported transceiver logging:

```
switch(config)# allow-unsupported-transceiver log-interval none
```

Disallowing unsupported transceivers with follow-up confirmation:

```
switch(config)# no allow-unsupported-transceivers
Warning: Unsupported transceivers, DACs, and AOCs will be disabled,
which could impact network connectivity. Use 'show allow-unsupported-transceiver'
to identify unsupported transceivers, DACs, and AOCs.

Ccontinue (y/n)? y
```

Disallowing unsupported transceivers with confirmation in command syntax:

```
switch(config)# no allow-unsupported-transceiver confirm
Warning: Unsupported transceivers, DACs, and AOCs will be disabled,
which could impact network connectivity. Use 'show allow unsupported-transceiver'
to identify unsupported transceivers, DACs, and AOCs.

switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6100	`config`	Administrators or local user group members with execution rights for this command.

default interface

`default interface <INTERFACE-ID>`

Description

Sets an interface (or a range of interfaces) to factory default values.

Parameter	Description
<INTERFACE-ID>	Specifies the ID of a single interface or range of interfaces. Format: **member/slot/port** or **member/slot/port-member/slot/port** to specify a range.

Examples

Resetting an interface:

```
switch(config)# default default interface 1/1/1
```

Resetting an range of interfaces:

```
switch(config)# default default interface 1/1/1-1/1/10
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

description

```
description <DESCRIPTION>
no description
```

Description

Associates descriptive information with an interface to help administrators and operators identify the purpose or role of an interface.

The **no** form of this command removes a description from an interface.

Parameter	Description
<DESCRIPTION>	Specify a description for the interface. Range: 1 to 64 ASCII characters (including space, excluding question mark).

Examples

Setting the description for an interface to **DataLink 01**:

```
switch(config-if)# description DataLink 01
```

Removing the description for an interface.

```
switch(config-if)# no description
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution rights for this command.

energy-efficient-ethernet

```
energy-efficient-ethernet
```

Description

Enables auto-negotiation of Energy-Efficient Ethernet (EEE) on an interface. EEE Negotiation is established only on auto-link negotiation with supported link partners.

Examples

Configuring an interface:

```
switch(config)# interface 1/1/1
switch(config-if)# energy-efficient-ethernet
```

Disabling Energy Efficient Ethernet on an interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no energy-efficient-ethernet
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config`	Administrators or local user group members with execution rights for this command.

flow-control

```
flow-control rxtx
no flow-control rxtx
```

Description

Command **flow-control** enables negotiation of IEEE 802.3x link-level flow control on the current interface. The switch advertises link-level flow control support to the link partner. The final configuration is determined based on the capabilities of both partners.

Each invocation of this command replaces the previous configuration.

The **no** form of these commands disables any configured flow control on the selected interface.

Parameter	Description
`rxtx`	Enables the ability to honor received and to transmit IEEE 802.3x LLFC pause frames to the remote device.

Usage (flow control)

- For interfaces that auto-negotiate, link-level flow control is subject to negotiation, plus speed and other parameters. Both ends of the link must negotiate the same flow control mode for it to be applied.
- For interfaces that do not auto-negotiate, the configured link-level flow control mode is always applied and the user is responsible for ensuring that both ends of the link are configured for the same mode.
- All members of a LAG must have the same flow control configuration.
- Lossless flow control is only supported for single destination unicast traffic. Replicated traffic (for example, broadcast, multicast, mirroring) cannot be guaranteed to be lossless.
- Lossless behavior is not supported when operating in a VSF stack configuration.
- Lossless flow control will only operate correctly when both the ingress and egress interfaces have flow control enabled.

Examples

Enabling support for RXTX flow control:

```
switch(config)# interface 1/1/1
switch(config-if)# flow-control txrx
```

Disabling support for RXTX flow control:

```
switch(config)# interface 1/1/1
switch(config-if)# no flow-control txrx
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution rights for this command.

interface

`interface <PORT-NUM>`

Description

Switches to the **config-if** context for a physical port. This is where you define the configuration settings for the logical interface associated with the physical port.

Parameter	Description
<PORT-NUM>	Specifies a physical port number. Format: `member/slot/port`.

Examples

Configuring an interface:

```
switch(config)# interface 1/1/1
switch(config-if)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

interface vlan

```
interface vlan <VLAN-ID>
no interface vlan <VLAN-ID>
```

Description

Creates an interface VLAN also know as an SVI (switched virtual interface) and changes to the **config-if-vlan** context. The specified VLAN must already be defined on the switch.

The **no** form of this command deletes an interface VLAN.

Parameter	Description
<VLAN-ID>	Specifies the interface ID.
none	Do not reserve any internal VLANs.

Examples

```
switch# config
switch(config)# vlan 10
switch(config-vlan-10)# exit
switch(config)# interface vlan 10
switch(config-if-vlan)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

ip address

```
ip address <IPV4-ADDR>/<MASK>
no ip address <IPV4-ADDR>/<MASK>
```

Description

Sets an IP/IPv6 address on the interface VLAN.

The **no** form of this command removes the IP/IPv6 address from the interface.

Parameter	Description
<IPV4-ADDR>	Specifies an IP address in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255. You can remove leading zeros. For example, the address **192.169.005.100** becomes **192.168.5.100**.
<MASK>	Specifies the number of bits in the address mask in CIDR format (**x**), where **x** is a decimal number from 0 to 128.

Examples

Assigning the IP address **192.168.199.1** with a mask of **24** bits to interface VLAN **10**:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# ip address 192.168.199.1/24
```

Removing the IP address **192.168.199.1** with a mask of **24** bits from interface VLAN **10**:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# no ip address 192.168.199.1/24
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-if-vlan`	Administrators or local user group members with execution rights for this command.

ip mtu

`ip mtu <VALUE>`

`no ip mtu`

Description

Sets the IP MTU (maximum transmission unit) for an interface. This defines the largest IP packet that can be sent or received by the interface. This value should be less than or equal to the overall MTU for the interface.

The **no** form of this command sets the IP MTU to the default value 1500.

Parameter	Description
<VALUE>	Specifies the IP MTU in bytes. Range: 68 to 9198. Default: 1500.

Examples

Setting the IP MTU to 576 bytes:

```
switch(config-if-vlan)# ip mtu 576
```

Command History

Release	Modification
10.08	Subinterface support added.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-if-vlan`	Administrators or local user group members with execution rights for this command.

ip source-interface

```
ip source-interface {sflow |  tftp |  radius | tacacs | ntp | syslog | simplivity | dns |
all} {interface <IFNAME> | <IPV4-ADDR>} [vrf <VRF-NAME>]
no ip source-interface {sflow |  tftp |  radius | tacacs | ntp | syslog | simplivity |
dns | all} [interface <IFNAME> | <IPV4-ADDR>] [vrf <VRF-NAME>]
```

Description

Sets a single source IP address for a feature on the switch. This ensures that all traffic sent the feature has the same source IP address regardless of how it egresses the switch. You can define a single global address that applies to all supported features, or an individual address for each feature.

This command provides two ways to set the source IP addresses: either by specifying a static IP address, or by using the address assigned to a switch interface. If you define both options, then the static IP address takes precedence.

The `no` form of this command deletes the single source IP address for all supported services, or a specific service.

Parameter	Description
`sflow | tftp | radius | tacacs | ntp | syslog | simplivity | dns | all`	Sets a single source IP address for a specific service. The `all` option sets a global address that applies to all protocols that do not have an address set.
`interface` *<IFNAME>*	Specifies the name of the interface from which the specified service obtains its source IP address. The interface must have a valid IP address assigned to it. If the interface has both a primary and secondary IP address, the primary IP address is used.
<IPV4-ADDR>	Specifies the source IP address to use for the specified service. The IP address must be defined on the switch, and it must exist on the specified VRF (which is the `default` VRF, if the `vrf` option is not used). Specify the address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.

Examples

Setting the IPv4 address `10.10.10.5` as the global single source address:

```
switch# config
switch(config)# ip source-interface all 10.10.10.5
```

Clearing the global single source IP address **10.10.10.5**:

```
switch(config)# no ip source-interface all 10.10.10.5
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

ipv6 address

`ipv6 address` *<IPV6-ADDR>*/*<MASK>*`{eui64 | [tag` *<ID>*`]}`

```
no ipv6 address <IPV6-ADDR>/<MASK>
```

Description

Sets an IPv6 address on the interface.

The **no** form of this command removes the IPv6 address on the interface.

This command automatically creates an IPv6 link-local address on the interface. However, it does not add the **ipv6 address link-local** command to the running configuration. If you remove the IPv6 address, the link-local address is also removed. To maintain the link-local address, you must manually execute the **ipv6 address link-local** command.

Parameter	Description
<IPV6-ADDR>	Specifies the IP address in IPv6 format (**xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address **2222:0000:3333:0000:0000:0000:4444:0055** becomes **2222:0:3333::4444:55**.
<MASK>	Specifies the number of bits in the address mask in CIDR format (**x**), where **x** is a decimal number from 0 to 128.
eui64	Configure the IPv6 address in the EUI-64 bit format.
tag *<ID>*	Configure route tag for connected routes. Range: 0 to 4294967295. Default: 0.

Examples

Setting the IPv6 address **2001:0db8:85a3::8a2e:0370:7334** with a mask of 24 bits:

```
switch(config-if)# ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

Removing the IP address **2001:0db8:85a3::8a2e:0370:7334** with mask of 24 bits:

```
switch(config-if)# no ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

ipv6 source-interface

```
ipv6 source-interface {sflow |  tftp |  radius | tacacs | ntp | syslog | simplivity | dns
| all} {interface <IFNAME> | <IPV6-ADDR>}
no ipv6 source-interface {sflow |  tftp |  radius | tacacs | ntp | syslog | simplivity |
dns | all} [interface <IFNAME> | <IPV6-ADDR>]
```

Description

Sets a single source IP address for a feature on the switch. This ensures that all traffic sent the feature has the same source IP address regardless of how it egresses the switch. You can define a single global address that applies to all supported features, or an individual address for each feature.

This command provides two ways to set the source IP addresses: either by specifying a static IP address, or by using the address assigned to a switch interface. If you define both options, then the static IP address takes precedence.

The `no` form of this command deletes the single source IP address for all supported protocols, or a specific protocol.

Parameter	Description
`sflow | tftp | radius | tacacs | ntp | syslog | simplivity | dns | all`	Sets a single source IP address for a specific protocol. The `all` option sets a global address that applies to all protocols that do not have an address set.
`interface <IFNAME>`	Specifies the name of the interface from which the specified protocol obtains its source IP address.
`<IPV6-ADDR>`	Specifies the source IP address to use for the specified protocol. The IP address must be defined on the switch, and it must exist on the specified VRF (which is the `default` VRF). Specify the IP address in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

Examples

Configuring the IPv6 address `2001:DB8::1` as the global single source address:

```
switch# config
switch(config)# ip source-interface all 2001:DB8::1/32
```

Stop the source IP address from using the IP address on interface **1/1/1** on VRF `default`.

```
switch(config)# no ip source-interface all interface 1/1/1 vrf default
```

Clear the source IP address `2001:DB8::1`.

```
switch(config)# no ip source-interface all 2001:DB8::1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

mtu

```
mtu <VALUE>
no mtu
```

Description

Sets the MTU (maximum transmission unit) for an interface. This defines the maximum size of a layer 2 (Ethernet) frame. Frames larger than the MTU (1500 bytes by default) are dropped.

To support jumbo frames (frames larger than 1522 bytes), increase the MTU as required by your network. A frame size of up to 9198 bytes is supported.

The largest possible layer 1 frame will be 18 bytes larger than the MTU value to allow for link layer headers and trailers.

The **no** form of this command sets the MTU to the default value 1500.

Parameter	Description
<VALUE>	Specifies the MTU in bytes. Range: 46 to 9198. Default: 1500.

Examples

Setting the MTU on interface **1/1/1** to 1000 bytes:

```
switch(config)# interface 1/1/1
switch(config-if)# no routing
switch(config-if)# mtu 1000
```

Setting the MTU on interface **1/1/1** to the default value:

```
switch(config)# interface 1/1/1
switch(config-if)# no routing
switch(config-if)# no mtu
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution rights for this command.

persona

```
persona {access | uplink | custom <PERSONA-NAME>} [copy | attach]
no persona {access | uplink | custom <PERSONA-NAME>} [copy | attach]
```

Description

Associates one of three persona types with an interface to classify the purpose or role of an interface. On the 10000 Switch Series, “access” persona ports are typically connected to workloads / VMs, and the “uplink” (fabric) persona ports are connected to the core / spine.

The **no** form of this command removes the interface persona.

Parameter	Description
`access`	Selects the **access** persona type.
`uplink`	Selects the **uplink** persona type.
`custom <PERSONA-NAME>`	Selects the **custom** persona type with a user-provided name. Range: 1 to 64 printable ASCII characters including space.
`copy`	Specifies the mode: copies settings from the persona interface of the same name.
`attach`	Specifies the mode: attaches the specified interface to the persona interface of the same name.

Usage

- If the mode is specified, either copy or attach, the interface configuration is dependent on the interface template whose name is "access", "uplink", or "<*PERSONA-NAME*>". On the other hand, if the mode is not specified, then the persona is just a label in the interface, and its configuration is not modified even if the interface persona exists. When configuring the mode, one of the following options is possible:
  - The **copy** option performs a one-time copy of the template interface. Subsequent changes to the template are not copied and the 'persona' setting is just a label. If the mode is set to **copy** and the interface persona does not exist, then the CLI command fails with the message "Interface persona not found".
  - The **attach** option performs a copy of the template interface, and subsequent changes to the template interface configuration are immediately applied to all attached interfaces. The template interface does not need to exist before attaching other interfaces to it. After attaching a template, the copied settings can be modified for an individual interface. However, any change in the attached template will overwrite the modified values with the new template values.
- When a mode is specified, it should match an interface created with the command interface persona <*PERSONA-NAME*>. The only exception to this rule is when the mode is set to `attach` and the persona does not already exist.

- The mode is only available to be configured for an interface that meets the following conditions:
  - IS a physical interface
  - IS NOT a LAG member
  - IS NOT a persona interface

Examples

Configuring an access persona:

```
switch(config)# interface 1/1/1
switch(config-if)# persona access
```

Configuring an uplink persona:

```
switch(config)# interface 1/1/1
switch(config-if)# persona uplink
```

Configuring a custom persona named "mypersona":

```
switch(config)# interface 1/1/1
switch(config-if)#persona custom mypersona
```

Removing the persona setting.

```
switch(config-if)# no persona
```

Copying a predefined persona name configuration to an interface:

1. Configuring the interface persona:

```
switch(config)# interface persona uplink
switch(config-if)# no shutdown
switch(config-if)# no routing
switch(config-if)# vlan access 100
switch(config-if)# exit
```

2. Applying the configuration from the persona named "mypersona" with **copy** mode:

```
switch(config)# interface 1/1/1
switch(config-if)# persona custom mypersona copy
switch(config-if)# exit
```

Attaching a custom persona name named "mypersona" to several interfaces simultaneously:

1. Configuring an interface persona named "mypersona":

```
switch(config)# interface persona mypersona
switch(config-if)# no shutdown
switch(config-if)# vrf attach upstream
switch(config-if)# exit
```

2. Applying the "mypersona" configuration with **attach** mode:

```
switch(config)# interface 1/1/1-1/1/24
switch(config-if)# persona custom mypersona attach
switch(config-if)# exit
```

Command History

Release	Modification
10.10	Added optional parameters: attach, copy.
10.09	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution rights for this command.

rate-interval

`rate-interval <VALUE>`

`no rate-interval`

Description

This command sets the time interval to calculate interface rates. Lower intervals are more useful for detecting traffic bursts, but may increase computation load to the overall system. Intervals must be a multiple of five seconds. The command-line interface will not accept a rate interval value that is not a multiple of five.

The **no** form of this command sets the rate collection interval to the default value of 300 seconds.

Parameter	Description
<VALUE>	The statistics rate collection interval in seconds. The supported range is 5-300 seconds, where the number of seconds is a multiple of five.

Examples

Setting the rate collection interval to 50 seconds

```
switch(config)# interface 1/1/1
switch(config-if)# rate-interval 50
```

Setting the rate collection interval to the default value:

```
switch(config-if)# no rate-interval
```

The following example shows the command-line interface warning that appears while configuring an invalid rate-interval.

```
switch(config)# interface 1/1/1
switch(config-if)# rate interval 6
The interval must be a multiple of 5.
```

Usage

The rate collection interval must be configured in the multiples of 5. Any other value will be rejected and the CLI will display the error message, **The interval must be a multiple of 5.**

Command History

Release	Modification
10.12.1000	Command supported on all platforms.
10.12	Command Introduced on 6300 and 8360 Switch series.

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution rights for this command.

show allow-unsupported-transceiver

`show allow-unsupported-transceiver`

Description

Displays configuration and status of unsupported transceivers.

Examples

Showing unallowed unsupported transceivers:

```
switch(config)# show allow-unsupported-transceiver
Allow unsupported transceivers : no
Logging interval               : 1440 minutes
---------------------------------------------
Port       Type           Status
---------------------------------------------
1/1/31     SFP-SX         unsupported
1/1/32     SFP-1G-BXD     unsupported
1/1/2      SFP28DAC3      unsupported
```

Showing allowed unsupported transceivers:

```
switch# show allow-unsupported-transceiver
Allow unsupported transceivers : yes
Logging interval               : 1440 minutes
---------------------------------------------
```

```
Port        Type            Status
---------------------------------------------
1/1/31      SFP-SX          unsupported-allowed
1/1/32      SFP-1G-BXD      unsupported-allowed
1/1/2       SFP28DAC3       unsupported
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution rights for this command.

show interface

```
show interface [<IFNNAME>|<IFRANGE>] [brief | physical]
show interface [<IFNNAME>|<IFRANGE>] [extended [non-zero] | [human-readable]]
show interface [<IFNNAME>] monitor [human-readable]
show interface [lag | vlan ] [<ID>] [brief]
show interface lag [<LAG-ID>] [extended [non-zero] | [human-readable]]
show interface lag [<LAG-ID>] monitor [human-readable]
```

Description

Shows active configurations and operational status information for interfaces.

Parameter	Description
<IFNAME>	Specifies a interface name.
<IFRANGE>	Specifies the port identifier range.
`brief`	Shows brief info in tabular format.
`physical`	Shows the physical connection info in tabular format.
`extended`	Shows additional statistics, including the **tx filtered** and **rx filtered** counters. ■ Rx filter packets are protocol packets received when the protocol is disabled on the switch and there is only one port in the VLAN. Protocols include OSPF, PIM, RIP, LACP, and LLDP. ■ An example of a Tx filtered packet would be a multicast packet being filtered from going out of the ingress port.
`human-readable`	Shows statistics rounded to the nearest power of 1000, for example, 1K, 345M, 2G. This is available only in the CLI interface output.
`non-zero`	Shows only non zero statistics.

Parameter	Description
LAG	Shows LAG interface information.
monitor	Continuously monitor interface statistics.
VLAN	Shows VLAN interface information.
<LAG-ID>	Specifies the LAG number. Range: 1-256
<VLAN-ID>	Specifies the VLAN ID. Range: 1-4094

Examples

Showing information when interface 1/1/1 is configured:

```
MDI mode: MDIX
VLAN Mode: native-untagged
Native VLAN: 1
Allowed VLAN List: all
Rate collection interval: 300 seconds
Rates                               RX                   TX         Total (RX+TX)
------------- -------------------- -------------------- --------------------
Mbits / sec                       0.00                 0.00                 0.00
KPkts / sec                       0.00                 0.00                 0.00
Unicast                           0.00                 0.00                 0.00
Multicast                         0.00                 0.00                 0.00
Broadcast                         0.00                 0.00                 0.00
Utilization %                     0.00                 0.00                 0.00
Statistics                          RX                   TX                Total
------------- -------------------- -------------------- --------------------
Packets                              0                    0                    0
Unicast                              0                    0                    0
Multicast                            0                    0                    0
Broadcast                            0                    0                    0
Bytes                                0                    0                    0
Jumbos                               0                    0                    0
Dropped                              0                    0                    0
Filtered                             0                    0                    0
Pause Frames                         0                    0                    0
Errors                               0                    0                    0
CRC/FCS                              0                  n/a                    0
Collision                          n/a                    0                    0
Runts                                0                  n/a                    0
Giants                               0                  n/a                    0
```

Showing information when the interface is currently linked at a downshifted speed:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is up
...
Auto-negotiation is on with downshift active
```

Showing information when the interface is currently linked with energy-efficient-ethernet negotiated:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is up
...
Energy-Efficient Ethernet is enabled and active
```

Showing information when the interface is configured with EEE and the EEE has auto-negotiated:

```
switch(config-if)# show interface 1/1/1 physical
---------------------------------------------------------------------------------
---------------------------------------------------------
                            Link    Admin         Speed           Flow-Control
     EEE       PoE Power                      Port
Port         Type          Status   Config   Status | Config    Status | Config
Status | Config  (Watts)    State Information  Description
---------------------------------------------------------------------------------
---------------------------------------------------------
1/1/1       1GbT          up       up       1G       auto      off      off
on       on       --        10M/100M/1G        --
```

Showing the monitor information:

In monitor mode, the CLI refreshes data automatically until it is exited by entering **q**. Pressing **?** opens the help menu to display which options are available in this context.

```
Interface 1/1/1 is up
Rate                                RX                   TX        Total (RX+TX)
---------------- -------------------- -------------------- --------------------
MBits / sec                   30196.43             30196.43             60392.85
MPkts / sec                   58977.39             58977.40            117954.79
Unicast                           0.00                 0.00                 0.00
Multicast                     58977.39             58977.40            117954.79
Broadcast                         0.00                 0.00                 0.00
Utilization %                    75.49                75.49               150.98
Statistic                           RX                   TX        Total (RX+TX)
---------------- -------------------- -------------------- --------------------
Packets                     4756527649           4756527865           9513055514
Unicast                              0                    0                    0
Multicast                   4756527649           4756527865           9513055514
Broadcast                            2                    0                    2
Bytes                     304417778668         304417795428         608835574096
Jumbos                               0                    0                    0
Dropped                              0          19028847730          19028847730
Pause Frames                         0                    0                    0
Errors                               0                    0                    0
CRC/FCS                              0                  n/a                    0
help: ?, quit: q
```

```
Help for Interface Monitor
h  Toggle human-readable mode
c  Clear interface statistics
Does not apply to rates
Arrows, PgUp, PgDn, Home, End
Navigate interface statistics
Delay: 2
help: ?, quit: q
```

Showing the output for interface 1/1/1 in human-readable format:

In human-readable format, the **< 1** symbol for **Utilization** indicates that the amount of packets is between zero and one. This is true in cases where the number of bytes increases but the number of packets and the **Utilization** value is not displayed even in the normal output, where the human-readable parameter is not included in the command.

```
switch(config-if)# show interface 1/1/1 human-readable
Interface 1/1/1 is up
Rate                               RX                   TX       Total (RX+TX)
---------------- -------------------- -------------------- --------------------
Bits / sec                         3M                   3M                   6M
Pkts / sec                        316                  316                  633
Unicast                           319                  319                  638
Multicast                           0                    0                    0
Broadcast                           0                    0                    0
Utilization %                     < 1                  < 1                  < 1
Statistic                          RX                   TX                Total
---------------- -------------------- -------------------- --------------------
Packets                          577K                 577K                   1M
Unicast                          577K                 577K                   1M
Multicast                           0                   51                   51
Broadcast                           0                   15                   15
Bytes                            744M                 745M                   1G
Jumbos                              0                    0                    0
Dropped                             0                    0                    0
Filtered                            0                    0                    0
Pause Frames                        0                    0                    0
Errors                              0                    0                    0
CRC/FCS                             0                  n/a                    0
Collision                         n/a                    0                    0
Runts                               0                  n/a                    0
Giants                              0                  n/a                    0
```

Showing information about extended counters:

The output of the `show interface extended` command varies depending on the switch model and configuration.

```
switch(config-if)# show interface 1/1/17 extended
-------------------------------------------------------------------
Interface 1/1/17
-------------------------------------------------------------------
Statistics                                      Value
-------------------------------------------------------------------
Dot1d Tp Port In Frames                         547
Dot1d Tp Port Out Frames                        608
Dot3 In Pause Frames                            0
Dot3 Out Pause Frames                           0
Ethernet Stats Broadcast Packets                19
Ethernet Stats Bytes                            40162
Ethernet Stats Packets                          342
...
-------------------------------------------------------------------
Error-Statistics                                Value
-------------------------------------------------------------------
Dot1d Base Port MTU Exceeded Discards           0
```

```
Dot3 Control In Unknown Opcodes                0
Dot3 Stats Alignment Errors                    0
Dot3 Stats FCS Errors                          0
Dot3 Stats Frame Too Longs                     0
Dot3 Stats Internal Mac Transmit Errors        0
Ethernet RX Oversize Packets                   0
...
```

Command History

Release	Modification
10.11	Added `monitor` parameter.
10.10	Added `human-readable` parameter.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface dom

`show interface [<INTERFACE-ID>] dom [detail]`

Description

Shows diagnostics information and alarm/warning flags for the optical transceivers (SFP, SFP+). This information is known as DOM (Digital Optical Monitoring). DOM information also consists of vendor determined thresholds which trigger high/low alarms and warning flags.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: **member/slot/port**.
`detail`	Show detailed information.

Example

```
switch# show interface dom
----------------------------------------------------------------------------------
Port    Type      Channel  Temperature Voltage  Tx Bias  Rx Power      Tx Power
                            (Celsius)   (Volts)  (mA)     (mW/dBm)      (mW/dBm)
```

```
--------------------------------------------------------------------------------
1/1/1   SFP+SR             47.65       3.31     8.40     0.08, -10.96  0.63, -2.49
1/1/2   SFP+SR             n/a         n/a      n/a      n/a           n/a
1/1/3   SFP+DA3            42.10       3.24     n/a      n/a           n/a
1/1/4   QSFP+SR4  1        44.46       3.30     6.12     0.08, -10.96  0.63, -1.95
2                          44.46       3.30     6.04     0.08, -10.96  0.63, -2.00
3                          44.46       3.30     6.51     0.08, -10.96  0.60, -2.16
4                          44.46       3.30     6.19     0.08, -10.96  0.63, -1.94
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface energy-efficient ethernet

`show interface [<IFNAME>|<IFRANGE>] energy-efficient-ethernet`

Description

Displays Energy-Efficient Ethernet information for the interface.

Parameter	Description
<IFNAME>	Specifies the name of an interface on the switch. Use the format **member/slot/port** (for example, **1/1/1**).
<IFRANGE>	Specifies the port identifier range of an interface on the switch. Use the format **member/slot/port** (for example, **1/1/1**).

Example

The following example shows when the interfaces are Energy-Efficient Ethernet capable.

```
switch# show interface energy-efficient-ethernet
------------------------------------------------------------------
Port     Enabled    Negotiated    Speed        TX Wake     RX Wake
                                  (MB/s)       Time(us)    Time (us)
------------------------------------------------------------------
1/1/1    no         no            --           --          --
1/1/2    yes        yes           100          36          36
1/1/3    yes        yes           1000         17          17
1/1/4    no         no            --           --          --
1/1/5    yes        no            1000         --          --
```

```
switch# show interface 1/1/1 energy-efficient-ethernet
--------------------------------------------------------------------------
Port        Enabled     Negotiated    Speed       TX Wake       RX Wake
                                      (Mb/s)      Time (us)     Time (us)
--------------------------------------------------------------------------
1/1/1       no          no            1000        --            --
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface flow-control

`show interface [<IFNNAME>|<IFRANGE>] flow-control [detail]`

Description

Shows the flow control configuration, status, and statistics of the specified interface for interfaces on which flow control is enabled.

If **detail** is not specified, this command shows a summary of all flow controlled interfaces with one interface per line. If **detail** is specified, this command shows flow control detailed statistics.

As of AOS-CX 10.10, the separate **show flow-control** command has been removed, with it being effectively replaced by this command.

Parameter	Description
`<IFNNAME>|<IFRANGE>`	Specifies the interface (port) name or range. When no interface range is specified, only interfaces with flow control enabled in the configuration or status are shown.
`detail`	Shows detailed information.

Examples

Showing summary flow control information:

```
switch# show interface flow-control
----------- -------------------------------------
Port        Flow
            Control
----------- -------------------------------------
1/1/1       config: llfc rx
            status: llfc rx
1/1/2       config: llfc rx
            status: none
```

Showing summary flow control information with PFC:

```
switch# show interface flow-control
----------- -------------------------------------
Port        Flow
            Control
----------- -------------------------------------
1/1/1       config: pfc rxtx-1,2
            status: pfc rxtx-1,2
1/1/2       config: pfc rxtx-5
            status: none
```

Showing summary flow control information with PFC:

```
switch# show interface flow-control
  Flow Control Watchdog Settings
      Trigger Timeout:  100 milliseconds
      Resume Time:      100 milliseconds
----------- ------------------------------------- ------------- --------
Port        Flow                                  Watchdog      Watchdog
            Control                               Status        Timeouts
----------- ------------------------------------- ------------- --------
1/1/1       config: llfc rx
            status: llfc rx
1/1/2       config: llfc rx                       incompatible         0
            status: llfc rx
1/1/10      config: pfc rxtx-1,2                  enabled           1234
            status: pfc rxtx-1,2
1/1/12      config: pfc rxtx-1,2                  error                0
            status: pfc rxtx-1,2
1/1/32:4    config: pfc rxtx-5
            status: pfc rxtx-5
```

Showing summary flow control information where the configuration does not match status due to a reboot required to apply PFC configuration in hardware:

```
switch# show interface flow-control

 Flow Control Watchdog Settings
   Trigger Timeout:  100 milliseconds (actual: not applied)
   Resume Time:      100 milliseconds (actual: not applied)

 ----------- ------------------------------------- ------------- --------
 Port        Flow                                  Watchdog      Watchdog
             Control                               Status        Timeouts
 ----------- ------------------------------------- ------------- --------
```

```
1/1/1        config: llfc rx
             status: llfc rx
1/1/2        config: llfc rx                       incompatible         0
             status: llfc rx
1/1/10       config: pfc rxtx-1,2                  pending           1234
             status: none
1/1/12       config: pfc rxtx-1,2                  pending              0
             status: none
1/1/32:4     config: pfc rxtx-5
             status: none
```

Showing detailed flow control information with RX flow control enabled:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
 Admin state is up
 Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
 Flow-control: llfc rx

 Statistics                             RX
 -------------------- --------------------
 Dot3 Pause Frames                       0
```

Showing detailed flow control information with RX flow control enabled:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
 Admin state is up
 Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
 Flow-control: llfc rx
 Flow-control watchdog: disabled

 Statistics                             RX
 -------------------- --------------------
 Dot3 Pause Frames                       0
```

Showing detailed flow control information with RXTX flow control enabled:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
 Admin state is up
 Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
 Flow-control: llfc rxtx

 Statistics                             RX                   TX
 -------------------- -------------------- --------------------
 Dot3 Pause Frames                       0                    0
```

Showing detailed flow control information with PFC enabled:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
 Admin state is up
 Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
 Flow-control: pfc rxtx-4,5

 Statistics                             RX                   TX
 -------------------- -------------------- --------------------
 Priority 0 Pauses                       0                    0
 Priority 1 Pauses                       0                    0
 Priority 2 Pauses                       0                    0
 Priority 3 Pauses                       0                    0
 Priority 4 Pauses                       0                    0
 Priority 5 Pauses                       0                    0
 Priority 6 Pauses                       0                    0
 Priority 7 Pauses                       0                    0
 Total Pause Frames                      0                    0
```

Showing detailed flow control information with PFC enabled and flow control watchdog disabled:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
 Admin state is up
 Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
 Flow-control: pfc rxtx-4,5
 Flow-control watchdog: disabled

 Statistics                             RX                   TX
 -------------------- -------------------- --------------------
 Priority 0 Pauses                       0                    0
 Priority 1 Pauses                       0                    0
 Priority 2 Pauses                       0                    0
 Priority 3 Pauses                       0                    0
 Priority 4 Pauses                       0                    0
 Priority 5 Pauses                       0                    0
 Priority 6 Pauses                       0                    0
 Priority 7 Pauses                       0                    0
 Total Pause Frames                      0                    0
Interface 1/1/1 is up
 Admin state is up
 Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
 Flow-control: pfc rxtx-4,5
 Flow-control watchdog: disabled

 Statistics                             RX                   TX
 -------------------- -------------------- --------------------
 Priority 0 Pauses                       0                    0
 Priority 1 Pauses                       0                    0
 Priority 2 Pauses                       0                    0
 Priority 3 Pauses                       0                    0
 Priority 4 Pauses                       0                    0
 Priority 5 Pauses                       0                    0
 Priority 6 Pauses                       0                    0
 Priority 7 Pauses                       0                    0
 Total Pause Frames                      0                    0
```

Showing detailed flow control information with both PFC and flow control watchdog enabled:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
 Admin state is up
 Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
 Flow-control: pfc rxtx-4,5
 Flow-control watchdog: enabled

 Statistics                              RX                   TX
 -------------------- -------------------- --------------------
 Priority 0 Pauses                        0                    0
 Priority 1 Pauses                        0                    0
 Priority 2 Pauses                        0                    0
 Priority 3 Pauses                        0                    0
 Priority 4 Pauses                        0                    0
 Priority 5 Pauses                        0                    0
 Priority 6 Pauses                        0                    0
 Priority 7 Pauses                        0                    0
 Total Pause Frames                       0                    0

 Queue        Watchdog Timeouts
 ------------ -----------------
 Queue 0                      0
 Queue 1                      0
 Queue 2                      0
 Queue 3                      0
 Queue 4                      0
 Queue 5                      0
 Queue 6                      0
 Queue 7                      0
```

Showing detailed flow control information when flow control watchdog is enabled in the configuration but it could not be applied because the configured flow control mode is not compatible with watchdog:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
 Admin state is up
 Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
 Flow-control: llfc rx
 Flow-control watchdog: incompatible
```

Showing detailed flow control information when flow control watchdog is enabled in the configuration but could not be applied because a compatible flow control mode first requires a reboot:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
 Admin state is up
 Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
 Flow-control: off
 Flow-control watchdog: pending
```

Command History

Release	Modification
10.10	Examples updated with new and changed output elements.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface statistics

```
show interface [<IFNAME>|<IFRANGE>] statistics [non-zero] [human-readable]
show interface [<IFNAME>|<IFRANGE>] statistics monitor [non-zero] [human-readable]
show interface [<IFNAME>|<IFRANGE>] error-statistics [non-zero] [human-readable]
show interface [<IFNAME>|<IFRANGE>] error-statistics monitor [non-zero] [human-readable]
show interface lag [<LAG-ID>] statistics [non-zero] [human-readable]
show interface lag [<LAG-ID>] statistics monitor [non-zero] [human-readable]
show interface lag [<LAG-ID>] error-statistics [non-zero] [human-readable]
show interface lag [<LAG-ID>] error-statistics monitor [non-zero] [human-readable]
```

Description

Shows statistics for switch interfaces such as packets transmitted and received, bytes transmitted and received, broadcast and multicast packets.

Parameter	Description
<IFNAME>	Specifies a interface name.
<IFRANGE>	Specifies the port identifier range.
`LAG`	Shows LAG interface information.
<LAG-ID>	Specifies the LAG number. Range: 1-256
`monitor`	Continuously monitor interface statistics.
`human-readable`	Shows statistics rounded to the nearest power of 1000, for example, 1K, 345M, 2G.
`non-zero`	Shows only non zero statistics.

Examples

Showing statistics of all interfaces:

```
show interface statistics
------------------------------------------------------------------------------------------------------------------------------------------------------------------
Interface                RX Bytes   RX Packets   RX Drops  TX Bytes  TX Packets  TX Drops  RX Broadcast RX Multicast  TX Broadcast TX Multicast  RX Pause   TX Pa
------------------------------------------------------------------------------------------------------------------------------------------------------------------
1/1/1                     2727136         1975          0     17796         195         0            82         1788            96           54          0
1/1/10                          0            0          0         0           0         0             0            0             0            0          0
1/1/11                          0            0          0         0           0         0             0            0             0            0          0
1/1/12                          0            0          0         0           0         0             0            0             0            0          0
...
1/1/30       - lag1             0            0          0     11271          92         0             0            0             0           51          0
1/1/31       - lag2          2360           25         50   2732119        2040         0             0            0           178         1839          0
1/1/32       - lag2             0            0          0     11373          93         0             0            0             0           51          0
vlan1                           0            0          0         0           0         0             0            0             0            0          0
```

Showing statistics of all interfaces with only non-zero statistics:

```
show interface statistics non-zero
------------------------------------------------------------------------------------------------------------------------------------------------------------
Interface                       RX Bytes  RX Packets  RX Drops  TX Bytes  TX Packets  TX Drops  RX Broadcast  RX Multicast  TX Broadcast  TX Multicast  RX Pause  TX Pause
------------------------------------------------------------------------------------------------------------------------------------------------------------
1/1/1                           2727136        1975         0     17796         195         0            82          1788            96            54         0         0
1/1/30       - lag1                   0           0         0     11271          92         0             0             0             0            51         0         0
1/1/31       - lag2                2360          25        50   2732119        2040         0             0             0           178          1839         0         0
1/1/32       - lag2                   0           0         0     11373          93         0             0             0             0            51         0         0
```

Showing statistics of all interfaces in the human-readable format:

```
show interface statistics human-readable
--------------------------------------------------------------------------------------------------------------------------
Interface             RX Bytes   RX Pkts  RX Drops  TX Bytes   TX Pkts  TX Drops  RX Bcast  RX Mcast  TX Bcast  TX Mcast  RX Pause  TX Pause
--------------------------------------------------------------------------------------------------------------------------
1/1/1                     744M      577K         0      745M      578K         0         0         0        73       287         0         0
1/1/2                     474M      367K         0      475M      369K         0         0         0        73       288         0         0
1/1/3                        0         0         0         0         0         0         0         0         0         0         0         0
```

Showing statistics of a single interfaces:

```
show interface 1/1/2 statistics
------------------------------------------------------------------------------------------------------------------------------------------------------------
Interface                      RX Bytes  RX Packets  RX Drops  TX Bytes  TX Packets  TX Drops  RX Broadcast  RX Multicast  TX Broadcast  TX Multicast  RX Pause  TX Pause
------------------------------------------------------------------------------------------------------------------------------------------------------------
1/1/2                           2725080        1931         0     25877         253         0            21          1788            65            55         0         0
```

Showing statistics of all members of a LAG interface:

```
show interface lag1 statistics
------------------------------------------------------------------------------------------------------------------------------------------------------------
Interface                      RX Bytes  RX Packets  RX Drops  TX Bytes  TX Packets  TX Drops  RX Broadcast  RX Multicast  TX Broadcast  TX Multicast  RX Pause  TX Pause
------------------------------------------------------------------------------------------------------------------------------------------------------------
1/1/3        - lag1                2424          26         0   2734082        2062         0             0             0           191          1848         0         0
1/1/30       - lag1                   0           0         0     12383         100         0             0             0             0            59         0         0
lag1                               2424          26         0   2746465        2162         0             0             0           191          1907         0         0
```

Showing error statistics of all interfaces:

```
show interface error-statistics
--------------------------------------------------------------------------------------------------
Interface                           RX Errors   TX Errors    Giants    Runts    CRC/FCS   Collisions
--------------------------------------------------------------------------------------------------
1/1/1                                     190          20    100647        0          0            0
1/1/10                                      0           0       100      290       7165          949
1/1/11                                      0           0         0        0          0            0
1/1/12                                      0           0         0        0          0            0
...
1/1/30 - lag1                            1500         500     45800        0          0            0
1/1/31 - lag2                               0           0        11       27          0            0
1/1/32 - lag2                               0           0         0        0          6           18
```

Showing monitor statistics:

The rows and columns of show interface monitor statistics depends on the length of width of the client terminal. The CLI can be navigated using the arrow keys as well as the PageUp, PageDown, Home, and End keys.

```
show interface statistics monitor
----------------------------------------------------------------
Interface                         RX Bytes          RX Packets >>
----------------------------------------------------------------
1/1/1                        3440525421984        53758209526
1/1/2                        3440526607008        53758228042
1/1/3                        3440527785312        53758246453
1/1/30                       3440559671264        53758744653
1/1/31                       3440560851680        53758763098
1/1/32                       3440562028704        53758781489

                                                     help: ?, quit: q
```

```
Help for Interface Monitor

f     Toggle full statistics
h     Toggle human-readable mode
n     Toggle non-zero mode
r     Toggle rate display

c     Clear interface statistics
         Does not apply to rates

Arrows, PgUp, PgDn, Home, End
         Navigate interface statistics

Delay:2
                                                                    help: ?, qui
```

Showing monitor error statistics in human-readable format:

```
show interface 1/1/1-1/1/3,1/1/30-1/1/32 error-statistics monitor human-readable
---------------------------------------------------------------------------------------------
Interface                RX Errors    TX Errors    RX Giants     RX Runts      CRC/FCS  Collisions
---------------------------------------------------------------------------------------------
1/1/1                            0            0            0            0            0            0
1/1/2                            0            0            0            0            0            0
1/1/3                            0            0            0            0            0            0
1/1/30                           0            0            0            0            0            0
1/1/31                           0            0            0            0            0            0
1/1/32                           0            0            0            0            0            0


Human-readable                                                               help: ?, quit: q
```

```
Help for Interface Monitor

h    Toggle human-readable mode
n    Toggle non-zero mode

c    Clear interface statistics
        Does not apply to rates

Arrows, PgUp, PgDn, Home, End
        Navigate interface statistics

Delay:2
                                                                       help: ?, quit: q
```

Command History

Release	Modification
10.11	Added `moitor` parameter.
10.10	Added `human-readable` parameter.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface transceiver

`show interface [<INTERFACE-ID>] transceiver [detail | threshold-violations]`

Description

Displays information about transceivers present in the switch. The information shown varies for different transceiver types and manufacturers. Only basic information is shown for unsupported HPE and third-party transceivers installed in the switch and they are also identified with an asterisk in the output.

Parameter	Description
<INTERFACE-ID>	Specifies the name or range of an interface on the switch. Use the format **member/slot/port** (for example, **1/3/1**).
`detail`	Show detailed information for the interfaces.
`threshold-violations`	Show threshold violations for transceivers.

Example

Showing summary transceiver information with identification of unsupported transceivers:

```
switch(config)# show interface transceiver
------------------------------------------------------------------
Port      Type           Product      Serial          Part
                         Number       Number          Number
------------------------------------------------------------------
1/1/15    SFP+SR         J9150D       xxxxxxxxxx      1990-4634
1/1/16    SFP+SR         J9150D       xxxxxxxxxx      1990-4634
```

Showing detailed transceiver information:

```
switch(config)# show interface transceiver detail
Transceiver in 1/1/15
 Interface Name       : 1/1/15
 Type                 : SFP+SR
```

```
 Connector Type      : LC
 Wavelength          : 850nm
 Transfer Distance   : 0.00km (SMF), 20m (OM1), 80m (OM2), 300m (OM3)
 Diagnostic Support  : DOM
 Product Number      : J9150D
 Serial Number       : xxxxxxxxxx
 Part Number         : 1990-4634

Status
 Temperature : 30.38C
 Voltage     : 3.26V
 Tx Bias     : 5.54mA
 Rx Power    : 0.56mW, -2.52dBm
 Tx Power    : 0.62mW, -2.08dBm
 Recent Alarms:
 Recent Errors:

Transceiver in 1/1/16
 Interface Name      : 1/1/16
 Type                : SFP+SR
 Connector Type      : LC
 Wavelength          : 850nm
 Transfer Distance   : 0.00km (SMF), 20m (OM1), 80m (OM2), 300m (OM3)
 Diagnostic Support  : DOM
 Product Number      : J9150D
 Serial Number       : xxxxxxxxxx
 Part Number         : 1990-4634

Status
 Temperature : 30.62C
 Voltage     : 3.28V
 Tx Bias     : 5.64mA
 Rx Power    : 0.61mW, -2.15dBm
 Tx Power    : 0.59mW, -2.29dBm

Recent Alarms:
Recent Errors:
```

Showing detailed transceiver information with identification of unsupported transceivers:

```
Transceiver in 1/1/2
 Interface Name      : 1/1/2
 Type                : SFP+ER (unsupported)
 Connector Type      : LC
 Wavelength          : 3590nm
 Transfer Distance   : 80m (SMF), 0m (OM1), 0m (OM2), 0m (OM3)
 Diagnostic Support  : DOM
 Vendor Name         : INNOLIGHT
 Vendor Part Number  : TR-PX15Z-NHP
 Vendor Part Revision: 1A
 Vendor Serial number: MYxxxxxxx

Status
 Temperature : 28.88C
 Voltage     : 3.30V
 Tx Bias     : 65.53mA
 Rx Power    : 0.00mW, -inf
 Tx Power    : 1.47mW, 1.67dBm

Recent Alarms:
 Rx Power low alarm
```

```
Rx Power low warning
Recent Errors:
```

Showing transceiver threshold-violations:

```
switch(config)# show interface transceiver threshold-violations
-----------------------------------------------------------------
Port      Type           Channel#  Recent Threshold Violations
-----------------------------------------------------------------
1/1/15    SFP+SR                   none
1/1/16    SFP+SR                   none

switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface utilization

`show interface [<IFNNAME>|<IFRANGE>] utilization [non-zero]`

Description

Displays physical port throughput and utilization.

Parameter	Description
<IFNAME>	Specifies an interface name.
<IFRANGE>	Specifies the port identifier range.
`utilization`	Displays utilization statistics.
`non-zero`	Displays non-zero statistics

Examples

The following example shows port utilization of all interfaces:

```
switch# show interface utilization
-------------------------|------------------------|------------------------|----------------
-----------|----------------------
Interval |             RX          |             TX          |    Total  (RX+TX)          |
Interface        seconds |   Mbps  KPkt/s  Util % |   Mbps  KPkt/s  Util % |    Mbps
KPkt/s   Util % | Description
-------------------------|------------------------|------------------------|----------------
-----------|----------------------
1/1/1                300  9578.02  788.70   95.78    25.70   45.89    0.26   9603.72
834.59    96.04   Aruba-AP
1/1/2                300    25.71   45.90    0.26  9581.09  788.96   95.81   9606.80
834.86    96.07   Aruba2530-AP-conce...
1/1/3  - lag123      300     0.00    0.00    0.00     0.00    0.00    0.00      0.00
0.00     0.00   ISL: SWRTS-0064-1
1/1/4                300  9261.79  804.52   92.62  9496.70  823.97   94.97  18758.50
1628.48   187.58   Backup data center...
1/1/5                300  9496.70  823.97   94.97  9261.79  804.52   92.62  18758.50
1628.48   187.58   --
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ip interface

`show ip interface <INTERFACE-ID>`

Description

Shows status and configuration information for an IPv4 interface.

Parameter	Description
<INTERFACE-ID>	Specifies the name of an interface. Format: **member/slot/port**.

Example

```
switch# show ip interface vlan1
Interface vlan1 is up
Admin state is up
Hardware: Ethernet, MAC Address: f8:60:f0:c9:11:60
IP MTU 1500
IPv4 address 10.120.3.8/26
```

```
switch# show interface <intfid>.id
Interface 1/1/14.1 is up
Admin state is up
IP MTU 1500
Description:
Hardware: Ethernet, MAC Address: b8:6a:97:22:2f:42
Encapsulation dot1q ID: 20
IPv4 address 30.0.0.1/24
L3 Counters: Rx Disabled, Tx Disable
```

```
switch# show interface lag2.1
Interface lag2.1 is up
Admin state is up
IP MTU 1500
Description:
Hardware: Ethernet, MAC Address: b8:6a:97:22:2f:42
Encapsulation dot1q ID: 30
L3 Counters: Rx Disabled, Tx Disabled
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ip source-interface

`show ip source-interface {sflow | tftp | radius | tacacs | all} [vrf <VRF-NAME>]`

Description

Shows single source IP address configuration settings.

Parameter	Description
`sflow	tftp	radius	tacacs	all`	Shows single source IP address configuration settings for a specific protocol. The **all** option shows the global setting that applies to all protocols that do not have an address set.

Examples

```
Showing single source IP address configuration settings for sFlow:
switch# show ip source-interface sflow
```

```
Source-interface Configuration Information
-----------------------------------------
Protocol        Source Interface
--------        ----------------
sflow            10.10.10.1
```

Showing single source IP address configuration settings for all protocols:

```
switch# show ip source-interface all
Source-interface Configuration Information
-----------------------------------------------------------------
Protocol        Src-Interface       Src-IP                   VRF
-----------------------------------------------------------------
all             vlan2               2.2.2.2             default
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show ipv6 interface

`show ipv6 interface <INTERFACE-ID>`

Description

Shows status and configuration information for an IPv6 interface.

Parameter	Description
<INTERFACE-ID>	Specifies an interface ID. Format: **member/slot/port**.

Examples

```
switch# show ipv6 interface vlan2
Interface vlan2 is up
  Admin state is up
    IPv6 address:
      2001::1/64 [VALID]
    IPv6 link-local address: fe80::883a:3080:247:c1c0/64 [VALID]
    IPv6 Forwarding feature: enabled
    IPv6 multicast groups locally joined:
      ff02::1  ff02::1:ff00:1  ff02::1:ff47:c1c0  ff02::1:ff00:0
      ff02::2
    IPv6 MTU 1500
```

```
    IPv6 unicast reverse path forwarding: none
    IPv6 load sharing: none
switch#
```

```
switch# show ipv6 interface <intfid>.id
Interface 1/1/14.1 is up
  Admin state is up
    IPv6 address:
      30::1/64 [VALID]
    IPv6 link-local address: fe80::b86a:97c0:122:2f42/64 [VALID]
    IPv6 virtual address configured: none
    IPv6 multicast routing: disable
    IPv6 Forwarding feature: enabled
    IPv6 multicast groups locally joined:
      ff02::1  ff02::1:ff22:2f42  ff02::1:ff00:1  ff02::1:ff00:0
      ff02::2
    IPv6 multicast (S,G) entries joined: none
    IPv6 MTU 1500
    IPv6 unicast reverse path forwarding: none
    IPv6 load sharing: none
Encapsulation dot1q ID: 20
```

```
switch# show ipv6 interface lag2.1
Interface lag2.1 is up
  Admin state is up
    IPv6 address:
      40::1/64 [VALID]
    IPv6 link-local address: fe80::b86a:97c0:122:2f42/64 [VALID]
    IPv6 virtual address configured: none
    IPv6 multicast routing: disable
    IPv6 Forwarding feature: enabled
    IPv6 multicast groups locally joined:
      ff02::1  ff02::1:ff22:2f42  ff02::1:ff00:1  ff02::1:ff00:0
      ff02::2
    IPv6 multicast (S,G) entries joined: none
    IPv6 MTU 1500
    IPv6 unicast reverse path forwarding: none
    IPv6 load sharing: none
  Encapsulation dot1q ID: 30
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

```
switch# show ipv6 interface vlan2
Interface vlan2 is up
  Admin state is up
    IPv6 address:
      2001::1/64 [VALID]
    IPv6 link-local address: fe80::883a:3080:247:c1c0/64 [VALID]
    IPv6 Forwarding feature: enabled
    IPv6 multicast groups locally joined:
      ff02::1  ff02::1:ff00:1  ff02::1:ff47:c1c0  ff02::1:ff00:0
      ff02::2
    IPv6 MTU 1500
    IPv6 unicast reverse path forwarding: none
    IPv6 load sharing: none
switch#
```

show ipv6 source-interface

`show ipv6 source-interface {sflow |  tftp |  radius | tacacs | all} [vrf <VRF-NAME>]`

Description

Shows single source IP address configuration settings.

Parameter	Description
`sflow	tftp	radius	tacacs	all`	Shows single source IP address configuration settings for a specific protocol. The **all** option shows the global setting that applies to all protocols that do not have an address set.
`vrf <VRF-NAME>`	Specifies the name of a VRF.

Examples

Showing single source IP address configuration settings for sFlow:

```
switch# show ipv6 source-interface sflow

Source-interface Configuration Information
----------------------------------------
Protocol        Source Interface
--------        ----------------
 sflow            2001:DB8::1
```

Showing single source IP address configuration settings for all protocols:

```
switch# show ipv6 source-interface all

Source-interface Configuration Information
---------------------------------------------------------------
Protocol        Src-Interface       Src-IP         VRF
---------------------------------------------------------------
 all             vlan2               2001::1       default

switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

shutdown

```
shutdown
no shutdown
```

Description

Disables an interface. Interfaces are disabled by default when created.

The **no** form of this command enables an interface.

Examples

Disabling an interface:

```
switch(config-if)# shutdown
```

Enabling an interface:

```
switch(config-if)# no shutdown
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

speed

```
speed {<SPEED> | <SPEED-DUPLEX> | auto [<SPEED>] }
no speed
```

Description

Configures the link speed, duplex, and auto-negotiation settings for an interface.

The no form of this command removes the configurations and returns to the defaults.

Parameter	Description
`Speed`	Configures interface speed, duplex, and auto-negotiation.
`10-full`	10 Mbps, full duplex, no auto-negotiation
`10-half`	10 Mbps, half duplex, no auto-negotiation
`100-full`	100 Mbps, full duplex, no auto-negotiation
`100-half`	100 Mbps, half duplex, no auto-negotiation
`1000-full`	1000 Mbps, full duplex, no auto-negotiation
`10g`	10 Gbps, full duplex, no auto-negotiation
`25g`	25 Gbps, full duplex, no auto-negotiation
`40g`	40 Gbps, full duplex, no auto-negotiation
`50g`	50 Gbps, full duplex, no auto-negotiation
`100g`	100 Gbps, full duplex, no auto-negotiation
`200g`	200 Gbps, full duplex, no auto-negotiation **NOTE:** Not applicable for override.
`400g`	400 Gbps, full duplex, no auto-negotiation **NOTE:** Not applicable for override.
`auto`	Auto-negotiate speed and duplex. More than one speed can be set at a time.
`10m`	Allow interface to link at 10 Mbps.
`100m`	Allow interface to link at 100 Mbps.
`1g`	Allow interface to link at 1 Gbps.
`2.5g`	Allow interface to link at 2.5 Gbps.
`5g`	Allow interface to link at 5 Gbps.
`10g`	Allow interface to link at 10 Gbps.
`25g`	Allow interface to link at 25 Gbps.
`40g`	Allow interface to link at 40 Gbps.
`50g`	Allow interface to link at 50 Gbps.
`100g`	Allow interface to link at 100 Gbps.

Parameter	Description
`200g`	Allow interface to link at 200 Gbps.
`400g`	Allow interface to link at 400 Gbps.

Usage

The following options can be configured for an interface. The option available is based on the interface type.

`speed <SPEED-DUPLEX>`

Uses a fixed speed and duplex mode with no auto-negotiation. Half-duplex is only supported for 10 Mbps and 100 Mbps link speeds.

`speed <SPEED>`

Uses a fixed speed with no auto-negotiation. If the currently installed transceiver does not support the speed, the setting is ignored and the port will use the highest speed that is supported.

`speed auto`

Uses auto-negotiation and offers all speeds supported by the port and transceiver. This is the default. If the link technology does not support auto-negotiation this setting is ignored, and the port uses the highest possible fixed speed.

`speed auto <SPEED>`

Uses auto-negotiation and offers the specified speeds only. For ports that support pluggable transceivers, only speeds supported by the transceiver are offered and other speeds are ignored. If the link technology does not support auto-negotiation, this setting is ignored and the port uses the highest possible fixed speed.

Examples

Configuring an interface to operate at a fixed speed of 1000 Mbps with full duplex and no auto-negotiation:

```
switch(config)# interface 1/1/1

switch(config-if)# speed 1000-full
```

Configuring an interface to operate at a fixed speed of 10 Gbps with no auto-negotiation:

```
switch(config)# interface 1/1

switch(config-if)# speed 10g
```

Configuring an interface to auto-negotiate and advertise only 1 Gbps and 2.5 Gbps speeds:

```
switch(config)# interface 1/1/1
switch(config-if)# speed auto 1g 2.5g
```

Configuring an interface to override the detected transceiver speed and use the configured speed if the installed transceiver does not support auto-negotiation:

```
switch(config)# interface 1/1/1
switch(config-if)#speed auto 50g override
```

Configuring an interface to use default settings for speed, duplex, and auto-negotiation:

```
switch(config)# interface 1/1/1
switch(config-if)#no speed
```

Command History

Release	Modification
10.09.0001	Speeds not supported by hardware hidden by CLI.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config-if`	Administrators or local user group members with execution rights for this command.

Chapter 6

Source interface selection

The source IP address is determined by the system and is typically the IP address of the outgoing interface in the routing table. However, routing switches may have multiple routing interfaces and outgoing packets can potentially be sent by different paths at different times. This results in different source IP addresses.

AOS-CX provides a configuration model that allows the selection of an IP address to use as the source address for all outgoing traffic. This allows unique identification of the software application on the server site regardless of which local interface has been used to reach the destination server. The source interface selection supports selecting an IP address or interface name.

If the source interface and source IP are configured, Source IP will have priority.

Source-interface selection commands

ip source-interface (protocol <ip-addr>)

```
ip source-interface <PROTOCOL> <IP-ADDR> [vrf <VRF-NAME>]
no ip source-interface <PROTOCOL> <IP-ADDR> [vrf <VRF-NAME>]
```

Description

Configures the source-interface IPv4 address to use for the specified protocol. If a VRF is not given, the default VRF applies. If no interface option is given, the device floods through interfaces and VRFs to reach Aruba Central. Whichever reaches Aruba Central will be picked automatically.

The `no` form of this command removes all configurations.

Parameter	Description

<PROTOCOL>	Specifies the protocol to configure.
	`all`
	Selects all protocols that can be configured by this command.
	`central`
	Selects Aruba Central.
	`dns`
	Selects DNS.
	`ntp`
	Selects NTP.
	`radius`
	Selects radius.
	`sflow`
	Selects sFLow.
	`simplivity`

Parameter	Description

	Selects simplivity.
	`syslog`
	Selects syslog.
	`tacacs`
	Selects TACACS.
	`tftp`
	Selects TFTP.

<IP-ADDR>	Specifies the IPv4 address.
vrf *<VRF-NAME>*	Specifies the VRF name.

Examples

Configuring source-interface IPv4 10.1.1.1 to use for the TFTP protocol:

```
switch(config)# ip source-interface tftp 10.1.1.1
```

Configuring source-interface IPv4 10.1.1.2 to use for the TFTP protocol on VRF green :

```
switch(config)# ip source-interface tftp 10.1.1.2 vrf green
```

Removing source-interface IPv4 10.1.1.1 configuration for the TFTP protocol:

```
switch(config)# no ip source-interface tftp 10.1.1.1
```

Removing source-interface IPv4 10.1.1.2 configuration for TFTP protocol on VRF green:

```
switch(config)# no ip source-interface tftp 10.1.1.2 vrf green
```

Configuring source-interface IPv4 10.1.1.1 to use for the DNS protocol:

```
switch(config)# ip source-interface dns 10.1.1.1
```

Configuring source-interface IPv4 10.1.1.2 to use for the DNS protocl on VRF green :

```
switch(config)# ip source-interface dns 10.1.1.2 vrf green
```

Removing source-interface IPv4 10.1.1.1configuration for the DNS protocol:

```
switch(config)# no ip source-interface tftp 10.1.1.1
```

Removing source-interface IPv4 10.1.1.2 configuration for the DNS protocol on VRF green:

```
switch(config)# no ip source-interface dns 10.1.1.2 vrf green
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

ip source-interface

`ip source-interface <PROTOCOL> {<IP-ADDR>|interface <IFNAME>} [vrf <VRF-NAME>]`
`no ip source-interface <PROTOCOL> interface <IFNAME> [vrf <VRF-NAME>]`

Description

Configures the IPv4 source-interface interface to use for the specified protocol. If a VRF is not given, the default VRF applies.

The **no** form of this command removes the specified configuration.

Parameter	Description

<PROTOCOL>	Specifies the protocol to configure.
	`all`
	Selects the source for all protocols covered by this command.
	`central`
	Selects Aruba Central.
	`dhcp_relay`
	Selects DHCP relay. When you configure a dhcp_relay source interface, you must also enable DHCP relay Option 82 using the **dhcp-relay option 82 source-interface** command.
	`dns`
	Selects DNS.
	`http`
	Selects HTTP.
	`ntp`
	Selects NTP.
	`radius`
	Selects RADIUS.
	`sflow`
	Selects sFLow.
	`sftp-scp`

Parameter	Description

	Selects SFTP and SCP.
	`ssh-client`
	Selects SSH Client.
	`syslog`
	Selects the source for syslog packets.
	`tacacs`
	Selects the source for TACACS packets.
	`tftp`
	Selects TFTP.
	`ubt`
	Selects UBT.

	Specifies the VRF name.
<IFNAME>	Specifies the interface name.
<IP-ADDR>	Specifies the IPv4 address.
vrf *<VRF-NAME>*	Specifies the VRF name.

Examples

Configuring IPv4 source-interface interface 1/1/1 to use for the TFTP protocol:

```
switch(config)# ip source-interface tftp interface 1/1/1
```

Configuring IPv4 source-interface interface 1/1/2 to use for the TFTP protocol on VRF green :

```
switch(config)# ip source-interface tftp interface 1/1/2 vrf green
```

Removing IPv4 source-interface 1/1/1configuration for the TFTP protocol:

```
switch(config)# no ip source-interface tftp interface 1/1/1
```

Removing source-interface interface 1/1/2 configuration for TFTP protocol on VRF green:

```
switch(config)# no ip source-interface tftp interface 1/1/2 vrf green
```

Configuring source-interface IPv4 10.1.1.1 to use for the TFTP protocol:

```
switch(config)# ip source-interface tftp 10.1.1.1
```

Configuring source-interface IPv4 10.1.1.2 to use for the TFTP protocol on VRF green :

```
switch(config)# ip source-interface tftp 10.1.1.2 vrf green
```

Removing source-interface IPv4 10.1.1.1 configuration for the TFTP protocol:

```
switch(config)# no ip source-interface tftp 10.1.1.1
```

Removing source-interface IPv4 10.1.1.2 configuration for TFTP protocol on VRF green:

```
switch(config)# no ip source-interface tftp 10.1.1.2 vrf green
```

Configuring source-interface IPv4 10.1.1.1 to use for the DNS protocol:

```
switch(config)# ip source-interface dns 10.1.1.1
```

Configuring source-interface IPv4 10.1.1.2 to use for the DNS protocl on VRF green :

```
switch(config)# ip source-interface dns 10.1.1.2 vrf green
```

Removing source-interface IPv4 10.1.1.1configuration for the DNS protocol:

```
switch(config)# no ip source-interface tftp 10.1.1.1
```

Removing source-interface IPv4 10.1.1.2 configuration for the DNS protocol on VRF green:

```
switch(config)# no ip source-interface dns 10.1.1.2 vrf green
```

Command History

Release	Modification
10.12.1000	Added `cental`, `sftp-scp`, and `ssh-client` parameters.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

ipv6 source-interface

```
ipv6 source-interface <PROTOCOL> <IPV6-ADDR> [vrf <VRF-NAME>]
no source-interface <PROTOCOL> <IPV6-ADDR> [vrf <VRF-NAME>]
```

Description

Configures the source-interface IPv6 address to use for the specified protocol. If a VRF is not given, the default VRF applies.

The `no` form of this command removes the specified protocol configuration.

Parameter	Description
<PROTOCOL>	Specifies the protocol to configure. ■ all:Selects all protocols supported by this command. ■ central:Selects Aruba Central. ■ ntp:Selects NTP. ■ radius:Selects radius. ■ sflow:Selects sFLow. ■ syslog:Selects syslog. ■ tacacs:Selects TACACS. ■ tftp:Selects TFTP.
<IPV6-ADDR>	Specifies the IPv6 address.
vrf *<VRF-NAME>*	Specifies the VRF name.

Examples

Configuring source-interface IPv6 1111:2222 to use for the TFTP protocol:

```
switch(config)# ipv6 source-interface tftp 1111:2222
```

Configuring source-interface IPv6 1111:3333 to use for TFTP protocol on VRF green :

```
switch(config)# ipv6 source-interface tftp 1111:3333 vrf green
```

Removing source-interface IPv6 1111:2222 configuration for TFTP protocol:

```
switch(config)# no ipv6 source-interface tftp 1111:2222
```

Removing source-interface IPv6 1111:3333 configuration for TFTP protocol on VRF green:

```
switch(config)# no ipv6 source-interface tftp 1111:3333 vrf green
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

ipv6 source-interface dns

```
ipv6 source-interface {dns | all} {interface | X:X::X:X} [vrf <VRF-NAME>]
[no] ipv6 source-interface {dns | all} {interface | X:X::X:X} [vrf <VRF-NAME>]
```

Description

Configures the IPv6 source-interface or source IP for IPv6 DNS clients.

The **no** form of this command removes all configurations.

Parameter	Description

<PROTOCOL>	Specifies the protocol to configure.
	`all`
	Selects all protocols supported by this command.
	`central`
	Selects Aruba Central.
	`dhcp_relay`
	Selects DHCP relay.
	`dns`
	Selects DNS packet source.
	`http`
	Selects HTTP.
	`ntp`
	Selects NTP.
	`radius`
	Selects radius.
	`sftp-scp`
	Selects SFTP and SCP.
	`sflow`
	Selects sFLow.
	`ssh-client`
	Selects SSH Client.
	`syslog`
	Selects syslog.
	`tacacs`
	Selects TACACS.
	`tftp`
	SelectsTFTP.
	`ubt`
	SelectsUBT.

<IPV6-ADDR>	Specifies the IPv6 address.
vrf *<VRF-NAME>*	Specifies the VRF name.

Examples

Configuring IPv6 source-interface dns :

```
switch(config)# ipv6 source-interface
        all         All protocols
        central     Aruba Central protocol
        dhcp_relay  DHCP_RELAY protocol
        dns         DNS protocol
        http        HTTP protocol
        ntp         NTP protocol
        radius      RADIUS protocol
        sflow       sFlow protocol
        sftp-scp    SFTP and SCP protocols
        ssh-client  SSH Client protocol
        syslog      syslog protocol
        tacacs      TACACS protocol
        tftp        TFTP protocol
```

Configuring IPv6 source -interface dns:

```
switch(config)# ipv6 source-interface dns
X:X::X:X   Specify an IPv6 address
interface  Interface information
```

Configuring IPv6 source-interface dns on 1: :1:

```
switch(config)# ipv6 source-interface dns 1::1
vrf   VRF Configuration
<cr>
```

Configuring IPv6 source-interface dns on 1: :1: vrf:

```
switch(config)#  ipv6 source-interface dns 1::1 vrf
VRF_NAME  VRF name
```

Configuring IPv6 source-interface dns on 1: :1 vrf BLUE

```
switch(config)# ipv6 source-interface dns 1::1 vrf BLUE
switch(config)# ipv6 source-interface dns interface vlan10
vrf   VRF Configuration
<cr>
```

Configuring IPv6 source-interface dns on vlan10 vrf BLUE:

```
switch(config)# ipv6 source-interface dns interface vlan10 vrf BLUE
```

Command History

Release	Modification
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

ipv6 source-interface

```
ipv6 source-interface <PROTOCOL> {<IPV6-ADDR>|interface <IFNAME>} [vrf <VRF-NAME>]
no ipv6 source-interface <PROTOCOL> {<IPV6-ADDR>|interface <IFNAME>} [vrf <VRF-NAME>]
```

Description

Configures the IPv6 source-interface interface to use for the specified protocol. If a VRF is not given, the default VRF applies.

The **no** form of this command removes all configurations.

Parameter	Description

<PROTOCOL>	Specifies the protocol to configure.
	`all`
	Selects all protocols supported by this command.
	`central`
	Selects Aruba Central.
	`dhcp_relay`
	Selects DHCP relay.
	`dns`
	Selects DNS packets
	`http`
	Selects HTTP.
	`ntp`
	Selects NTP.
	`radius`
	Selects radius.
	`sftp-scp`
	Selects SFTP and SCP.
	`sflow`
	Selects sFLow.
	`ssh-client`
	Selects SSH Client.
	`syslog`
	Selects syslog.
	`tacacs`
	Selects TACACS.
	`tftp`
	SelectsTFTP.

<IPV6-ADDR>	Specifies the IPv6 address.
<IFNAME>	Specifies the interface name.
vrf *<VRF-NAME>*	Specifies the VRF name.

Examples

Configuring IPv6 source-interface interface 1/1/1 to use for the TFTP protocol :

```
switch(config)# ipv6 source-interface tftp interface 1/1/1
```

Configuring IPv6 source-interface interface 1/1/2 to use for the TFTP protocol on VRF green :

```
switch(config)# ipv6 source-interface tftp interface 1/1/2 vrf green
```

Removing IPv6 source-interface interface 1/1/1 configuration for the TFTP protocol:

```
switch(config)# no ipv6 source-interface tftp interface 1/1/1
```

Removing IPv6 source-interface interface 1/1/2 configuration for the TFTP protocol on VRF green:

```
switch(config)# no ipv6 source-interface tftp interface 1/1/2 vrf green
```

Configuring source-interface IPv6 1111:2222 to use for the TFTP protocol:

```
switch(config)# ipv6 source-interface tftp 1111:2222
```

Configuring source-interface IPv6 1111:3333 to use for TFTP protocol on VRF green :

```
switch(config)# ipv6 source-interface tftp 1111:3333 vrf green
```

Removing source-interface IPv6 1111:2222 configuration for TFTP protocol:

```
switch(config)# no ipv6 source-interface tftp 1111:2222
```

Removing source-interface IPv6 1111:3333 configuration for TFTP protocol on VRF green:

```
switch(config)# no ipv6 source-interface tftp 1111:3333 vrf green
```

Command History

Release	Modification
10.13.0001	Added the **dns** protocol parameter.
10.12.1000	Added **central, sftp-scp, dhcp_relay** and **ssh-client** parameters.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

show ip source-interface

`show ip source-interface <PROTOCOL> [vrf <VRF-NAME> | all-vrfs]`

Description

Displays the source interface information for all VRFs or a specific VRF.

If a VRF is not specified, the default is displayed.

Parameter	Description

<PROTOCOL>	Specifies the protocol to show.
	`all`
	Shows the source interface configuration for all other protocols.
	`central`
	Shows the source interface configuration for Aruba Central.
	`dhcp relay`
	Shows the source interface configuration for DHCP relay.
	`dns`
	Shows the source interface configuration for DNS.
	`http`
	Shows the source interface configuration for HTTP.
	`ntp`
	Shows the source interface configuration for NTP.
	`radius`
	Shows the source interface configuration for radius.
	`sflow`
	Shows the source interface configuration for sFLow.
	`sftp-scp`
	Shows source interface configuration for SFTP and SCP.
	`ssh-client`
	Shows source interface configuration for SSH Client.
	`syslog`
	Shows the source interface configuration for syslog.
	`tacacs`
	Shows the source interface configuration for TACACS.
	`tftp`
	Shows the source interface configuration for TFTP.
	`ubt`
	Shows the source interface configuration for PTP.

vrf *<VRF-NAME>*	Specifies the VRF name.
all-vrfs	Shows the source interface configuration for all VRFs.

Examples

Displaying all source-interface protocol configurations for VRF red:

```
switch# show ip source-interface all vrf red
Source-interface Configuration Information
---------------------------------------------------------------
Protocol        Src-Interface       Src-IP                 VRF
---------------------------------------------------------------
all             1/1/1                                      red
switch#
```

Displaying all source-interface protocol configurations for default VRF:

```
switch# show ip source-interface all
Source-interface Configuration Information
-------------------------------------------------------------------
Protocol        Src-Interface       Src-IP                 VRF
-------------------------------------------------------------------
all                                 1.1.1.1                default
switch#
```

Displaying all source-interface protocol configurations for all VRFs:

```
switch# show ip source-interface all all-vrfs
Source-interface Configuration Information
-------------------------------------------------------------------
Protocol        Src-Interface       Src-IP                 VRF
-------------------------------------------------------------------
all                                 2.2.2.2                all-vrfs
all                                 1.1.1.1                default
all              1/1/1/1                                   red
switch#
```

Command History

Release	Modification
10.12.1000	Added `central`, `sftp-scp`, and `ssh-client` parameters.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show ipv6 source-interface

`show ipv6 source-interface <PROTOCOL> [detail] [vrf <VRF-NAME> | all-vrfs]`

Description

Displays the IPV6 source interface information configured in the router for all VRFs or a specific VRF.

If a VRF is not specified, the default is displayed.

Parameter	Description

<PROTOCOL>	Specifies the protocol to show.
	`all`
	Shows the source interface configuration for all other protocols.
	`central`
	Shows the source interface configuration for Aruba Central.
	`dhcp_relay`
	Shows the source interface configuration for DHCP realy.
	`dns`
	Shows the source interface configuration for DNS.
	`http`
	Shows the source interface configuration for HTTP.
	`ntp`
	Shows the source interface configuration for NTP.
	`radius`
	Shows the source interface configuration for radius.
	`sflow`
	Shows the source interface configuration for sFLow.
	`sftp-scp`
	Shows source interface configuration for SFTP and SCP.
	`ssh-client`
	Shows source interface configuration for SSH Client.
	`syslog`
	Shows the source interface configuration for syslog.
	`tacacs`
	Shows the source interface configuration for TACACS.
	`tftp`
	Shows the source interface configuration for TFTP.

`vrf` *<VRF-NAME>*	Specifies the VRF name.
all-vrfs	Shows the source interface configuration for all VRF.

Examples

Displaying all IPv6 source-interface protocol configurations for default VRF:

```
switch# show ipv6 source-interface all
Source-interface Configuration Information
-----------------------------------------------------------------
Protocol        Src-Interface       Src-IP                 VRF
-----------------------------------------------------------------
all                                 1111::2222              default
switch#
```

Displaying all IPv6 source-interface protocol configuration for VRF red:

```
switch# show ipv6 source-interface all vrf red
Source-interface Configuration Information
---------------------------------------------------------------
Protocol        Src-Interface       Src-IP                 VRF
---------------------------------------------------------------
all             1/1/1              2005::2                        red
switch#
```

Displaying all IPv6 source-interface protocol configurations for all VRFs:

```
switch# show ipv6 source-interface all all-vrfs
Source-interface Configuration Information
-------------------------------------------------------------------
Protocol        Src-Interface       Src-IP                VRF
-------------------------------------------------------------------
all                                2222::3333           all-vrfs
all                                1111::2222           default
all             1/1/1              2::2                 red
```

Displaying all IPv6 source-interface protocol confirgurations for dns all VRFs:

```
switch# show ipv6 source-interface dns all-vrfs
Source-interface Configuration Information
----------------------------------------------------------------------------------
--
Protocol        Src-Interface       Src-IP                  VRF
----------------------------------------------------------------------------------
--
dns                                 1::3                    blue
dns                                 1::4                    default
dns                                 1::2                    red
```

Command History

Release	Modification
10.13	Added `dns` parameters.
10.12.1000	Added `central`, `sftp-scp`, and `ssh-client` parameters.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show running-config

`show running-config`

Description

Displays the current running configuration.

Examples

Displaying the running configuration (only items of interest to source interface selection are shown in this example output command):

Aruba Central is the priority agent. If no command is specified for ip source-interface, Central will choose the command automatically if it is reachable on any of the known ports.

```
switch# show running-config
vrf green
ip source-interface tftp interface 1/1/2 vrf green
ip source-interface radius interface 1/1/2 vrf green
ip source-interface ntp interface 1/1/2 vrf green
ip source-interface tacacs interface 1/1/2 vrf green
ip source-interface dns interface 1/1/2 vrf green
ip source-interface central interface 1/1/2 vrf green
ip source-interface all interface 1/1/2 vrf green
ipv6 source-interface tftp 2222::3333 vrf green
ipv6 source-interface radius 2222::3333 vrf green
ipv6 source-interface ntp 2222::3333 vrf green
ipv6 source-interface tacacs 2222::3333 vrf green
ipv6 source-interface central 2222::3333 vrf green
ipv6 source-interface all 2222::3333 vrf green
ip source-interface tftp 10.20.3.1
ip source-interface radius 10.20.3.1
ip source-interface ntp 10.20.3.1
ip source-interface tacacs 10.20.3.1
ip source-interface dns 10.20.3.1
ip source-interface central 10.20.3.1
ip source-interface all 10.20.3.1
interface 1/1/1
     no shutdown
     ip address 10.20.3.1/24
interface 1/1/2
     vrf attach green
     ip address 20.1.1.1/24
     ipv6 address 2222::3333/64
interface 1/1/45
     no shutdown
     ip address 100.1.0.1/24
     ipv6 address 1111::2222/64
ip route 100.2.0.0/24 10.20.3.2
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Chapter 7
VLANs

VLANs are primarily used to provide network segmentation at layer 2. VLANs enable the grouping of users by logical function instead of physical location. They make managing bandwidth usage within networks possible by:

- Allowing grouping of high-bandwidth users on low-traffic segments
- Organizing users from different LAN segments according to their need for common resources and individual protocols
- Improving traffic control at the edge of networks by separating traffic of different protocol types.
- Enhancing network security by creating subnets to control in-band access to specific network resources

VLANs are generally assigned on an organizational basis rather than on a physical basis. For example, a network administrator could assign all workstations and servers used by a particular workgroup to the same VLAN, regardless of their physical locations.

Hosts in the same VLAN can directly communicate with one another. A router or a Layer 3 switch is required for hosts in different VLANs to communicate with one another.

VLANs help reduce bandwidth waste, improve LAN security, and enable network administrators to address issues such as scalability and network management.

Refer to the Layer 2 Bridging Guide for VLAN configuration and commands.

Chapter 8

Configuration and firmware management

Upgrade and downgrade scenarios

Upgrade and downgrade scenarios are provided by the Configuration Migration Framework (CMF).

Upgrades

All upgrade scenarios are supported and configuration is migrated.

Downgrades

The following scenario is supported:

Create a checkpoint before upgrading.

Upgrade.

Perform config changes.

Set the startup configuration to the checkpoint created in step 1.

Downgrade

Configuration changes that occur after the checkpoint in step 1 will be lost during the downgrade.

Limitations

The following are the limitations of upgrade and downgrade scenarios provided by CMF.

Table 1: *Configuration from and to the same build*

From/to	Checkpoint	Running	Startup
Checkpoint	N/A	Supported	Supported
Running	Supported	N/A	Supported
Startup	Supported	Supported	N/A
Config from URL in CLI Format	Supported	Supported	Supported
Config from URL in JSON Format	Supported	Supported	Supported

Table 2: *Configuration from an older build to a newer build (upgrade)*

From/to	Checkpoint	Running	Startup
Config from URL in CLI Format	Not supported	Not supported	Not supported

From/to	Checkpoint	Running	Startup
Config from URL in JSON Format	Not supported	Not supported	Not supported
Startup	Not supported	Supported	Supported

Table 3: *Configuration from a newer build to an older build (downgrade)*

From/to	Checkpoint	Running	Startup
Config from URL in CLI Format	Not supported	Not supported	Not supported
Config from URL in JSON Format	Not supported	Not supported	Not supported
Startup	Not supported	Not supported	Not supported

Hot-patch software

Overview

Hot-patching provides users of AOS-CX with a way to update running software without rebooting their system. Hot-patches are distributed as signed .patch files and are applied on top of an official AOS-CX image.

A hot-patch can be downloaded from a remote server onto an AOS-CX switch then applied without rebooting the switch. By default, the software patch remains on the switch after it is applied, and is automatically reapplied when the switch reboots. In order to revert an applied hot-patch update, the patch must be disabled. When the hot-patch is disabled, rebooting the box is not necessary, though the hot-patch will still remain on the system. The hot-patch can be removed from the system after disabling the hot-patch.

Once a downloaded hot-patch is removed, it must be downloaded again before it can be reapplied.

Hot-patches are cumulative, meaning that each subsequent patch is a superset of the previous patch(es) for a given software image. When multiple hot-patches are required, the first should be removed after the second has been applied.

Prerequisites

- AOS-CX hot-patch software can be obtained from Aruba customer support, and is identified with a **.patch** extension. The name of the patch indicates the version of AOS-CX to which it can be applied. For example, the patch **FL.10.12.XXXX-YYYY.patch** indicates that the patch can be applied on top of switch image **FL.10.12.XXXX.swi** and brings software up to date with the **FL.10.12.YYYY** release.
- Hot-patch software can be applied only after it has been downloaded to the switch.

Caveats and limitations

- Hot-patch files cannot be managed when the switch is loaded with a .swi image that does not support the hot-patch feature.
- Only restartable daemons can be updated with a hot-patch file.

- Each hot-patch applies to only one release image and each subsequent hot-patch for that release image is cumulative.This means that applying the most recent hot-patch provides all the same fixes included in previous hot-patches. As such, only one hot-patch can be applied at a time.
- incompatible and un-applied hot-patches are allowed to remain on the system and in the config to support rollback to the previous version of the software, and to automatically apply any hot-patches that have been previously applied.
- A maximum of ten hot-patches are allowed to be present or preconfigured on the system at a time.
- A daemon/service may restart if it has a hard dependency on another service that gets restarted by being hot-patched. Upon rebooting, hot-patches are reapplied after daemons have started.
- The **dpsed** and **hpe-cardd** daemon cannot have a hot-patch applied while a switchover or failover is in progress.

To apply a hot-patch update

1. Request a hot-patch from Aruba customer support, then place the hot-patch on an FTP-SFTP server.
2. Use the **copy** command to copy the hot-patch from the remote server to the standalone switch or VSF.
3. Use the **hot-patch apply** command to apply the patch.

To unapply hot-patch update

This option only unapplies the patch, but does not remove the patch from the device. The patch can be reapplied again since it is still on the device. If an unapplied patch needs to be removed/deleted from the switch, it can be removed with the command **hot-patch remove <patch>.patch**.

```
switch(config)# show hot-patch
Name                       Status
------------------------   ------
FL_10_12_0001-0002.patch   Not applied
switch(config)# hot-patch apply FL_10_12_0001-0002.patch
switch(config)#
switch(config)# show hot-patch
Name                      Status
------------------------  ------
FL_10_12_0001-0002.patch  Applied

switch(config)# no hot-patch apply FL_10_12_0001-0002.patch
switch(config)# show hot-patch
Name                      Status
------------------------  ------
FL_10_12_0001-0002.patch  Not applied

switch(config)#
switch(config)# hot-patch remove FL_10_12_0001-0002.patch
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.
switch(config)#
switch(config)# show hot-patch
No hot-patch found or configured
switch(config)#
```

To remove/delete a patch:

An applied hot-patch can be both unapplied and removed from the device with the command **hot-patch remove <patch-name>.patch**. The patch cannot be reapplied unless it is downloaded again.

Checkpoints

A checkpoint is a snapshot of the running configuration of a switch and its relevant metadata during the time of creation. Checkpoints can be used to apply the switch configuration stored within a checkpoint whenever needed, such as to revert to a previous, clean configuration. Checkpoints can be applied to other switches of the same platform. A switch is able to store multiple checkpoints.

Checkpoint types

The switch supports two types of checkpoints:

- **System generated checkpoints**: The switch automatically generates a system checkpoint whenever a configuration change occurs.
- **User generated checkpoints**: The administrator can manually generate a checkpoint whenever required.

Maximum number of checkpoints

- Maximum checkpoints: 64 (including the startup configuration)
- Maximum user checkpoints: 32
- Maximum system checkpoints: 32

User generated checkpoints

User checkpoints can be created at any time, as long as one configuration difference exists since the last checkpoint was created. Checkpoints can be applied to either the running or startup configurations on the switch.

All user generated checkpoints include a time stamp to identify when a checkpoint was created.

A maximum of 32 user generated checkpoints can be created.

System generated checkpoints

System generated checkpoints are automatically created by default. Whenever a configuration change occurs, the switch starts a timeout counter (300 seconds by default). For each additional configuration change, the timeout counter is restarted. If the timeout expires with no additional configuration changes being made, the switch generates a new checkpoint.

System generated checkpoints are named with the prefix `CPC` followed by a time stamp in the format `<YYYYMMDDHHMMSS>`. For example: `CPC20170630073127`.

System checkpoints can be applied using the checkpoint rollback feature or copy command.

A maximum of 32 system checkpoints can be created. Beyond this limit, the newest system checkpoint replaces the oldest system checkpoint.

Supported remote file formats

You can restore a switch configuration by copying a switch configuration stored on a USB drive or a remote network device through SFTP/TFTP. The remote file formats that the switch supports depends on where you plan to restore the checkpoint.

Restoring a checkpoint to a...	File type supported
Running configuration	▪ CLI ▪ JSON ▪ Checkpoint
Startup configuration	▪ JSON ▪ Checkpoint
Specified checkpoint	Specified checkpoint

Rollback

The term rollback is used to refer to when a switch configuration is reverted to a pre-existing checkpoint.

For example, the following command applies the configuration from checkpoint `ckpt1`. All previous configurations are lost after the execution of this command: `checkpoint rollback ckpt1`

You can also specify the rollback of the running configuration or of the startup configuration with a specified checkpoint, as shown with the following command: `copy checkpoint <checkpoint-name> {running-config | startup-config}`

Checkpoint auto mode

Checkpoint auto mode configures the switch with failover support, causing it to automatically revert to a previous configuration if it becomes inoperable or inaccessible due to configuration changes that are being made.

After entering checkpoint auto mode, you have a set amount of time to add, remove, or modify the existing switch configuration. To save your changes, you must execute the `checkpoint auto confirm` command before the auto mode timer expires. If you do not execute the `checkpoint auto confirm` command within the specified time, all configuration changes you made are discarded and the running configuration reverts to the state it was before entering checkpoint auto mode.

Testing a switch configuration in checkpoint auto mode

Process overview:

1. Enable the checkpoint auto mode.
2. To save the configuration, enter the `checkpoint auto confirm` command before the specified time set in step 1.

Checkpoint commands

checkpoint auto

`checkpoint auto <TIME-LAPSE-INTERVAL>`

Description

Starts auto checkpoint mode. In auto checkpoint mode, the switch temporarily saves the runtime configuration as a checkpoint only for the specified time lapse interval. Configuration changes must be saved before the interval expires, otherwise the runtime configuration is restored from the temporary checkpoint.

Parameter	Description
<TIME-LAPSE-INTERVAL>	Specifies the time lapse interval in minutes. Range: 1 to 60.

Usage

To save the runtime checkpoint permanently, run the **checkpoint auto confirm** command during the time lapse interval. The filename for the saved checkpoint is named **AUTO <YYYYMMDDHHSS>**. If the **checkpoint auto confirm** command is not entered during the specified time lapse interval, the previous runtime configuration is restored.

Examples

Confirming the auto checkpoint:

```
switch# checkpoint auto 20
Auto checkpoint mode expires in 20 minute(s)
switch# WARNING  Please "checkpoint auto confirm" within 2 minutes
switch# checkpoint auto confirm
checkpoint AUTO20170801011154 created
```

In this example, the runtime checkpoint was saved because the **checkpoint auto confirm** command was entered within the value set by the **time-lapse-interval** parameter, which was 20 minutes.

Not confirming the auto checkpoint:

```
switch# checkpoint auto 20
Auto checkpoint mode expires in 20 minute(s)
switch# WARNING  Please "checkpoint auto confirm" within 2 minutes
WARNING: Restoring configuration. Do NOT add any new configuration.
Restoration successful
```

In this example, the runtime checkpoint was reverted because the **checkpoint auto confirm** command was not entered within the value set by the **time-lapse-interval** parameter, which was 20 minutes.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

checkpoint auto confirm

`checkpoint auto confirm`

Description

Signals to the switch to save the running configuration used during the auto checkpoint mode. This command also ends the auto checkpoint mode.

Usage

To save the runtime checkpoint permanently, run the **checkpoint auto confirm** command during the time lapse value set by the **checkpoint auto TIME-LAPSE-INTERVAL** command. The generated checkpoint name will be in the format **AUTO <YYYYMMDDHHSS>**. If the **checkpoint auto confirm** command is not entered during the specified time lapse interval, the previous runtime configuration is restored.

Examples

Confirming the auto checkpoint:

```
switch# checkpoint auto confirm
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

checkpoint diff

```
checkpoint diff {<CHECKPOINT-NAME1> | running-config | startup-config}
    {<CHECKPOINT-NAME2> | running-config | startup-config}
```

Description

Shows the difference in configuration between two configurations. Compare checkpoints, the running configuration, or the startup configuration.

Parameter	Description
`{<CHECKPOINT-NAME1>	running-config	startup-config}`	Selects either a checkpoint, the running configuration, or the startup configuration as the baseline.
`{<CHECKPOINT-NAME2>	running-config	startup-config}`	Selects either a checkpoint, the running configuration, or the startup configuration to compare.

Usability

The output of the **checkpoint diff** command has several symbols:

- The plus sign (+) at the beginning of a line indicates that the line exists in the comparison but not in the baseline.
- The minus sign (-) at the beginning of a line indicates that the line exists in the baseline but not in the comparison.

Examples

In the following example, the configurations of checkpoints **cp1** and **cp2** are displayed before the `checkpoint diff` command, so that you can see the context of the **checkpoint diff** command.

```
switch# show checkpoint cp1
Checkpoint configuration:
!
!Version AOS-CX XL.10.00.0002
!Schema version 0.1.8
module 1/1 product-number jl363a
!
!
!
!
!
!
!
vlan 1,200
interface 1/1/1
    no shutdown
    ip address 1.0.0.1/24
interface 1/1/2
    no shutdown
    ip address 2.0.0.1/24

switch# show checkpoint cp2
Checkpoint configuration:
!
!Version AOS-CX XL.10.00.0002
!Schema version 0.1.8
module 1/1 product-number jl363a
!
!
!
!
!
!
!
vlan 1,200,300
interface 1/1/1
    no shutdown
    ip address 1.0.0.1/24
interface 1/1/2
    no shutdown
    ip address 2.0.0.1/24

switch# checkpoint diff cp1 cp2
--- /tmp/chkpt11501550258421    2017-08-01 01:17:38.420514016 +0000
+++ /tmp/chkpt21501550258421    2017-08-01 01:17:38.420514016 +0000
@@ -9,7 +9,7 @@
 !
 !
 !
-vlan 1,200
+vlan 1,200,300
```

```
interface 1/1/1
    no shutdown
    ip address 1.0.0.1/24
```

```
switch# checkpoint diff chkpt01 chkpt02
--- /tmp/chkpt011607564301327
+++ /tmp/chkpt021607564301353
@@ -1,7 +1,7 @@
 !
 !Version AOS-CX PL.10.06.0100V
 !export-password: default
-hostname Switch
+hostname Switch1
 user admin group administrators password ciphertext
AQBapTyg9tpaiAaTfSVV5eNdFzOORRvZ6CMpglh1P+LQUHQLYgAAAGAhmRqFbkNvrgy2SBVk7H8C5hvg/I
ib8rWYFZLEaSCrobNP9EwMu+hLNM0xmsh45yG8dncP7WkxjwrW4p4Qra6dVfr0EW8xh/lpQf8F/2Wki20L
c9JLXiYge7ti0H6cVn+G
 radius-server tracking interval 60
 no usb

switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

checkpoint post-configuration

```
checkpoint post-configuration

no checkpoint post-configuration
```

Description

Enables creation of system generated checkpoints when configuration changes occur. This feature is enabled by default.

The **no** form of this command disables system generated checkpoints.

Usage

System generated checkpoints are automatically created by default. Whenever a configuration change occurs, the switch starts a timeout counter (300 seconds by default). For each additional configuration change, the timeout counter is restarted. If the timeout expires with no additional configuration changes being made, the switch generates a new checkpoint.

System generated checkpoints are named with the prefix **CPC** followed by a time stamp in the format **<YYYYMMDDHHMMSS>**. For example: **CPC20170630073127**.

System checkpoints can be applied using the checkpoint rollback feature or copy command.

A maximum of 32 system checkpoints can be created. Beyond this limit, the newest system checkpoint replaces the oldest system checkpoint.

Examples

Enabling system checkpoints:

```
switch(config)# checkpoint post-configuration
```

Disabling system checkpoints:

```
switch(config)# no checkpoint post-configuration
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

checkpoint post-configuration timeout

`checkpoint post-configuration timeout <TIMEOUT>`

`no checkpoint post-configuration timeout <TIMEOUT>`

Description

Sets the timeout for the creation of system checkpoints. The timeout specifies the amount of time since the latest configuration for the switch to create a system checkpoint.

The **no** form of this command resets the timeout to 300 seconds, regardless of the value of the **<TIMEOUT>** parameter.

Parameter	Description
`timeout <TIMEOUT>`	Specifies the timeout in seconds. Range: 5 to 600. Default: 300.

Examples

Setting the timeout for system checkpoints to 60 seconds:

```
switch(config)# checkpoint post-configuration timeout 60
```

Resetting the timeout for system checkpoints to 300 seconds:

```
switch(config)# no checkpoint post-configuration timeout 1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

checkpoint rename

`checkpoint rename <OLD-CHECKPOINT-NAME> <NEW-CHECKPOINT-NAME>`

Description

Renames an existing checkpoint.

Parameter	Description
`<OLD-CHECKPOINT-NAME>`	Specifies the name of an existing checkpoint to be renamed.
<NEW-CHECKPOINT-NAME>	Specifies the new name for the checkpoint. The checkpoint name can be alphanumeric. It can also contain underscores (_) and dashes (-). **NOTE:** Do not start the checkpoint name with **CPC** because it is used for system-generated checkpoints.

Examples

Renaming checkpoint **ckpt1** to **cfg001**:

```
switch# checkpoint rename ckpt1 cfg001
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

checkpoint rollback

```
checkpoint rollback {<CHECKPOINT-NAME> | startup-config}
```

Description

Applies the configuration from a pre-existing checkpoint or the startup configuration to the running configuration.

Parameter	Description
<CHECKPOINT-NAME>	Specifies a checkpoint name.
startup-config	Specifies the startup configuration.

Examples

Applying a checkpoint named **ckpt1** to the running configuration:

```
switch# checkpoint rollback ckpt1
Success
```

Applying a startup checkpoint to the running configuration:

```
switch# checkpoint rollback startup-config
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy checkpoint *<CHECKPOINT-NAME> <REMOTE-URL>*

```
copy checkpoint <CHECKPOINT-NAME> <REMOTE-URL> [vrf <VRF-NAME>]
```

Description

Copies a checkpoint configuration to a remote location as a file. The configuration is exported in checkpoint format, which includes switch configuration and relevant metadata.

Parameter	Description
<CHECKPOINT-NAME>	Specifies the name of a checkpoint.

<REMOTE-URL>	Specifies the remote destination and filename using the syntax:
	TFTP format:
	`tftp://<IP-ADDR>[:<PORT-NUM>]`
	`[;blocksize=<Value>]/<FILENAME>`
	SFTP format:
	`sftp://<USERNAME>@<IP-ADDR>`
	`[:<PORT-NUM>]/<FILENAME>`
	SCP format:
	`scp://USER@{IP|HOST}[:PORT]/FILE`

vrf *<VRF-NAME>*	Specifies a VRF name.

Examples

Copying checkpoint configuration to remote file through TFTP:

```
switch# copy checkpoint ckpt1 tftp://192.168.1.10/ckptmeta vrf default
######################################################################### 100.0%
Success
```

Copying checkpoint configuration to remote file through SFTP:

```
switch# copy checkpoint ckpt1 sftp://root@192.168.1.10/ckptmeta vrf default
The authenticity of host '192.168.1.10 (192.168.1.10)' can't be established.
ECDSA key fingerprint is SHA256:FtOm6Uxuxumil7VCwLnhz92H9LkjY+eURbdddOETy50.
Are you sure you want to continue connecting (yes/no)? yes
root@192.168.1.10's password:
sftp> put /tmp/ckptmeta ckptmeta
Uploading /tmp/ckptmeta to /root/ckptmeta
Warning: Permanently added '192.168.1.10' (ECDSA) to the list of known hosts.
Connected to 192.168.1.10.
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy checkpoint <*CHECKPOINT-NAME*> {running-config | startup-config}

`copy checkpoint <CHECKPOINT-NAME> {running-config | startup-config}`

Description

Copies an existing checkpoint configuration to the running configuration or to the startup configuration.

Parameter	Description
<CHECKPOINT-NAME>	Specifies the name of an existing checkpoint.
	Selects whether the running configuration or the startup configuration receives the copied checkpoint configuration. If the startup configuration is already present, the command overwrites the startup configuration.

Examples

Copying **ckpt1** checkpoint to the running configuration:

```
switch# copy checkpoint ckpt1 running-config
Success
```

Copying **ckpt1** checkpoint to the startup configuration:

```
switch# copy checkpoint ckpt1 startup-config
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy checkpoint *<CHECKPOINT-NAME> <STORAGE-URL>*

`copy checkpoint <CHECKPOINT-NAME> <STORAGE-URL>`

Description

Copies an existing checkpoint configuration to a USB drive. The file format is defined when the checkpoint was created.

Parameter	Description
<CHECKPOINT-NAME>	Specifies the name of the checkpoint to copy. The checkpoint name can be alphanumeric. It can also contain underscores (_) and dashes (-).

Parameter	Description
<STORAGE-URL>>	Specifies the name of the target file on the USB drive using the following syntax: `usb:/<FILE>` The USB drive must be formatted with the FAT file system.

Examples

Copying the **test** checkpoint to the **testCheck** file on the USB drive:

```
switch# copy checkpoint test usb:/testCheck
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy *<REMOTE-URL>* checkpoint *<CHECKPOINT-NAME>*

```
copy <REMOTE-URL> checkpoint <CHECKPOINT-NAME> [vrf <VRF-NAME>]
```

Description

Copies a remote configuration file to a checkpoint. The remote configuration file must be in checkpoint format.

Parameter	Description

<REMOTE-URL>	Specifies a remote file using the following syntax:
	TFTP format:
	`tftp://<IP-ADDR>[:<PORT-NUM>]`
	`[;blocksize=<Value>]/<FILENAME>`
	SFTP format:
	`sftp://<USERNAME>@<IP-ADDR>`
	`[:<PORT-NUM>]/<FILENAME>`
	SCP format:
	`scp://USER@{IP|HOST}[:PORT]/FILE`

<CHECKPOINT-NAME>	Specifies the name of the target checkpoint. The checkpoint name can be alphanumeric. It can also contain underscores (_) and dashes (-). Required. **NOTE:** Do not start the checkpoint name with `CPC` because it is used for system-generated checkpoints.

Parameter	Description
vrf *<VRF-NAME>*	Specifies a VRF name. Default: `default`.

Examples

Copying a checkpoint format file to checkpoint **ckpt5** on the default VRF:

```
switch# copy tftp://192.168.1.10/ckptmeta checkpoint ckpt5
######################################################################## 100.0%
100.0%
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy *<REMOTE-URL>* {running-config | startup-config}

`copy <REMOTE-URL> {running-config | startup-config } [vrf <VRF-NAME>]`

Description

Copies a remote file containing a switch configuration to the running configuration or to the startup configuration.

Parameter	Description

<REMOTE-URL>	Specifies a remote file with the following syntax:
	TFTP format:
	`tftp://<IP-ADDR>[:<PORT-NUM>]`
	`[;blocksize=<Value>]/<FILENAME>`
	SFTP format:
	`sftp://<USERNAME>@<IP-ADDR>`
	`[:<PORT-NUM>]/<FILENAME>`
	SCP format:
	`scp://USER@{IP|HOST}[:PORT]/FILE`

{running-config | startup-config}	Selects whether the running configuration or the startup configuration receives the copied checkpoint configuration. If the startup configuration is already present, the command overwrites the startup configuration.
vrf *<VRF-NAME>*	Specifies the name of a VRF. Default: default.

Usage

The switch copies only certain file types. The format of the file is automatically detected from contents of the file. The **startup-config** option only supports the JSON file format and checkpoints, but the **running-config** option supports the JSON and CLI file formats and checkpoints.

When a file of the CLI format is copied, it overwrites the running configuration. The CLI command does not clear the running configuration before applying the CLI commands. All of the CLI commands in the file are applied line-by-line. If a particular CLI command fails, the switch logs the failure and it continues to the next line in the CLI configuration. The event log (**show events -d hpe-config**) provides information as to which command failed.

Examples

Copying a JSON format file to the running configuration:

```
switch# copy tftp://192.168.1.10/runjson running-config
######################################################################### 100.0%
Configuration may take several minutes to complete according to configuration file
size
 --0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%--
Success
```

Copying a CLI format file to the running configuration with an error in the file:

```
switch# copy tftp://192.168.1.10/runcli running-config
######################################################################### 100.0%
Configuration may take several minutes to complete according to configuration file
size
 --0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%--
Some of the configuration lines from the file were NOT applied. Use 'show
events -d hpe-config' for more info.
######################################################################### 100.0%
Configuration may take several minutes to complete according to configuration file
size
 --0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%--
Some of the configuration lines from the file were NOT applied. Use 'show
events -d hpe-config' for more info.
```

Copying a CLI format file to the startup configuration:

```
switch# copy tftp://192.168.1.10/startjson startup-config
######################################################################### 100.0%
100.0%
Success
```

Copying an unsupported file format to the startup configuration:

```
switch# copy tftp://192.168.1.10/startfile startup-config
######################################################################### 100.0%
100.0%
unsupported file format
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy running-config {startup-config | checkpoint <*CHECKPOINT-NAME*>}

`copy running-config {startup-config | checkpoint <CHECKPOINT-NAME>}`

Description

Copies the running configuration to the startup configuration or to a new checkpoint. If the startup configuration is already present, the command overwrites the existing startup configuration.

Parameter	Description
`startup-config`	Specifies that the startup configuration receives a copy of the running configuration.
`checkpoint <CHECKPOINT-NAME>`	Specifies the name of a new checkpoint to receive a copy of the running configuration. The checkpoint name can be comprised of alphanumeric character, underscores (_) and dashes (-), and must be 32 characters or fewer. **NOTE:** Do not start the checkpoint name with **CPC** because it is used for system-generated checkpoints.

Examples

Copying the running configuration to the startup configuration:

```
switch# copy running-config startup-config
Success
```

Copying the running configuration to a new checkpoint named **ckpt1**:

```
switch# copy running-config checkpoint ckpt1
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy {running-config | startup-config} *<REMOTE-URL>*

`copy {running-config | startup-config} <REMOTE-URL> {cli | json} [vrf <VRF-NAME>]`

Description

Copies the running configuration or the startup configuration to a remote file in either CLI or JSON format.

Parameter	Description
`{running-config | startup-config}`	Selects whether the running configuration or the startup configuration is copied to a remote file.

`<REMOTE-URL>`	Specifies the remote file using the syntax:
	TFTP format:
	`tftp://<IP-ADDR>[:<PORT-NUM>] [;blocksize=<Value>]/<FILENAME>`
	SFTP format:
	`sftp://<USERNAME>@<IP-ADDR> [:<PORT-NUM>]/<FILENAME>`
	SCP format:
	`scp://USER@{IP|HOST}[:PORT]/FILE`

`{cli | json}`	Selects the remote file format: P: CLI or JSON.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

Examples

Copying a running configuration to a remote file in CLI format:

```
switch# copy running-config tftp://192.168.1.10/runcli cli
######################################################################### 100.0%
Success
```

Copying a running configuration to a remote file in JSON format:

```
switch# copy running-config tftp://192.168.1.10/runjson json
######################################################################### 100.0%
Success
```

Copying a startup configuration to a remote file in CLI format:

```
switch# copy startup-config sftp://root@192.168.1.10/startcli cli
root@192.168.1.10's password:
sftp> put /tmp/startcli startcli
Uploading /tmp/startcli to /root/startcli
Connected to 192.168.1.10.
```

```
Success
```

Copying a startup configuration to a remote file in JSON format:

```
switch# copy startup-config sftp://root@192.168.1.10/startjson json
root@192.168.1.10's password:
sftp>
root@192.168.1.10's password:
sftp> put /tmp/startjson startjson
Uploading /tmp/startjson to /root/startjson
Connected to 192.168.1.10.
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy {running-config | startup-config} *<STORAGE-URL>*

`copy {running-config | startup-config} <STORAGE-URL> {cli | json}`

Description

Copies the running configuration or a startup configuration to a USB drive.

Parameter	Description
`{running-config	startup-config}`	Selects the running configuration or the startup configuration to be copied to the switch USB drive.
<STORAGE-URL>	Specifies a remote file with the following syntax: **usb:/*<file>***
`{cli	json}`	Selects the format of the remote file: CLI or JSON.

Usage

The switch supports JSON and CLI file formats when copying the running or starting configuration to the USB drive. The USB drive must be formatted with the FAT file system.

The USB drive must be enabled and mounted with the following commands:

```
switch(config)# usb
switch(config)# end
switch# usb mount
```

Examples

Copying a running configuration to a file named **runCLI** on the USB drive:

```
switch# copy running-config usb:/runCLI cli
Success
```

Copying a startup configuration to a file named **startCLI** on the USB drive:

```
switch# copy startup-config usb:/startCLI cli
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy startup-config running-config

`copy startup-config running-config`

Description

Copies the startup configuration to the running configuration.

Examples

```
switch# copy startup-config running-config
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy <*STORAGE-URL*> running-config

```
copy <STORAGE-URL> {running-config | startup-config | checkpoint <CHECKPOINT-NAME>}
```

Description

This command copies a specified configuration from the USB drive to the running configuration, to a startup configuration, or to a checkpoint.

Parameter	Description
<STORAGE-URL>	Specifies the name of a configuration file on the USB drive with the syntax: **usb:/*<FILE>***
running-config	Specifies that the configuration file is copied to the running configuration. The file must be in CLI, JSON, or checkpoint format or the copy will fail. the copy will not work.
startup-config	Specifies that the configuration file is copied to the startup configuration. The switch stores this configuration between reboots. The startup configuration is used as the operating configuration following a reboot of the switch. The file must be in JSON or checkpoint format or the copy will fail.
checkpoint *<CHECKPOINT-NAME>*	Specifies the name of a new checkpoint file to receive a copy of the configuration. The configuration file on the USB drive must be in checkpoint format. **NOTE:** Do not start the checkpoint name with **CPC** because it is used for system-generated checkpoints.

Usage

This command requires that the USB drive is formatted with the FAT file system and that the file be in the appropriate format as follows:

- **running-config**: This option requires the file on the USB drive be in CLI, JSON, or checkpoint format.
- **startup-config**: This option requires the file on the USB drive be in JSON or checkpoint format.
- **checkpoint *<checkpoint-name>***: This option requires the file on the USB drive be in checkpoint format.

Examples

Copying the file **runCli** from the USB drive to the running configuration:

```
switch# copy usb:/runCli running-config
Configuration may take several minutes to complete according to configuration
file size
--0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%--
Success
```

Copying the file **startUp** from the USB drive to the startup configuration:

```
switch# copy usb:/startUp startup-config
Success
```

Copying the file **testCheck** from the USB drive to the **abc** checkpoint:

```
switch# copy usb:/testCheck checkpoint abc
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

erase

```
erase
  checkpoint <checkpont-name>
  core-dump all|daemon|dsm|kernel
  startup-config
  all
```

Description

Deletes an existing checkpoint, startup configuration, or core-dump.

Parameter	Description
`checkpoint <CHECKPOINT-NAME>`	Specifies the name of a checkpoint.
`core-dump all|daemon <daemon-name> |kernel|vsf`	Erase one of the following sets of core-dump files: ■ all: Erase all core-dump files. ■ daemon <daemon-name>: Erase daemon core-dump files. ■ kerne:l Erase the kernel core-dump. ■
startup-config	Specifies the startup configuration.
all	Specifies all checkpoints.

Examples

Erasing checkpoint **ckpt1**:

```
switch# erase checkpoint ckpt1
```

Erasing the startup configuration:

```
switch# erase startup-config
```

Erasing all checkpoints:

```
switch# erase checkpoint all
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show checkpoint <*CHECKPOINT-NAME*>

```
show checkpoint <CHECKPOINT-NAME> [json]
```

Description

Shows the configuration of a checkpoint.

Parameter	Description
<CHECKPOINT-NAME>	Specifies the name of a checkpoint.
[json]	Specifies that the output is displayed in JSON format.

Examples

Showing the configuration of the **ckpt1** checkpoint in CLI format:

```
switch# show checkpoint ckpt1
Checkpoint configuration:
!
!Version AOS-CX PL.10.07.0000K-75-g55e5193
!export-password: default
lacp system-priority 65535
user admin group administrators password ciphertext
AQBapQjwipebv36io0jFfde7ZzrHckncal1D+3n8XFTZKQdmYgAAADEtYOeHSme93xzdD0uz6Vr9Kl+XBz
B+2GB0UBxSF7rvgN2x8KSgkqv7iqXVQ0Te6LkSMnH4BdNaT3Bf25qyvOqmr4YakO1V3rg8zAOADkPktQD8
joTHXflzwomoIzcmv/uX
cli-session
    timeout 0
!
!
!
!
ssh server vrf default
vlan 1
spanning-tree
interface lag 1
    no shutdown
    vlan access 1
```

```
interface lag 128
    no shutdown
    vlan access 1
interface lag 129
    shutdown
    vlan access 1
    lacp mode active
interface 1/1/1
    no shutdown
    lag 128
    lacp port-id 65535
interface 1/1/2
    no shutdown
    vlan access 1
interface 1/1/3
    no shutdown
    vlan access 1
interface 1/1/4
    no shutdown
    vlan access 1
interface 1/1/5
    no shutdown
    vlan access 1
interface 1/1/6
    no shutdown
    vlan access 1
interface 1/1/7
    no shutdown
    vlan access 1
interface 1/1/8
    no shutdown
    vlan access 1
interface 1/1/9
    no shutdown
    vlan access 1
interface 1/1/10
    no shutdown
    vlan access 1
interface 1/1/11
    no shutdown
    vlan access 1
interface 1/1/12
    no shutdown
    vlan access 1
interface 1/1/13
    no shutdown
    vlan access 1
interface 1/1/14
    no shutdown
    vlan access 1
interface 1/1/15
    no shutdown
    vlan access 1
interface 1/1/16
    no shutdown
    vlan access 1
interface vlan 1
    ip dhcp
snmp-server vrf default
!
!
!
```

```
!
!
https-server vrf default
```

Showing the configuration of the **ckpt1** checkpoint in JSON format:

```
switch# show checkpoint ckpt1 json
Checkpoint configuration:
{
    "AAA_Server_Group": {
        "local": {
            "group_name": "local"
        },
        "none": {
            "group_name": "none"
        }
    },
...
...
...
...
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show checkpoint <*CHECKPOINT-NAME*> hash

```
show checkpoint <CHECKPOINT-NAME> hash [cli | json]
```

Description

Shows a configuration checkpoint hash calculated with the SHA-256 algorithm. When the output format is not specified, the CLI format is used. This enables you to determine whether there has been a configuration change since a previous hash was calculated.

Parameter	Description
`<CHECKPOINT-NAME>`	Specifies an existing checkpoint name.
`[cli	json]`	Selects either the CLI or JSON format.

Examples

Showing a checkpoint SHA-256 hash in JSON format:

```
switch# show checkpoint ckpt1 hash json
Calculating the hash: [Success]

The SHA-256 hash of the checkpoint in JSON format, created in image XX.10.08.xxxx:

cc7a57a9bbb4e6600d3b4180296a35f6af9e797ce9c439955dfe5de58b06da9e

This hash is only valid for comparison to a baseline hash if the configuration has
not been explicitly changed (such as with a CLI command, REST operation, etc.)
or implicitly changed (such as by changing a hardware module, upgrading the
SW version, etc.).
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show checkpoint post-configuration

```
show checkpoint post-configuration
```

Description

Shows the configuration settings for creating system checkpoints.

Examples

```
switch# show checkpoint post-configuration

Checkpoint Post-Configuration feature
-------------------------------------

 Status                : enabled
 Timeout (sec) : 300
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show checkpoint

`show checkpoint`

Description

Shows a detailed list of all saved checkpoints.

Examples

Showing a detailed list of all saved checkpoints:

```
switch# show checkpoint

NAME              TYPE        WRITER   DATE(YYYY/MM/DD)       IMAGE VERSION
ckpt1             checkpoint  User     2017-02-23T00:10:02Z  XX.01.01.000X
ckpt2             checkpoint  User     2017-03-08T18:10:01Z  XX.01.01.000X
ckpt3             checkpoint  User     2017-03-09T23:11:02Z  XX.01.01.000X
ckpt4             checkpoint  User     2017-03-11T00:00:03Z  XX.01.01.000X
ckpt5             latest      User     2017-03-14T01:12:27Z  XX.01.01.000X
```

Command History

Release	Modification
10.08	Command syntax `show checkpoint list all` is replaced with `show checkpoint`.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show checkpoint date

`show checkpoint date <START-DATE> <END-DATE>`

Description

Shows detailed list of all saved checkpoints created within the specified date range.

Parameter	Description
<START-DATE>	Specifies the starting date for the range of saved checkpoints to show. Format: **YYYY-MM-DD**.
<END-DATE>	Specifies the endingdate for the range of saved checkpoints to show. Format: **YYYY-MM-DD**.

Examples

Showing a detailed list of saved checkpoints for a specific date range:

```
switch# show checkpoint date 2017-03-08 2017-03-12

NAME                 TYPE        WRITER  DATE(YYYY/MM/DD)      IMAGE VERSION
ckpt2                checkpoint  User    2017-03-08T18:10:01Z  XX.01.01.000X
ckpt3                checkpoint  User    2017-03-09T23:11:02Z  XX.01.01.000X
ckpt4                checkpoint  User    2017-03-11T00:00:03Z  XX.01.01.000X
```

Command History

Release	Modification
10.08	Command syntax `show checkpoint list date <START-DATE> <END-DATE>` is replaced with `show checkpoint date <START-DATE> <END-DATE>`
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show running-config hash

`show running-config hash [cli | json]`

Description

Shows the running-config checkpoint hash, calculated with the SHA-256 algorithm. When the output format is not specified, the CLI format is used. This enables you to determine whether there has been a configuration change since a previous hash was calculated.

Parameter	Description
`[cli	json]`	Selects either the CLI or JSON format.

Examples

Showing the running-config checkpoint SHA-256 hash in CLI format:

```
switch# show running-config hash cli
Calculating the hash: [Success]

SHA-256 hash of the config in CLI format:

8db4e7e10f4b7f1a6ab17ad2b4efe0e72f1849103eaf43da62aa1d715075b89e

This hash is only valid for comparison to a baseline hash if the configuration has
not been explicitly changed (such as with a CLI command, REST operation, etc.)
or implicitly changed (such as by changing a hardware module, upgrading the
SW version, etc.).
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show startup-config hash

`show startup-config hash [cli | json]`

Description

Shows the startup-config checkpoint hash, calculated with the SHA-256 algorithm. When the output format is not specified, the CLI format is used. This enables you to determine whether there has been a configuration change since a previous hash was calculated.

Parameter	Description
`[cli	json]`	Selects either the CLI or JSON format.

Examples

Showing the startup-config checkpoint SHA-256 hash in CLI format:

```
switch# show startup-config hash cli
Calculating the hash: [Success]

SHA-256 hash of the config in CLI format:

8db4e7e10f4b7f1a6ab17ad2b4efe0e72f1849103eaf43da62aa1d715075b89e

This hash is only valid for comparison to a baseline hash if the configuration has
not been explicitly changed (such as with a CLI command, REST operation, etc.)
or implicitly changed (such as by changing a hardware module, upgrading the
SW version, etc.).
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

write memory

`write memory`

Description

Saves the running configuration to the startup configuration. It is an alias of the command **copy running-config startup-config**. If the startup configuration is already present, this command overwrites the startup configuration.

Examples

```
switch# write memory
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Boot commands

boot set-default

`boot set-default {primary | secondary}`

Description

Sets the default operating system image to use when the system is booted.

Parameter	Description
`primary`	Selects the primary network operating system image.
`secondary`	Selects the secondary network operating system image.

Example

Selecting the primary image as the default boot image:

```
switch# boot set-default primary
Default boot image set to primary.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

boot system

`boot system [primary | secondary | serviceos]`

Description

Reboots all modules on the switch. By default, the configured default operating system image is used. Optional parameters enable you to specify which system image to use for the reboot operation and for future reboot operations.

Parameter	Description
`primary`	Selects the primary operating system image for this reboot and sets the configured default operating system image to **primary** for future reboots.
secondary	Selects the secondary operating system image for this reboot and sets the configured default operating system image to **secondary** for future reboots.
`serviceos`	Selects the service operating system for this reboot. Does not change the configured default operating system image. The service operating system acts as a standalone bootloader and recovery OS for switches running the AOS-CX operating system and is used in rare cases when troubleshooting a switch.

Usage

This command reboots the entire system. If you do not select one of the optional parameters, the system reboots from the configured default boot image.

You can use the **show images** command to show information about the primary and secondary system images.

Choosing one of the optional parameters affects the setting for the default boot image:

- If you select the **primary** or **secondary** optional parameter, that image becomes the configured default boot image for future system reboots. The command fails if the switch is not able to set the operating system image to the image you selected.

You can use the **boot set-default** command to change the configured default operating system image.

- If you select **serviceos** as the optional parameter, the configured default boot image remains the same, and the system reboots all management modules with the service operating system.

If the configuration of the switch has changed since the last reboot, when you execute the **boot system** command you are prompted to save the configuration and you are prompted to confirm the reboot operation.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot system** command is aborted.

Examples

Rebooting the system from the configured default operating system image:

```
switch# boot system
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
The system is going down for reboot.
```

Rebooting the system from the secondary operating system image, setting the secondary operating system image as the configured default boot image:

```
switch# boot system secondary
Default boot image set to secondary.

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Canceling a system reboot:

```
switch# boot system

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? n
Reboot aborted.
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show boot-history

```
show boot-history [all|{vsf member <1-10>}]
```

Description

Shows boot history information. When no parameters are specified, shows the most recent information about the current boot operation, and the three previous boot operations for the switch. When the **all** parameter is specified, the output of this command shows the boot information for the active management module.

.

To view boot-history on a standby, the command must be sent on the conductor console.

Parameter	Description
`all`	Optional. Shows boot information for the active management module.
`vsf member <1-10>`	Optional. Display boot history for the specified VSF member

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

The output of this command includes the following information:

Parameter	Description
`Index`	The position of the boot in the history file. Range: 0 to 3.
`Boot ID`	A unique ID for the boot . A system-generated 128-bit string.
`Current Boot, up for <time>`	For the current boot, the **show boot-history** command shows the number of seconds the module has been running on the current software.
`<Timestamp>: boot reason`	For previous boot operations, the **show boot-history** command shows the time at which the operation occurred and the reason for the boot. The reason for the boot is one of the following values: ■ ***<DAEMON-NAME>* crash**: The daemon identified by *<DAEMON-NAME>* caused the module to boot. ■ **Kernel crash**: The operating system software associated with the module caused the module to boot. ■ **Uncontrolled reboot**: The reason for the reboot is not known. ■ **Reboot requested through database**: The reboot occurred because of a request made

Parameter	Description
	through the CLI or other API.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=================

Index : 2
Boot ID : c34a2c2499004a02bbeeff4992e1fdbd
Current Boot, up for 1 days 13 hrs 13 mins 27 secs

Index : 1
Boot ID : bfba9bc486304e57904ac717a0ccbdcd
02 Sep 23 02:55:33 : CPU request reset with 0x20201, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:55:33 : Switch boot count is 2

Index : 0
Boot ID : a88a71b7ca9a4574af7e3b811ddfdc7e
02 Sep 23 02:49:26 : Reboot requested by user, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:50:02 : Switch boot count is 1

Index : 3
Boot ID : f00ba10c8c44457f83fee303d014a89a
25 Aug 23 10:27:42 :  Power on reset with 0x1, Version: FL.10.14.0000-1465-
g9df95249d06b0~dirty
25 Aug 23 10:28:18 :  Switch boot count is 3
25 Aug 23 10:29:02 :  Primary overtemperature fault detected with 0x2 in PSU 1/1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Firmware management commands

copy {primary | secondary} *<REMOTE-URL>*

`copy {primary | secondary} <REMOTE-URL> [vrf <VRF-NAME>]`

Description

Uploads a firmware image to a TFTP or SFTP server.

Parameter	Description
`{primary | secondary}`	Selects the primary or secondary image profile to upload. Required

`<REMOTE-URL>`	Specifies the URL to receive the uploaded firmware using SFTP , TFTP or SCP.
	TFTP format:
	`tftp://<IP-ADDR>[:<PORT-NUM>] [;blocksize=<Value>]/<FILENAME>`
	SFTP format:
	`sftp://<USERNAME>@<IP-ADDR> [:<PORT-NUM>]/<FILENAME>`
	SCP format:
	`scp://USER@{IP|HOST}[:PORT]/FILE`

`vrf <VRF-NAME>`	Specifies a VRF name. Default: default.

Examples

TFTP upload:

```
switch# copy primary tftp://192.0.2.0/00_10_00_0002.swi
######################################################################## 100.0%
Verifying and writing system firmware...
```

SFTP upload:

```
switch# copy primary sftp://swuser@192.0.2.0/00_10_00_0002.swi
swuser@192.0.2.0's password:
Connected to 192.0.2.0.
sftp> put primary.swi XL_10_00_0002.swi
Uploading primary.swi to /users/swuser/00_10_00_0002.swi
primary.swi                           100%  179MB  35.8MB/s   00:05
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy {primary | secondary} *<FIRMWARE-FILENAME>*

`copy {primary | secondary} <FIRMWARE-FILENAME>`

Description

Copies a firmware image to USB storage.

Parameter	Description
`{primary | secondary}`	Selects the primary or secondary image from which to copy the firmware. Required
<FIRMWARE-FILENAME>	Specifies the name of the firmware file to create on the USB storage device. Prefix the filename with **usb:/**. For example: **usb:/firmware_v1.2.3.swi** For information on how to format the path to a firmware file on a USB drive, see USB URL.

Examples

```
switch# copy primary usb:/11.10.00.0002.swi
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy primary secondary

`copy primary secondary`

Description

Copies the firmware image from the primary to the secondary location.

Examples

```
switch# copy primary secondary
The secondary image will be deleted.

Continue (y/n)? y
Verifying and writing system firmware...
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy *<REMOTE-URL>*

`copy <REMOTE-URL> {hot-patch|primary|secondary} [vrf <VRF-NAME>]`

Description

Downloads a hot-patch or firmware image from a TFTP or SFTP server.

Parameter	Description

`<REMOTE-URL>`	Specifies the URL from which to download the firmware using SFTP or TFTP.
	TFTP format:
	`tftp://<IP-ADDR>[:<PORT-NUM>]`
	`[;blocksize=<Value>]/<FILENAME>`
	SFTP format:
	`sftp://<USERNAME>@<IP-ADDR>`
	`[:<PORT-NUM>]/<FILENAME>`
	SCP format:
	`scp://USER@{IP|HOST}[:PORT]/FILE`

`{hot-patch|primary|secondary}`	Select a hot-patch or a primary or secondary image profile for receiving the downloaded firmware. Required. **NOTE:** For more information about hot-patch, see hot-patch.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

TFTP usage

To specify a URL with:

- an IPv4 address: **tftp://*192.0.2.1/a.txt***
- an IPv6 address: **tftp://*[2000::2]/a.txt***
- a hostname: **tftp://*hpe.com/a.txt***

To specify TFTP with:

- the port number of the server in the URL: **tftp://*192.0.2.1:12/a.txt***
- the blocksize in the URL: **tftp://*192.0.2.1*;blocksize=*1462/a.txt***

The valid blocksize range is 8 to 65464.

- the port number of the server and blocksize in the URL: **tftp://192.0.2.1*:12*;blocksize=*1462/a.txt***

To specify a file in a directory of URL: **tftp://*192.0.2.1/dir/a.txt***

SFTP usage

To specify:

- A URL with an IPv4 address: **sftp://*user@192.0.2.1/a.txt***
- A URL with an IPv6 address: **sftp://*user*@[*2000::2*]*/a.txt***

- A URL with a hostname: ***sftp://user@hpe.com/a.txt***
- SFTP port number of a server in the URL: ***sftp://user@192.0.2.1:12/a.txt***
- A file in a directory of URL: ***sftp://user@192.0.2.1/dir/a.txt***
- To specify a file with absolute path in the URL: ***sftp://user@192.0.2.1//home/user/a.txt***

SCP Usage

To specify:

- A username with an IP address: **scp://user@192.0.2.1:12/a.txt**
- A username with a remote host: **scp://user@hpe.com/a.txt**

Examples

TFTP download for a hot-patch:

```
switch# copy tftp://192.168.1.1/FL.10.12.0001-0002.patch hot-patch vrf vrf1

Fetching /users/swuser/FL.10.10.0001-0002.patch to hotpatch.dnld.uE2YT1
FL.10.12.0001-0002.patch                     100%   62KB  12.4MB/s   00:00

Verifying and writing hot-patch...
```

TFTP download for primary software image:

```
switch#  copy tftp://192.10.12.0/FL_10_12_0001.swi primary
The primary image will be deleted.

Continue (y/n)? y
######################################################################## 100.0%
Verifying and writing system firmware...
```

SFTP download:

```
switch# copy sftp://swuser@192.10.12.0/FL_10_12_0001.swi primary
The primary image will be deleted.

Continue (y/n)? y
The authenticity of host '192.10.12.0 (192.10.12.0)' can't be established.
ECDSA key fingerprint is SHA256:L64khLwlyLgXlARKRMiwcAAK8oRaQ8C0oWP+PkGBXHY.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.10.12.0' (ECDSA) to the list of known hosts.
swuser@192.10.12.0's password:
Connected to 192.10.12.0.
Fetching /users/swuser/ss.10.00.0002.swi to ss.10.00.0002.swi.dnld
/users/swuser/ss.10.00.0002.swi                   100%  179MB  25.6MB/s   00:07
Verifying and writing system firmware...
```

Command History

Release	Modification
10.12	The `hot-patch` parameter is supported on all platforms.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy secondary primary

```
copy secondary primary
```

Description

Copies the firmware image from the secondary to the primary location.

Examples

```
switch# copy secondary primary
The primary image will be deleted.

Continue (y/n)? y
Verifying and writing system firmware...
```

```
switch# copy sftp://stor@192.22.1.0/im-switch.swi primary vrf default
The primary image will be deleted.

Continue (y/n)? y
The authenticity of host '192.22.1.0 (192.22.1.0)' can't be established.
ECDSA key fingerprint is SHA256:MyI1xbdKnehYut0NLfL69gDpNzCmZqBVvBaRR46m7o8.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.22.1.0' (ECDSA) to the list of known hosts.
stor@192.22.1.0's password:
Connected to 192.22.1.0.
sftp> get c8d5b9f-topflite.swi c8d5b9f-topflite.swi.dnld
Fetching /home/dr/im-switch.swi to c8d5b9f-topflite.swi.dnld
/home/dr/im-switch.swi                100%  226MB  56.6MB/s   00:04

Verifying and writing system firmware...
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy <*STORAGE-URL*>

```
copy <STORAGE-URL> {hot-patch|primary|secondary}
```

Description

Copies, verifies, and installs a hot-patch or firmware image from a USB storage device connected to the active management module.

Parameter	Description
<STORAGE-URL>	Specifies the name of the firmware file to copy from the storage device. Required. **USB format:** `usb:/<FILENAME>`
`{hot-patch|primary|secondary}`	Select a hot-patch image or a primary or secondary profile for receiving the copied firmware. **NOTE:** For more information about hot-patch, see hot-patch.

USB usage

To specify a file:

- In a USB storage device: **usb:/*a.txt***
- In a directory of a USB storage device: **usb:/*dir*/*a.txt***

Examples

```
switch# copy usb:/FL.10.12.0001-0002.patch
```

```
switch# copy usb:/FL.10.12.0001.swi primary
The primary image will be deleted.

Continue (y/n)? y

Verifying and writing system firmware...
```

Command History

Release	Modification
10.12	The `hot-patch` parameter is supported on all platforms.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy hot-patch

`copy hot-patch <Word> {<REMOTE-URL>|<Storage-URL>} [vrf <VRF-NAME>]`

Description

Copies a hot-patch from a switch to the specified remote URL or storage URL.

Parameter	Description
<Word>	Name of the hot-patch software to upload.
<REMOTE-URL>	Specifies the URL to receive the uploaded patch using SFTP or TFTP. For information on how to format the remote URL, see URL formatting for copy commands.
vrf *<VRF-NAME>*	[Optional] specify the VRF instance to use for upload.
<STORAGE-URL>	Specifies the name of the patch file to create on the USB storage device. Prefix the filename with **usb:/**, for example, **usb:/firmware_FL_10_12_0001-0002.patch.**

Examples

```
switch# copy hot-patch FL_10_12_0001-0002.patch tftp:172.21.18.170/FL_10_12_0001-
0002.patch vrf vrf1
```

Related Commands

Command	Description
copy <REMOTE-URL>	Downloads a hot-patch image from a TFTP or SFTP server.
hot-patch	Apply a hot-patch image or remove it from the switch.

Command History

Release	Modification
10.12	Hot-patch is now supported on all platforms.
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

hot-patch

```
hot-patch apply|remove <name.patch>
no hot-patch apply <name.patch>
```

Description

Apply hot-patch software or remove it from the switch. The **no** form of the **hot-patch apply** command disables the hot-patch image, but does not remove it from the switch. Rebooting the system after disabling or removing the patch is not required.

Profile names	Description
`apply <name.patch>`	Apply the specified hot-patch image to a standalone switch or VSF stack. AOS-CX hot-patch software images can be obtained from Aruba customer support, and are identified with a **.patch** extension.
`remove <name.patch>`	Disables the hot-patch image and removes the patch from the switch. This removal will also disable the patch. Once removed, a hot-patch must be downloaded again in order to be applied.

Usage

A hot-patch can be downloaded from a remote server onto a switch then applied without rebooting the switch. When the hot-patch is disabled, the hot-patch will still remain on the system. The disabled hot-patch can be removed from the system without the need for a reboot of the system.

If a checkpoint configuration that does not contain a hot-patch is restored to a running configuration that does have a hot-patch, the patch is not deleted, it remains as not applied but is present in the device memory.

Examples

```
switch(config)# hot-patch apply FL_10_12_0001-0002.patch
```

Related Commands

Command	Description
`copy <REMOTE-URL>`	Downloads and installs a hot-patch image from a TFTP or SFTP server.
`copy hot-patch`	Copies a hot-patch software image from a switch to a specified remote URL or storage URL.

Command History

Release	Modification
10.12	Hot-patch is now supported on all platforms.
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

show hot-patch

`show hot-patch [detail]`

Parameter	Description
`detail`	Displays the detailed status of all hot-patches present on the system.

Description

the **show hot-patch** command displays the status of all hot-patches present on the system. The **show hot-patch detail** command displays detailed information for all hot patches present on the system.

Examples

```
switch# show hot-patch

Name                            Status
-----------------------         -------
FL_10_12_0001-0002.patch        Applied

switch# show hot-patch detail

Name                   : FL_10_12_0001-0002.patch
Status                 : Applied
Version                : FL_10_12_0001-0002.patch
Compatible Version     : FL.10.12.0001
Issues Fixed           : CR1234, CR2345
Patch Date             : 2022-03-29 20:46:15 UTC
Patch ID               : ArubaOS-CX:FL.10.12.0001-sp1-256-gd457e868d39:202204142009
Patch SHA              : a40438d06a82e5fe7e30d457e868d39e8526185b
```

Related Commands

Command	Description
copy <REMOTE-URL>	Downloads a hot-patch image from a TFTP or SFTP server.
hot-patch	Apply a hot-patch image or remove it from the switch.

Command History

Release	Modification
10.12	Command supported on all platforms.
10.10	Command introduced on 6300 Switch series.

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Chapter 9

Dynamic Segmentation

For information on dynamic segmentation, view the *Security Guide*.

Chapter 10
SNMP

Simple Network Management Protocol (SNMP) is an Internet-standard protocol used for managing and monitoring the devices connected to a network by collecting, organizing and modifying information about managed devices on IP networks.

Configuring SNMP

(The SNMP agent provides read-only access.)

Procedure

1. Enable SNMP on a VRF using the command `snmp-server vrf default`.
2. Set the system contact, location, and description for the switch with the following commands:
   - `snmp-server system-contact`
   - `snmp-server system-location`
   - `snmp-server system-description`
3. If required, change the default SNMP port on which the agent listens for requests with the command `snmp-server agent-port`.
4. By default, the agent uses the community string **public** to protect access through SNMPv1/v2c. Set a new community string with the command `snmp-server community`.
5. Configure the trap receivers to which the SNMP agent will send trap notifications with the command `snmp-server host`.
6. Create an SNMPv3 context and associate it with any available SNMPv3 user to perform context specific v3 MIB polling using the command `snmpv3 user` .
7. Create an SNMPv3 context and associate it with an available SNMPv1/v2c community string to perform context specific v1/v2c MIB polling using the command `snmpv3 context`.
8. Review your SNMP configuration settings with the following commands:
   - `show snmp agent-port`
   - `show snmp community`
   - `show snmp system`
   - `show snmpv3 context`
   - `show snmp trap`
   - `show snmp vrf`
   - `show snmpv3 users`
   - `show tech snmp`

Example 1

This example creates the following configuration:

- Sets the contact, location, and description for the switch to: **JaniceM**, **Building2**, **LabSwitch**.
- Sets the community string to **Lab8899X**.

```
switch(config)# snmp-server vrf default
switch(config)# snmp-server system-contact JaniceM
switch(config)# snmp-server system-location Building2
switch(config)# snmp-server system-description LabSwitch
switch(config)# snmp-server community Lab8899X
```

Example 2

This example creates the following configuration:

- Creates an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**.
- Associates the SNMPv3 user `Admin` with a context named `newContext`.

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des
     priv-pass plaintext myprivpass
switch(config)# snmpv3 user Admin context newContext
```

Refer to the SNMP Guide for SNMP Commands.

Chapter 11

Aruba Central integration

The Aruba Central network management solution, a software-as-a-service subscription in the cloud, provides streamlined management of multiple network devices. AOS-CX switches are able to talk to Aruba Central and utilize cloud-based management functionality. Cloud-based management functionality allows for the deployment of network devices at sites with no or few dedicated IT personnel (branch offices, retail stores, and so forth). AOS-CX switches utilize secure communication protocols to connect to the Aruba Central cloud portal, and can coexist with corporate security standards, such as those mandating the use of firewalls.

When Aruba Central manages AOS-CX switches, it functions as the single source of truth and the Web UI operates in read-only mode.

This feature provides:

- Zero-touch provisioning
- Network Management/Remote monitoring
- Events/alerts notification
- Switch Configuration using templates
- Firmware management

Connecting to Aruba Central

AOS-CX switch downloads the location of Aruba Central server using:

- Command-line interface (CLI).
- Aruba Activate server.
- DHCP options provided during ZTP.

DHCP servers are used to connect to Central on-premise management.

If switch is unable to connect to Activate server, it retries to establish connection in exponential back off of 1s, 2s, 8s, 16s, 32s, 64s, 128s, and 256s. After the maximum back off of 256s, switch retries happen for every 5 minutes.

If the Network Time Protocol (NTP) is not enabled on the switch, it will synchronize the system time with the Activate server.

When a switch is able to connect to Aruba Central, but is not registered in the Aruba Central inventory or does not have a proper license, the switch will get disconnected. If the Aruba Central feature is enabled using this command, the switch will then reconnect back to Aruba Central and will get disconnected again. This connect/disconnect process will continue until the switch is properly registered in Aruba Central. To avoid this unnecessary reconnection cycle, best practices is to disable Aruba Central until the switch is registered in Aruba Central, or a license is obtained for that device.

Custom CA certificate

To use a custom CA certificate to connect to Aruba Central, AOS-CX switch downloads the certificate from the Aruba Activate server.

- If there is no custom CA provided by Aruba Activate, the CA certificate present in the device is used.
- Duplicate CA certificates from Aruba Activate server will be ignored.
- If CA certificate is absent in consecutive responses from Aruba Activate server, the installed custom CA certificate in device will be removed.
- Switch will have only one custom CA certificate installed from Aruba Activate Server.
- The certificate installed from Aruba Activate server will not be displayed in the show commands.

Support mode in Aruba Central

When the AOS-CX switch is managed by Aruba Central, the switch configuration cannot be modified using other interfaces such as CLI or Web UI. The following command categories are blocked:

- auto-confirm
- boot
- checkpoint
- copy-in commands
- erase
- https-server
- mfgread
- mfgwrite
- port-access
- All configuration commands except the aruba-central command

In cases where a maintenance or troubleshooting activity requires configuration updates, aruba-central support-mode can be enabled to allow these operations.

The aruba-central support-mode enable or disable operation is effective only in the CLI session where it is executed and does not impact the other CLI sessions.

If the user tries to execute any command that is not allowed, an **Invalid input:** error message is displayed.

One Touch Provisioning

One Touch Provisioning (OTP) allows a switch to be provisioned on Aruba Central with the minimum user configuration.

Configuring One Touch Provisioning

To provision a switch to Aruba Central by using an OTP scenario, you can configure:

- The Aruba Central Location

Following configurations can be done optionally:

- The Domain Name System (DNS) server address.
- The HTTP proxy location.

Configuring Aruba Central location

1. Create the Aruba-Central context.

```
switch(config)# aruba-central
```

2. Configure the Aruba Central server location

```
switch(config-aruba-central)# location-override <LOCATION> vrf <VRF>
```

IPv4 or IPv6 address can be used.

3. Validate the connectivity status to Aruba Central

```
switch# show aruba-central
```

Configuring DNS server address

1. Configure the DNS server address.

```
switch(config)# ip dns server-address <ADDRESS> vrf <VRF>
```

IPv4 or IPv6 a ddress can be used.

2. Validate the DNS server status

```
switch# show ip dns
```

Configuring HTTP proxy location

1. Configure the HTTP Proxy location

```
switch(config)# http-proxy <LOCATION> vrf <VRF>
```

LOCATION can be either a FQDN, IPv4 address or IPv6 address

Limitations for Zero Touch Provisioning

The following are the limitations for OTP when provisioning a switch by using IPv6:

- Support for obtaning the Aruba Central location via Activate.
- Support for obtaning the Aruba Central location via DHCP.
- Support for connectivity to an Aruba Central on-premise instance.

Supportability

Debug logging can be enabled with the following command:

The central debug logs are helpful to determine connectivity issues with Aruba Central.

```
switch# debug central all
```

Aruba Central commands

aruba-central

Applies only to the 4100i switch.

```
aruba-central
no aruba-central
```

Description

Creates or enters the Aruba Central configuration context (`config-aruba-central`).

Example

Creating the Aruba Central configuration context:

```
switch(config)# aruba-central
switch(config-aruba-central)#
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
4100i	`config`	Administrators or local user group members with execution rights for this command.

aruba-central support-mode

Applies only to the 4100i switch.

```
aruba-central support-mode
```

```
no aruba-central support-mode
```

Description

Allows the device to be writable for all operations in Aruba Central lockout mode for troubleshooting. The no form of this command disables this activity.

> Support-mode is disabled by default when the switch is managed by Aruba Central. This command is only effective in the CLI session where it is executed.

Examples

Configuring the device to be writable for all operations in Aruba Central lockout mode:

```
switch# aruba-central support-mode
switch#
```

Removing the configuration that allows the device to be writable for all operations in Aruba Central lockout mode:

```
switch# no aruba-central support-mode
switch#
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
4100i	Manager (#)	Administrators or local user group members with execution rights for this command.

configuration-lockout central managed

> Applies only to the 4100i switch.

```
configuration-lockout central managed
no configuration-lockout central managed
```

Description

Configures the device to only be writable from Aruba Central. Aruba Central will be the only agent that can add, modify, or delete configurations on the device. The no form of this command disables this feature.

> The no form of this command is only available when the device is disconnected from Aruba Central.

Usage

The AOS-CX switch connects to Aruba Central in either of two modes: monitor or managed. When the device is connected in monitor mode, Aruba Central monitors the configurations on the switch. When the device is connected in managed mode, the **configuration-lockout central managed** command does not allow configuration changes from other interfaces such as CLI or Web UI.

Examples

Configuring the device to only be writable from Aruba Central :

```
switch(config)# configuration-lockout central managed
switch# show configuration-lockout
configuration lockout
---------------------
central: managed
switch# sh aruba-central
Central admin state                    :enable
Central location                       :20.0.0.2:8083
VRF for connection                     :default
Central connection status              :connected

Central source                         :cli
Central source connection status       :connected
Central source last connected on       :Tue Feb 9 17:53:13 UTC 2021

Activate Server URL                    :devices-v2.arubanetworks.com
CLI location                           :20.0.2:8083
CLI VRF                                :default
switch(config)# end
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
4100i	`config`	Administrators or local user group members with execution rights for this command.

disable

Applies only to the 4100i switch.

```
disable
```

Description

Disables connection to Aruba Central server.

When the connection is disabled, the switch does not attempt to connect to the Aruba Central server or fetch central location from any of the three sources (CLI/Aruba Activate/DHCP). It also disconnects any active connection to the Aruba Central server.

Example

```
switch(config-aruba-central)# disable
switch(config-aruba-central)#
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
4100i	`config-aruba-central`	Administrators or local user group members with execution rights for this command.

enable

`enable`

Description

Enables connection to Aruba Central server. When the connection is enabled, the switch attempts to download the location of the Aruba Central server in one of the following ways at startup and after the connection is lost:

- Using command-line interface (CLI).
- Connecting to Aruba Activate server.
- Using DHCP options provided during ZTP.

DHCP servers provide the options requested by the device to connect to Central, Central On-premise managment, or the TFTP server.

When a switch is able to connect to Aruba Central, but is not registered in the Aruba Central inventory or does not have a proper license, the switch will get disconnected. If the Aruba Central feature is enabled using this command, the switch will then reconnect back to Aruba Central and will get disconnected again. This connect/disconnect process will continue until the switch is properly registered in Aruba Central. To avoid this unnecessary reconnection cycle, best practices is to disable Aruba Central until the switch is registered in Aruba Central, or a license is obtained for that device.

Examples

```
switch(config-aruba-central)# enable
switch(config-aruba-central)#
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
4100i	`config-aruba-central`	Administrators or local user group members with execution rights for this command.

location-override

```
location-override <location> [vrf <VRF>]
no location-override
```

Description

When **location** and **vrf** are configured, the switch overrides existing connections to Aruba Central. The switch attempts to establish connection to Aruba Central with the specified location and VRF with highest priority.

Location can take one of the following values:

- A fully qualified domain name (FQDN) along with an optional port number.
- An IPv4 address with an optional port number.
- An IPv6 address with an optional port number.

If the port number is not specified, then port 443 is used by default. If the command is executed without the VRF parameter, the switch uses

the 'default' VRF.

The **no** form of this command removes location override values from the Aruba Central configuration context.

When you configure an IPv6 address with a port number, specify the address part inside square brackets, optionally followed by the port number, e.g. [2001:0db8:85a3:0000:0000:8a2e:0370:7334]:443.

Parameter	Description
<location>	Specifies one of these values: ■ *<FQDN>*: a fully qualified domain name. ■ *<IPV4>*: an IPv4 address. ■ *<IPV6>*: an IPv6 address.
`vrf` *<VRF-NAME>*	Specifies the VRF name to be used for communicating with the server. If no VRF name is provided, the default VRF named `default` is used.

Examples

Configuring location override with location and VRF:

```
switch(config-aruba-central)# location-override aruba-central.com vrf default
switch(config-aruba-central)#
switch(config-aruba-central)# location-override aruba-central.com vrf red
switch(config-aruba-central)# location-override 10.0.0.1 vrf red
switch(config-aruba-central)# location-override 10.0.0.1:443 vrf red
```

```
switch(config-aruba-central)# location-override
2001:0db8:85a3:0000:0000:8a2e:0370:7334 vrf red
switch(config-aruba-central)# location-override
[2001:0db8:85a3:0000:0000:8a2e:0370:7334]:443 vrf red
```

Configuring location override with location only:

```
switch(config-aruba-central)# location-override aruba-central.com
switch(config-aruba-central)#
```

Removing location override values from the Aruba Central configuration context:

```
switch(config-aruba-central)# no location-override
switch(config-aruba-central)#
```

Command History

Release	Modification
10.13.1000	Command updated to reflect OTP scenario..
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	`config-aruba-central`	Administrators or local user group members with execution rights for this command.

location-override-alternative

```
location-override-alternative <LOCATION> [vrf <VRF>]
no location-override-alternative <LOCATION> [vrf <VRF>]
```

Description

Configures information about Aruba Central connection when the alternative location is used.

The **no** form of this command removes the **location-override-alternative** configuration.

Parameter	Description
<LOCATION>	Specifies the Aruba-Central location.
vrf *<VRF>*	Specifies the VRF used to connect to Aruba-Central.

Usage

When the main and alternative Aruba Central server locations are specified, the switch attempts to connect to the main Aruba Central server. If there is connectivity failure with the main Aruba Central server location, it attempts to establish a connection with the alternative server location.

If the alternative location is configured without a main location, the user is prompted for confirmation. In this case, there is no redundancy and the switch attempts to connect to the alternative location.

Location can take one of the following values:

- A fully qualified domain name (FQDN) along with an optional port number.
- An IPv4 address with an optional port number.
- An IPv6 address with an optional port number.

If the port number is not specified, then port 443 is used by default. If the command is executed without the VRF parameter, the switch uses

the 'default' VRF.

An Aruba Central server location can only be a fully qualified domain name (FQDN) or a valid IP address. If the command is entered without the VRF parameter, the switch uses the default VRF.

Examples

Example of configuring with the aruba-central.com location and VRF red:

```
switch(config-aruba-central)# location-override-alternative aruba-central.com vrf
red
switch(config-aruba-central)#
```

Example of a configuration with location only:

```
switch(config-aruba-central)# location-override-alternative aruba-central.com
switch(config-aruba-central)#
```

Example of removing the override configuration:

```
switch(config-aruba-central)# no location-override-alternative
switch(config-aruba-central)# location-override-alternative 10.0.0.1 vrf red
switch(config-aruba-central)# location-override-alternative 10.0.0.1:443 vrf red
switch(config-aruba-central)# location-override-alternative
2001:0db8:85a3:0000:0000:8a2e:0370:7334 vrf red
switch(config-aruba-central)# location-override-alternative
[2001:0db8:85a3:0000:0000:8a2e:0370:7334]:443 vrf red
```

Command History

Release	Modification
10.13.1000	Command updated to reflect OTP scenario.
10.12.1000	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config-aruba-central`	Administrators or local user group members with execution rights for this command.

show aruba-central

Applies only to the 4100i switch.

```
show aruba-central
```

Description

Shows information about Aruba Central connection and the status of the Activate server connection.

Examples

Example of a switch that has the Aruba Central connection:

```
switch# show aruba-central
Central admin state                         : enabled
Central location                            : 10.0.0.1
VRF for connection                          : mgmt
Shared Token                                : N/A
Central connection status                   : connected
Central source                              : activate
Central source connection status            : connected
Central source last connected on            : Wed Jun 28 23:07:25 UTC 2023
Main location                               : 10.0.0.1
Main VRF                                    : mgmt
Alternative location                        : N/A
Alternative VRF                             : N/A
Activate Server URL                         : devices-v2.arubanetworks.com
System time synchronized from Activate      : N/A
Source IP                                   : N/A
Source IP Overridden                        : False
Central support mode                        : disabled
```

Example of a switch when the main CLI location is used:

```
switch# show aruba-central
Central admin state                        : enabled
Central location                           : 10.0.0.1
VRF for connection                         : mgmt
Shared secret                              : N/A
Central connection status                  : connected
Central source                             : cli
Central source connection status           : connected
Central source last connected on           : Wed Jun 28 23:07:25 UTC 2023
Main location                              : 10.0.0.1
Main VRF                                   : mgmt
Alternative location                       : 20.0.0.1
Alternative VRF                            : default
Activate server URL                        : devices-v2.arubanetworks.com
System time synchronized from Activate : N/A
Source IP                                  : N/A
Source IP Overridden                       : False
Central support mode                       : disabled
```

Example of a switch when the alternative CLI location is used:

```
switch# show aruba-central
Central admin state                        : enabled
```

```
Central location                        : 20.0.0.1
VRF for connection                      : default
Shared secret                           : N/A
Central connection status               : connected
Central source                          : cli
Central source connection status        : connected
Central source last connected on        : Wed Jun 28 23:07:25 UTC 2023
Main location                           : 10.0.0.1
Main VRF                                : mgmt
Alternative location                    : 20.0.0.1
Alternative VRF                         : default
Activate server URL                     : devices-v2.arubanetworks.com
System time synchronized from Activate : N/A
Source IP                               : N/A
Source IP Overridden                    : False
Central support mode                    : disabledswitch# show aruba-central
Central admin state                     : enabled
Central location                        : 20.0.0.1
VRF for connection                      : default
Shared secret                           : N/A
Central connection status               : connected
Central source                          : cli
Central source connection status        : connected
Central source last connected on        : Wed Jun 28 23:07:25 UTC 2023
Main location                           : 10.0.0.1
Main VRF                                : mgmt
Alternative location                    : 20.0.0.1
Alternative VRF                         : default
Activate server URL                     : devices-v2.arubanetworks.com
System time synchronized from Activate : N/A
Source IP                               : N/A
Source IP Overridden                    : False
Central support mode                    : disabled
```

Example of a switch when the location is obtained from DHCP options:

```
switch# show aruba-central
Central admin state                     : enabled
Central location                        : central-western-us.arubanetworks.com
VRF for connection                      : RED
Shared secret                           : N/A
Central connection status               : connected
Central source                          : DHCP
Central source connection status        : connected
Central source last connected on        : Fri Jun 30 20:22:33 UTC 2023
Main location                           : central-western-us.arubanetworks.com
Main VRF                                : mgmt
Alternative location                    : N/A
Alternative VRF                         : N/A
Activate server URL                     : devices-v2.arubanetworks.com
System time synchronized from Activate : N/A
Source IP                               : 100.0.0.1
Source IP Overridden                    : False
Central support mode                    : disabled
```

Example of a switch when Aruba Central is disabled:

```
switch# show aruba-central
Central admin state                     : disabled
```

```
Central location                        : N/A
VRF for connection                      : N/A
Shared secret                           : N/A
Central connection status               : N/A
Central source                          : none
Central source connection status        : N/A
Central source last connected on        : N/A
Main location                           : N/A
Main VRF                                : N/A
Alternative location                    : N/A
Alternative VRF                         : N/A
Activate server URL                     : devices-v2.arubanetworks.com
System time synchronized from Activate : N/A
Source IP                               : N/A
Source IP Overridden                    : False
Central support mode                    : disabledswitch# show aruba-central
Central admin state                     : disabled
Central location                        : N/A
VRF for connection                      : N/A
Shared secret                           : N/A
Central connection status               : N/A
Central source                          : none
Central source connection status        : N/A
Central source last connected on        : N/A
Main location                           : N/A
Main VRF                                : N/A
Alternative location                    : N/A
Alternative VRF                         : N/A
Activate server URL                     : devices-v2.arubanetworks.com
System time synchronized from Activate : N/A
Source IP                               : N/A
Source IP Overridden                    : False
Central support mode                    : disabled
```

Example of a switch when Aruba Central is not reachable:

```
switch# show aruba-central
Central admin state                     : enabled
Central location                        : N/A
VRF for connection                      : N/A
Shared secret                           : N/A
Central connection status               : not-reachable
Central source                          : activate
Central source connection status        : connected
Central source last connected on        : Fri Jun 30 20:22:33 UTC 2023
Main location                           : N/A
Main VRF                                : N/A
Alternative location                    : N/A
Alternative VRF                         : N/A
Activate server URL                     : devices-v2.arubanetworks.com
System time synchronized from Activate : N/A
Source IP                               : N/A
Source IP Overridden                    : False
Central support mode                    : disabled
```

Command History

Release	Modification
10.12.1000	Enhanced to support more scenarios
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show running-config current-context

Applies only to the 4100i switch.

```
show running-config current-context
```

Description

Shows the running configuration for the current-context. If user is in the context of Aruba-Central (**config-aruba-central**), then Aruba Central running configuration is displayed.

Examples

Shows the running configuration of Aruba Central:

```
switch(config-aruba-central)# show running-config current-context
aruba-central
       disable
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
4100i	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Chapter 12
DNS

The Domain Name System (DNS) is the Internet protocol for mapping domain names or hostnames into addresses. DNS allows users to enter more readily memorable and intuitive hostnames, rather than IP addresses, to identify devices connected to a network. It also allows a host to keep the same hostname even if it changes its IP address.

Hostname resolution can be either static or dynamic.

- In static resolution, a local table is defined on the switch that associates hostnames with their IP addresses. Static tables can be used to speed up the resolution of frequently queried hosts.
- Dynamic resolution requires that the switch query a DNS server located elsewhere on the network. Dynamic name resolution takes more time than static name resolution, but requires far less configuration and management.

Configuration

VxLAN IPv4 and IPv6 underlay support the DNS client. To ensure configuration, make sure the following are set up:

- VxLAN overlay
    - Must be configured to establish the VxLAN tunnel.
- VxLAN tunnel
    - Ensure the tunnel is established between the VTEPS via either static or EVPN using the following command: **show interface VxLAN VTEPS**.

For more information, refer to the AOS-CX VxLAN EVPN Guide.

DNS client

The DNS client resolves hostnames to IP addresses for protocols that are running on the switch. When the DNS client receives a request to resolve a hostname, it can do so in one of two ways:

- Forward the request to a DNS name server for resolution.
- Reply to the request without using a DNS name server, by resolving the name using a statically defined table of hostnames and their associated IP addresses.

Configuring the DNS client

Procedure

1. Configure one or more DNS name servers with the command `ip dns server`.
2. To resolve DNS requests by appending a domain name to the requests, either configure a single domain name with the command `ip dns domain-name`, or configure a list of up to six domain names with the command `ip dns domain-list`.
3. To use static name resolution for certain hosts, associate an IP address to a host with the command `ip dns host`.
4. Review your DNS configuration settings with the command `show ip dns`.

Examples

This example creates the following configuration:

- Defines the domain **switch.com** to append to all requests.
- Defines a DNS server with IPv4 address of **1.1.1.1**.
- Defines a static DNS host named **myhost1** with an IPv4 address of **3.3.3.3**.
- DNS client traffic is sent on the default VRF (named **default**).

```
switch(config)# ip dns domain-name switch.com
switch(config)# ip dns server-address 1.1.1.1
switch(config)# ip dns host myhost1 3.3.3.3
switch(config)# exit
switch# show ip dns

VRF Name : default
Domain Name: switch.com
Name Server(s): 1.1.1.1

Host Name                                                         Address
--------------------------------------------------------------------------------
myhost1                                                           3.3.3.3
switch#
```

DNS client commands

ip dns domain-list

```
ip dns domain-list <DOMAIN-NAME> [vrf default]
no ip dns domain-list <DOMAIN-NAME> [vrf default]
```

Description

Configures one or more domain names that are appended to the DNS request. The DNS client appends each name in succession until the DNS server replies. Domains can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF.

The **no** form of this command removes a domain from the list.

Parameter	Description
`list <DOMAIN-NAME>`	Specifies a domain name. Up to six domains can be added to the list. Length: 1 to 256 characters.

Examples

This example defines a list with two entries: **domain1.com** and **domain2.com**.

```
switch(config)# ip dns domain-list domain1.com
switch(config)# ip dns domain-list domain2.com
```

This example removes the entry **domain1.com**.

```
switch(config)# no ip dns domain-list domain1.com
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

ip dns domain-name

```
ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]
no ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]
```

Description

Configures a domain name that is appended to the DNS request. The domain can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF. If a domain list is defined with the command `ip dns domain-list`, the domain name defined with this command is ignored.

The **no** form of this command removes the domain name.

Parameter	Description
<DOMAIN-NAME>	Specifies the domain name to append to DNS requests. Length: 1 to 256 characters.
`vrf` *<VRF-NAME>*	Specifies a VRF name. Default: default.

Examples

Setting the default domain name to `domain.com`:

```
switch(config)# ip dns domain-name domain.com
```

Removing the default domain name `domain.com`:

```
switch(config)# no ip dns domain-name domain.com
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

ip dns host

```
ip dns host <HOST-NAME> <IP-ADDR> [ vrf <VRF-NAME> ]
no ip dns host <HOST-NAME> <IP-ADDR> [ vrf <VRF-NAME> ]
```

Description

Associates a static IP address with a hostname. The DNS client returns this IP address instead of querying a DNS server for an IP address for the hostname. Up to six hosts can be defined. If no VRF is defined, the default VRF is used.

The **no** form of this command removes a static IP address associated with a hostname.

Parameter	Description
`host <HOST-NAME>`	Specifies the name of a host. Length: 1 to 256 characters.
`<IP-ADDR>`	Specifies an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255, or IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.
`vrf <VRF-NAME>`	Specifies a VRF name. Default: default.

Examples

This example defines an IPv4 address of **3.3.3.3** for **host1**.

```
switch(config)# ip dns host host1 3.3.3.3
```

This example defines an IPv6 address of **b::5** for **host 1**.

```
switch(config)# ip dns host host1 b::5
```

This example defines removes the entry for **host 1** with address **b::5**.

```
switch(config)# no ip dns host host1 b::5
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

ip dns server address

```
ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]
no ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]
```

Description

Configures the DNS name servers that the DNS client queries to resolve DNS queries. Up to six name servers can be defined. The DNS client queries the servers in the order that they are defined. If no VRF is defined, the default VRF is used.

The **no** form of this command removes a name server from the list.

Parameter	Description
<IP-ADDR>	Specifies an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255, or IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.
`vrf` *<VRF-NAME>*	Specifies a VRF name. Default: default.

Examples

This example defines a name server at **1.1.1.1**.

```
switch(config)# ip dns server-address 1.1.1.1
```

This example defines a name server at **a::1**.

```
switch(config)# ip dns server-address a::1
```

This example removes a name server at **a::1**.

```
switch(config)# no ip dns server-address a::1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

show ip dns

`show ip dns [vrf <VRF-NAME>]`

Description

Shows all DNS client configuration settings or the settings for a specific VRF.

Parameter	Description
`vrf <VRF-NAME>`	Specifies the VRF for which to show information. If no VRF is defined, the default VRF is used.

Examples

DNS client arbitration on the MGMT interface on a MGMT VRF can be updated via three different methods.

1. Using the **domain-name <name>** or **nameservers <servers>** commands in the command-line interface.
2. Using the **ip dns domain-name <*DOMAIN-NAME*> vrf *MGMT*** or **ip dns server-address <SERVER> vrf MGMT** commands in the command-line interface.
3. Using the **ip dhcp** command in the command-line interface (dynamic enties).

AOS-CX gives the following priority levels to the these three update mothods.

- Priority 1 - standalone CLI configuration
- Priority 2 - static ip dns configuration
- Priority 3 - Dynamic config

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Chapter 13

Device discovery and configuration

The switch supports automatic discovery and configuration of other devices on the network.

Device profiles

Device profiles rely on role configurations. For information on role configurations, see the *Security Guide*.

Device profiles are used to dynamically assign port attributes based on the type of devices connected, without having to create a RADIUS infrastructure. You can map device profiles to device groups. A device group contains various match criteria, which can be obtained from multiple sources, such as LLDP, CDP, and local MAC match. Device profiles contain port attributes to be assigned to the port when a connected device matches a device group.

Device profiles are supported on different scenarios. It can be applied on interfaces that are configured with security (802.1X or MAC authentication), or applied based on L2 port (LLDP, CDP), or applied on standalone ports with the block-until-profile-applied command enabled. All the methods are mutually exclusive of each other. The block-until-profile-applied mode must be configured only when there is a standalone port where no security has been configured and when you want the port to be offline until at least one client is onboarded based on the match and ignore criteria that you configure. Local MAC match is supported when you configure block-until-profile-applied command or device profile with security.

Up to eight device profiles can be configured.

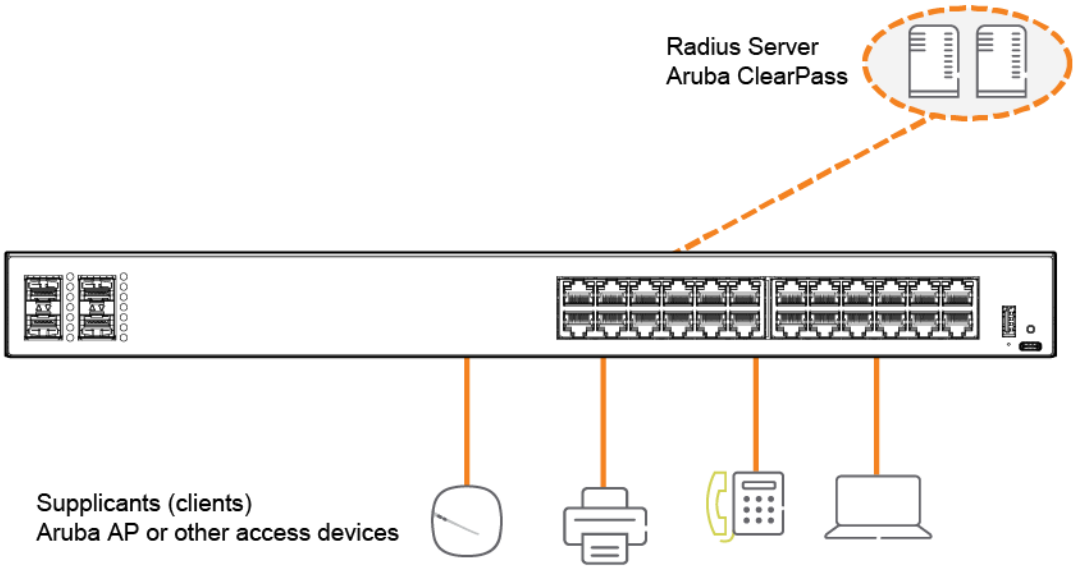
See the *Security Guide* for the following commands:

- The `port-access onboarding-method precedence` command—If you are configuring both security and device profile on the port, and you want to configure the order in which the methods will be executed.
- The `port-access fallback-role` command—If you want to configure a role that must be applied to devices when no other role exists or can be derived for that device.

If you configure a match criteria that matches across multiple device profiles, then the priority considered is LLDP, CDP, and then local MAC match. That is, LLDP precedes over CDP, which in turn precedes over local MAC match.

The following figure displays a simple configuration of device profile and AAA authentication with RADIUS server and Aruba ClearPass Policy Manager. Local MAC match feature is useful when you do not want to afford RADIUS infrastructure or when you want to use local authentication as a backup method in case the RADIUS server is unreachable.

Figure 1 *Example of device profile setup along with RADIUS infrastructure*



;" />

Configuring a device profile for LLDP

Procedure

1. Create an LLDP group with the command `port-access lldp-group`.
2. Define rules for adding devices to an LLDP group with the command `match`.
3. Define rules for ignoring devices so that they are not added to an LLDP group with the command `ignore`.
4. Create a device profile with the command `port-access device-profile`.
5. Add the LLDP group with the command `associate lldp-group`.
6. Add a role to a device profile with the command `associate role`. Make sure that the role is already created. For information on how to create a role, see port access role information in the *Security Guide*.
7. Enable the device profile with the command `enable`.

Configuring a device profile for CDP

1. Create a CDP group with the command `port-access cdp-group`.
2. Define rules for adding devices to a CDP group with the command `match`.
3. Define rules for ignoring devices so that they are not added to a CDP group with the command `ignore`.
4. Create a device profile with the command `port-access device-profile`.
5. Add a CDP group to a device profile with the command `associate cdp-group`.
6. Add a role to a device profile with the command `associate role`. Make sure that the role is already created. For information on how to create a role, see port access role information in the *Security Guide*.
7. Enable a device profile with the command `enable`.

Configuring a device profile for local MAC match

Procedure

1. Create a MAC group with the `mac-group` command.
2. Define rules for adding devices to a MAC group with the `match (for MAC groups)` command.
3. Define rules for ignoring devices so that they are not added to a MAC group with the `ignore (for MAC groups)` command.
4. Create a device profile with the `port-access device-profile` command.
5. Associate a MAC group with a device profile with the `associate mac-group` command.
6. Add a role to a device profile with the `associate role` command. Make sure that the role is already created. For information on how to create a role, see port access role information in the *Security Guide*.
7. Enable a device profile with the `enable` command.

Device Profile Usage Considerations

The following behaviors and limitations should be taken into consideration when configuring a device profile on a Link Aggregation Group (LAG).

Device Profiles on a LAG

- When using Device profile to onboard a device with multiple interfaces (such as a dual port AP) over a LAG, set the authentication mode in the role applied to the device to **devicemode**. The **multi-domain** authentication mode is not supported with a device profile on a LAG.
- It is not recommended to connect different devices to the same LAG when using a device profile on the LAG.
- When updating a LAG membership, all the device profile clients on the LAG are logged off automatically from the switch.
- Device profile is not supported on a multi-chassis LAG.
- If you issue the command **aaa authentication port-access allow-cdp-proxy-logoff** on the LAG, a proxy-logoff TLV received on any member interface of the LAG will log off all clients onboarded on the LAG.

Security Considerations

- Each individual interface of a LAG must successfully match a Device profile for the corresponding interface to be unblocked on the data plane.
- An onboarded client on the LAG is authorized to send and receive packets on any of the member interface of the LAG that is unblocked on the data plane. This is true even for a different client connected to the same LAG.
- When using LLDP packets to authenticate a device using a device profile on a LAG, issue the command **aaa authentication port-access allow-lldp-auth mac chassis-mac** to set the authentication MAC address as the chassis MAC address
- LLDP BPDUs are blocked by default. Issue the command **aaa authentication port-access allow-lldp-bpdu** to allow the LLDP BPDUs.
- If a Device profile is used to onboard multiple devices connected to the same LAG, the client limit on the LAG must be configured to match the number of clients onboarding the LAG.
- When using LLDP packets to authenticate a Dual Port AP using a device profile on a LAG, issue the command **aaa authentication port-access allow-lldp-auth mac source-mac** to set the authentication MAC address as the port/interface MAC address.

Supported Role Attributes

The following port-access role attributes are supported for clients that onboard via device profile on a LAG.

- Authentication mode
- Client inactivity timeout
- Session timeout
- Access and Trunk VLANs
- Port-Access policy
- MTU
- PoE Priority
- PoE Allocate-By
- Private-VLAN
- STP admin edge port
- Trust mode

Device profile commands

aaa authentication port-access allow-cdp-auth

```
aaa authentication port-access allow-cdp-auth
no aaa authentication port-access allow-cdp-auth
```

Description

Use this command to allow or block authentication on the CDP (Cisco Discovery Protocol) BPDU (Bridge Protocol Data Unit) . This is allowed by default. The **no** form of this command prevents authentication on CDP packets received on the port. This command can be issued from the interface (**config-if**) or Link Aggregation Group (**config-lag-if**) contexts.

Examples

Allowing authentication on a CDP CPDU on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# aaa authentication port-access allow-cdp-auth
```

Allowing authentication on a CDP CPDU on a LAG port:

```
switch(config)# interface lag 1
switch(config-lag-if)# aaa authentication port-access allow-cdp-auth
```

Command History

Release	Modification
10.13	This command can be issued from a Link Aggregation Group (LAG) context.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config-if` `config-lag-if`	Administrators or local user group members with execution rights for this command.

aaa authentication port-access allow-cdp-bpdu

```
aaa authentication port-access allow-cdp-bpdu
no aaa authentication port-access allow-cdp-bpdu
```

Description

Allows all packets related to the CDP (Cisco Discovery Protocol) BPDU (Bridge Protocol Data Unit) on a secure port or LAG. This command can be issued from the interface (**config-if**) or Link Aggregation Group (**config-lag-if**) contexts. The **no** form of this command blocks the CDP BPDU on a secure port. On a nonsecure port, the command has no effect.

Examples

Allowing a CDP BPDU on secure port **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# aaa authentication port-access allow-cdp-bpdu
switch(config-if)# do show running-config
Current configuration:
!
!Version AOS-CX 10.0X.0000
led locator on
!
!
vlan 1
aaa authentication port-access mac-auth
    enable
aaa authentication port-access dot1x authenticator
    enable
interface 1/1/1
    no shutdown
    vlan access 1
    aaa authentication port-access allow-cdp-bpdu
    aaa authentication port-access mac-auth
        enable
    aaa authentication port-access dot1x authenticator
    enable

switch(config-if)# do show port-access device-profile interface all
Port 1/1/1, Neighbor-Mac 00:0c:29:9e:d1:20
    Profile Name    : access_switches
    LLDP Group      :
    CDP Group       : aruba-ap_cdp
    Role            : test_ap_role
    Status          : In Progress
    Failure Reason  :
```

Blocking LLDP packet on secure port **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no aaa authentication port-access allow-cdp-bpdu
```

```
switch(config-if)# do show running-config
Current configuration:
!
!Version AOS-CX 10.0X.0000
led locator on
!
!
vlan 1
aaa authentication port-access mac-auth
    enable
interface 1/1/1
    no shutdown
    vlan access 1
    aaa authentication port-access mac-auth
        enable
```

Allowing a CDP BPDU on LAG **1**:

```
switch(config)# interface lag 1
switch(config-lag-if)# aaa authentication port-access allow-cdp-bpdu
```

Command History

Release	Modification
10.13	This command can be issued from a Link Aggregation Group (LAG) context.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config-if` `config-lag-if`	Administrators or local user group members with execution rights for this command.

aaa authentication port-access allow-cdp-proxy-logoff

```
aaa authentication port-access allow-cdp-proxy-logoff
no aaa authentication port-access allow-cdp-proxy-logoff
```

Description

Allows a client to be logged off from the system via a special TLV in the CDP packet. By default, proxy logoff via CDP packet support is disabled. When **allow-cdp-proxy-logoff** is enabled, TLV received from CDP packets corresponding to logoff processing will be read and logoff is issued to the clients. This only works on client authentication enabled ports and **aaa authentication port-access allow-cdp-bpdu** must be enabled to process. This command can be issued from the interface (**config-if**) or Link Aggregation Group (**config-lag-if**) contexts.

Examples

Allowing a client to be logged off from the system via a special TLV in the CDP packet:

```
switch(config)# interface 1/1/1
switch(config-if)# aaa authentication port-access allow-cdp-proxy-logoff
switch(config-if)# show running-config interface 1/1/1
interface 1/1/1
    no shutdown

    vlan access 1
    aaa authentication port-access allow-cdp-bpdu
    aaa authentication port-access allow-cdp-proxy-logoff
    aaa authentication port-access allow client-limit 2
    aaa authentication port-access dot1x authenticator
        enable
    aaa authentication port-accss mac-auth
        enable
    exit
```

The aaa authentication port-access allow-cdp-proxy-logoff command can also be issued from a LAG port context

```
switch(config)# interface lag 1
switch(config-lag if)# aaa authentication port-access allow-cdp-proxy-logoff
```

Command History

Release	Modification
10.13	This command can be issued from a Link Aggregation Group (LAG) context.
10.09.1000	Command introduced.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config-if` `config-lag-if`	Administrators or local user group members with execution rights for this command.

aaa authentication port-access allow-lldp-bpdu

```
aaa authentication port-access allow-lldp-bpdu
no aaa authentication port-access allow-lldp-bpdu
```

Description

Allows all packets related to the LLDP BPDU (Bridge Protocol Data Unit) on a secure port. This command can be issued from the interface (**config-if**) or Link Aggregation Group (**config-lag-if**) contexts.

The **no** form of this command blocks the LLDP BPDU on a secure port. On a nonsecure port, the command has no effect.

Examples

Allowing an LLDP BPDU on secure port **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# aaa authentication port-access allow-lldp-bpdu
switch(config-if)# do show running-config
Current configuration:
!
!Version AOS-CX 10.0X.0000
led locator on
!
!
vlan 1
aaa authentication port-access mac-auth
    enable
interface 1/1/1
    no shutdown

    vlan access 1
    aaa authentication port-access allow-lldp-bpdu
    aaa authentication port-access mac-auth
        enable

switch(config-if)# do show port-access device-profile interface all
Port 1/1/1, Neighbor-Mac 00:0c:29:9e:d1:20
    Profile Name    : access_switches
    LLDP Group      : 2920-grp
    CDP Group       :
    Role            : local_2920_role
    Status          : Profile Applied
    Failure Reason  :
```

Blocking LLDP BPDU on secure port **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no aaa authentication port-access allow-lldp-bpdu
switch(config-if)# do show running-config
Current configuration:
!
!Version AOS-CX 10.0X.0000led locator on
!
!
vlan 1
aaa authentication port-access mac-auth
    enable
interface 1/1/1
    no shutdown

    vlan access 1
    aaa authentication port-access mac-auth
    enable
```

Allowing an LLDP BPDU on a LAG port:

```
switch(config)# interface lag 1
switch(config-lag-if)#aaa authentication port-access allow-lldp-bpdu
```

Command History

Release	Modification
10.13	This command can be issued from a Link Aggregation Group (LAG) context.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config-if` `config-lag-if`	Administrators or local user group members with execution rights for this command.

associate cdp-group

```
associate cdp-group <GROUP-NAME>
no associate cdp-group <GROUP-NAME>
```

Description

Associates a CDP (Cisco Discovery Protocol) group with a device profile. A maximum of two CDP groups can be associated with a device profile.

The **no** form of this command removes a CDP group from a device profile.

Parameter	Description
<GROUP-NAME>	Specifies the name of the CDP group to associate with this device profile. Range: 1 to 32 alphanumeric characters.

Examples

Associating the CDP group **my-cdp-group** with the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# associate cdp-group my-cdp-group
```

Removing the CDP group **my-cdp-group** from the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no associate cdp-group my-cdp-group
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config-device-profile`	Administrators or local user group members with execution rights for this command.

associate lldp-group

```
associate lldp-group <GROUP-NAME>
no associate lldp-group <GROUP-NAME>
```

Description

Associates an LLDP group with a device profile. A maximum of two LLDP groups can be associated with a device profile

The **no** form of this command removes an LLDP group from a device profile.

Parameter	Description
<GROUP-NAME>	Specifies the name of the LLDP group to associate with the device profile. Range: 1 to 32 alphanumeric characters.

Examples

Associating the LLDP group **my-lldp-group** with the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# associate lldp-group my-lldp-group
```

Removing the LLDP group **my-lldp-group** from the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no associate lldp-group my-lldp-group
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config-device-profile`	Administrators or local user group members with execution rights for this command.

associate mac-group

```
associate mac-group <GROUP-NAME>
no associate mac-group <GROUP-NAME>
```

Description

Associates a MAC group with a device profile. A maximum of two MAC groups can be associated with a device profile.

The **no** form of this command removes a MAC group from a device profile.

Parameter	Description
<GROUP-NAME>	Specifies the name of the MAC group to associate with this device profile. Range: 1 to 32 alphanumeric characters.

Examples

Associating the MAC group **mac01-group** with the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# associate mac-group mac01-group
```

Removing the MAC group **mac01-group** from the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no associate mac-group mac01-group
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config-device-profile`	Administrators or local user group members with execution rights for this command.

associate role

```
associate role <ROLE-NAME>
no associate role <ROLE-NAME>
```

Description

Associates a role with a device profile. Only one role can be associated with a device profile. For information on how to configure a role, see the port access role information in the *Security Guide*.

The **no** form of this command removes a role from a device profile.

Parameter	Description
<ROLE-NAME>	Specifies the name of the role to associate with the device profile. Range: 1 to 64 alphanumeric characters.

Examples

Associating the role **my-role** with the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# associate role my-role
```

Removing the role **my-role** from the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no associate role my-role
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config-device-profile`	Administrators or local user group members with execution rights for this command.

disable

```
disable
no disable
```

Description

Disables a device profile.

The **no** form of this command enables a device profile.

Examples

Disabling a device profile:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# disable
```

Enabling a device profile named profile01:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config-device-profile`	Administrators or local user group members with execution rights for this command.

enable

```
enable
no enable
```

Description

Enables a device profile.

The **no** form of this command disables a device profile.

Examples

Enabling a device profile:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# enable
```

Disabling a device profile named profile01:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no enable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config-device-profile`	Administrators or local user group members with execution rights for this command.

ignore (for CDP groups)

```
ignore [seq <SEQ-NUM>] {platform <PLATFORM> | sw-version <SWVERSION> |
     voice-vlan-query <VLAN-ID>}
no ignore [seq <SEQ-ID>] {platform <PLATFORM> | sw-version <SWVERSION> |
     voice-vlan-query <VLAN-ID>}
```

Description

Defines a rule to ignore devices for a CDP (Cisco Discovery Protocol) group. Up to 64 match/ignore rules can be defined for a group.

The **no** form of this command removes a rule for ignoring devices from a CDP group.

Parameter	Description
`seq <SEQ-ID>`	Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.
`platform <PLATFORM>`	Specifies the hardware or model details of the neighbor. Range: 1 to 128 alphanumeric characters.
`sw-version <SWVERSION>`	Specifies the software version of the neighbor. Range: 1 to 128 alphanumeric characters.
`voice-vlan-query <VLAN-ID>`	Specifies the VLAN query value of the neighbor. Range: 1 to 65535.

Examples

Adding a rule to the CDP group **grp01** that ignores a device that transmits **PLATFORM01** in the platform TLV:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# ignore platform PLATFORM01
```

Adding a rule to the CDP group **grp01** that ignores a device that transmits **SWVERSION** in software version TLV:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# ignore sw-version SWVERSION
```

Removing the rule that matches the sequence number **25** from the CDP group named **grp01**.

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# no ignore seq 25
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config-cdp-group`	Administrators or local user group members with execution rights for this command.

ignore (for LLDP groups)

```
ignore [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |
     vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}
```

```
no ignore [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |
     vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}
```

Description

Defines a rule to ignore devices for an LLDP group. Up to 64 match/ignore rules can be defined for a group.

The **no** form of this command removes a rule for ignoring devices from an LLDP group.

Parameter	Description
`seq <SEQ-ID>`	Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.
`sys-desc <SYS-DESC>`	Specifies the LLDP system description type-length-value (TLV). Range: 1 to 256 alphanumeric characters.
sysname *<SYS-NAME>*	Specifies the LLDP system name TLV. Range: 1 to 64 alphanumeric characters.
`vendor-oui <VENDOR-OUI>`	Specifies the LLDP system vendor OUI TLV. Range: 1 to 6 alphanumeric characters.
`type <KEY>`	Specifies the vendor OUI subtype key. Optional.
`value <VALUE>`	Specifies the vendor OUI subtype value. Range: 1 to 256 alphanumeric characters.

Examples

Adding a rule to the LLDP group **grp01** that ignores a device that transmits **PLATFORM01** in the system description TLV:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# ignore sys-desc PLATFORM01
```

Removing the rule that matches the sequence number **25** from the LLDP group named **grp01**.

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# no match seq 25
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config-lldp-group`	Administrators or local user group members with execution rights for this command.

ignore (for MAC groups)

```
[seq <SEQ-ID>] ignore {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}
no [seq <SEQ-ID>] ignore {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}
```

Description

Defines a rule to ignore devices for a MAC group based on the criteria of MAC address, MAC address mask, or MAC Organizational Unique Identifier (OUI). Up to 64 ignore rules can be defined for a group.

The **no** form of this command removes a rule for ignoring devices from a MAC group.

Parameter	Description
`seq <SEQ-ID>`	Specifies the entry sequence ID of the rule to create or modify a MAC group. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence. Range: **1** to **4294967295**.
`mac <MAC-ADDR>`	Specifies the MAC address of the device to ignore.
`mac-mask <MAC-MASK>`	Specifies the MAC address mask to ignore devices in that range. Supported MAC address masks: **/32** and **/40**.
`mac-oui <MAC-OUI>`	Specifies the MAC OUI to ignore devices in that range. Supports MAC OUI address of maximum length of 24 bits.

Usage

To achieve the required configuration of matches for devices, it is recommended to first ignore the devices that you do not want to add. Then match the criteria for the rest of the devices that you want to add to the MAC group.

For example, if you want to ignore a specific device but add all the other devices that belong to a MAC OUI, then you must first configure the ignore criteria with a lower sequence number. And then configure match criteria with a higher sequence number.

Examples

Adding a rule to the MAC group **grp01** to ignore a device based on MAC address, but match all other devices belonging to a MAC OUI:

```
switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac 1a:2b:3c:4d:5e:6f
switch(config-mac-group)# match mac-oui 1a:2b:3c
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
```

````
!export-password: default
led locator on
!
!
!
!
ssh server vrf
!
!
!
!
!
vlan 1
interface vlan 1
    no shutdown
    ip dhcp
mac-group grp01
     seq 10 ignore mac 1a:2b:3c:4d:5e:6f
     seq 20 match mac-oui 1a:2b:3c

```
````

Adding a rule to the MAC group **grp01** to ignore devices based on MAC address mask, but match all other devices belonging to a MAC OUI:

````
switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac-mask 1a:2b:3c:4d/32
switch(config-mac-group)# match mac-oui 1a:2b:3c
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
!
!
ssh server vrf
!
!
!
!
!
vlan 1
interface vlan 1
    no shutdown
    ip dhcp
mac-group grp01
     seq 10 ignore mac-mask 1a:2b:3c:4d/32
     seq 20 match mac-oui 1a:2b:3c

```
````

Adding a rule to the MAC group **grp01** that ignores a device based on complete MAC address:

```
switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac 1a:2b:3c:4d:5e:6f
```

Adding a rule to the MAC group **grp02** that ignores devices based on MAC mask:

```
switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac-mask 1a:2b:3c:4d:5e/40
switch(config-mac-group)# ignore mac-mask 18:e3:ab:73/32
```

Adding a rule to the MAC group **grp03** that ignores devices based on MAC OUI:

```
switch(config)# mac-group grp03
switch(config-mac-group)# ignore mac-oui 81:cd:93
```

Adding a rule to the MAC group **grp01** that ignores devices with a sequence number and based on MAC address:

```
switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 ignore mac b2:c3:44:12:78:11
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
    no shutdown
    ip dhcp
mac-group grp01
    seq 10 ignore mac b2:c3:44:12:78:11

```
```

Removing the rule from the MAC group **grp01** based on sequence number:

```
switch(config)# mac-group grp01
switch(config-mac-group)# no ignore seq 10
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
    no shutdown
    ip dhcp
mac-group grp01

```
```

Adding a rule to the MAC group **grp01** that ignores devices with MAC entry sequence number and based on MAC OUI:

```
switch(config)# mac-group grp01
switch(config)#  mac-group grp01
switch(config-mac-group)# seq 10 ignore mac b2:c3:44:12:78:11
switch(config-mac-group)# seq 20 ignore mac-oui 1a:2b:3c
switch(config-mac-group)# seq 30 ignore mac-mask 71:14:89:42/32
switch(config-mac-group)# exit
switch(config)#
switch(config)# ^Z

switch#
switch#
switch# show running-config
Current configuration:
!
!Version AOS-CX PL.10.06.0002
!export-password: default
!
!
!
!
ssh server vrf default
vlan 1
spanning-tree
mac-group grp01
     seq 10 ignore mac b2:c3:44:12:78:11
     seq 20 ignore mac-oui 1a:2b:3c
     seq 30 ignore mac-mask 71:14:89:42/32
interface 1/1/1
    no shutdown
    vlan access 1
interface 1/1/2
    no shutdown
    vlan access 1
interface 1/1/3
    no shutdown
    vlan access 1
interface 1/1/4
    no shutdown
    vlan access 1
interface 1/1/5
    no shutdown
    vlan access 1
interface 1/1/6
    no shutdown
    vlan access 1
interface 1/1/7
    no shutdown
    vlan access 1
interface 1/1/8
    no shutdown
    vlan access 1
interface 1/1/9
    no shutdown
    vlan access 1
interface 1/1/10
    no shutdown
    vlan access 1
interface 1/1/11
    no shutdown
    vlan access 1
interface 1/1/12
    no shutdown
```

```
    vlan access 1
interface 1/1/13
    no shutdown
    vlan access 1
interface 1/1/14
    no shutdown
    vlan access 1
interface 1/1/15
    no shutdown
    vlan access 1
interface 1/1/16
    no shutdown
    vlan access 1
interface vlan 1
    ip dhcp
!
!
!
!
!
https-server vrf default
```

Removing the rule from the MAC group **grp01** based on sequence number and MAC OUI:

```
switch(config)# mac-group grp01
switch(config-mac-group)# no seq 20 ignore mac-oui 1a:2b:3c
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
    no shutdown
    ip dhcp
mac-group grp01
     seq 10 ignore mac b2:c3:44:12:78:11
     seq 30 ignore mac-mask 71:14:89:f3/32


```
```

Removing the rule that matches the sequence number **25** from the MAC group named **grp01**.

```
switch(config)# mac-group grp01
switch(config-mac-group)# no ignore seq 25
```

Command History

Release	Modification
10.07 or earlier	--

Command Information
```

Platforms	Command context	Authority
4100i 6000 6100	`config-mac-group`	Administrators or local user group members with execution rights for this command.

mac-group

```
mac-group <MAC-GROUP-NAME>
no mac-group <MAC-GROUP-NAME>
```

Description

Creates a MAC group or modifies an existing MAC group. A MAC group is used to classify connected devices based on the MAC address details, such as mask or OUI.

A maximum of 32 MAC groups can be configured on the switch. A maximum of 2 MAC groups can be associated with a device profile. Each group accepts 64 match or ignore commands.

The **no** form of this command removes a MAC group.

Parameter	Description
<MAC-GROUP-NAME>	Specifies the name of the MAC group to create or modify. The maximum number of characters supported is 32.

Examples

Creating a MAC group named **grp01**:

```
switch(config)# mac-group grp01
switch(config-mac-group)# exit
```

Removing a MAC group named **grp01**:

```
switch(config)# no mac-group grp01
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config`	Administrators or local user group members with execution rights for this command.

match (for CDP groups)

```
match [seq <SEQ-ID>] {platform <PLATFORM> | sw-version <SWVERSION> |
    voice-vlan-query <VLAN-ID>}
```

```
no match [seq <SEQ-ID>] {platform <PLATFORM> | sw-version <SWVERSION> |
     voice-vlan-query <VLAN-ID>}
```

Description

Defines a rule to match devices for a CDP group. A maximum of 32 CDP groups can be configured on the switch. Up to 64 match or ignore rules can be defined for each group.

The **no** form of this command removes a rule for adding devices to a CDP group.

Parameter	Description
`seq <SEQ-ID>`	Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.
`platform <PLATFORM>`	Specifies the hardware or model details of the neighbor. Range: 1 to 128 alphanumeric characters.
`sw-version <SWVERSION>`	Specifies the software version of the neighbor. Range: 1 to 128 alphanumeric characters.
`voice-vlan-query <VLAN-ID>`	Specifies the VLAN query value of the neighbor. Range: 1 to 65535.

Examples

Adding rules to match a Cisco device with a specific software version on VLAN **512** to the CDP group **grp01**:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# match platform CISCO
switch(config-cdp-group)# match sw-version 11.2(12)P
switch(config-cdp-group)# match voice-vlan-query 512
switch(config-cdp-group)# match seq 50 platform cisco sw-version 11.2(12)P voice-
vlan-query 512
switch(config-cdp-group)# exit
switch(config)# do show running-config

Current configuration:
!
!Version AOS-CX Virtual.10.0X.000
!export-password: default
led locator on
!
!
vlan 1
port-access cdp-group grp01
     seq 10 match platform CISCO
     seq 20 match sw-version 11.2(12)P
     seq 30 match voice-vlan-query 512
     seq 50 match platform cisco sw-version 11.2(12)P voice-vlan-query 512
```

Removing a rule that matches the sequence number **25** from the CDP group named **grp01**:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# no match seq 25
```

Adding a rule that matches the value of vendor-OUI **000b86** to the CDP group named **grp01**:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# match vendor-oui 000b86
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config-cdp-group`	Administrators or local user group members with execution rights for this command.

match (for LLDP groups)

```
match [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |
     vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}
no match [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |
     vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}
```

Description

Defines a rule to match devices for an LLDP group. Up to 64 match/ignore rules can be defined for a group.

The **no** form of this command removes a rule.

Parameter	Description
`seq <SEQ-ID>`	Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.
`sys-desc <SYS-DESC>`	Specifies the LLDP system description type-length-value (TLV). Range: 1 to 256 alphanumeric characters.
`sysname <SYS-NAME>`	Specifies the LLDP system name TLV. Range: 1 to 64 alphanumeric characters.
`vendor-oui <VENDOR-OUI>`	Specifies the LLDP system vendor OUI TLV. Range: 1 to 6 alphanumeric characters.
`type <KEY>`	Specifies the vendor OUI subtype key.
`value <VALUE>`	Specifies the vendor OUI subtype value. Range: 1 to 256 alphanumeric characters.

Examples

Adding rules that match the LLDP system description **ArubaSwitch** and system name **Aruba** to the LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# match sys-desc ArubaSwitch
switch(config-lldp-group)# match sysname Aruba
switch(config)# do show running-config

Current configuration:
!
!Version AOS-CX Virtual.10.0X.000
!export-password: default
led locator on
!
!
vlan 1
port-access lldp-group grp01
     seq 10 match sys-desc ArubaSwitch
     seq 20 match sysname Aruba
```

Removing a rule that matches the sequence number **25** from an LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# no match seq 25
```

Adding a rule that matches the value of vendor-OUI **000b86** with type of **1** to the LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# match vendor-oui 000b86 type 1
```

Adding a rule that matches the value of vendor-OUI **000c34** to the LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# match vendor-oui 000c34
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config-lldp-group`	Administrators or local user group members with execution rights for this command.

match (for MAC groups)

```
[seq <SEQ-ID>] match {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}
no [seq <SEQ-ID>] match {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}
```

Description

Defines a rule to match devices for a MAC group based on the criteria of MAC address, MAC address mask, or MAC Organizational Unique Identifier (OUI). Up to 64 match rules can be defined for a group.

You must not configure the following special MAC addresses:

- Null MAC—For example, 00:00:00:00:00:00 or 00:00:00/32
- Multicast MAC
- Broadcast MAC—For example, ff:ff:ff:ff:ff:ff
- System MAC

Although the switch accepts these addresses, it will not process these addresses for the local MAC match feature.

The **no** form of this command removes a rule for adding devices to a MAC group.

The number of clients that can onboard based on the match criteria is configured in the **aaa authentication port-access client-limit** command. For information about this command, see the *Security Guide* for your switch.

Parameter	Description
`seq <SEQ-ID>`	Specifies the entry sequence ID of the rule to create or modify a MAC group. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence. Range: **1** to **4294967295**.
`mac <MAC-ADDR>`	Specifies the MAC address of the device.
`mac-mask <MAC-MASK>`	Specifies the MAC address mask to add devices in that range. Supported MAC address masks: **/32** and **/40**.
`mac-oui <MAC-OUI>`	Specifies the MAC OUI to add devices in that range. Supports MAC OUI address of maximum length of 24 bits.

Examples

Adding a device to the MAC group **grp01** based on complete MAC address:

```
switch(config)# mac-group grp01
switch(config-mac-group)# match mac 1a:2b:3c:4d:5e:6f
switch(config-mac-group)# exit
```

Adding devices to the MAC group **grp02** based on MAC mask:

```
switch(config)# mac-group grp01
switch(config-mac-group)# match mac-mask 1a:2b:3c:4d:5e/40
switch(config-mac-group)# match mac-mask 18:e3:ab:73/32
switch(config-mac-group)# exit
```

Adding devices to the MAC group **grp03** based on MAC OUI:

````
switch(config)# mac-group grp03
switch(config-mac-group)# match mac-oui 81:cd:93
switch(config-mac-group)# exit
````

Adding devices to the MAC group **grp01** with MAC entry sequence number and based on MAC address:

````
switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 match mac b2:c3:44:12:78:11
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
    no shutdown
    ip dhcp
mac-group grp01
    seq 10 match mac b2:c3:44:12:78:11

```
````

Removing devices from the MAC group **grp01** based on sequence number:

````
switch(config)# mac-group grp01
switch(config-mac-group)# no match seq 10
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
    no shutdown
    ip dhcp
mac-group grp01

```
````

Adding devices to the MAC group **grp01** with MAC entry sequence number and based on MAC address, MAC address mask, and MAC OUI:

````
switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 match mac b2:c3:44:12:78:11
switch(config-mac-group)# seq 20 match mac-oui 1a:2b:3c
switch(config-mac-group)# seq 30 match mac-mask 71:14:89:f3/32
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
````

```
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
    no shutdown
    ip dhcp
mac-group grp01
     seq 10 match mac b2:c3:44:12:78:11
     seq 20 match mac-oui 1a:2b:3c
     seq 30 match mac-mask 71:14:89:f3/32

'''
```

Removing devices from the MAC group **grp01** based on MAC OUI:

```
switch(config)# mac-group grp01
switch(config-mac-group)# no seq 20 match mac-oui 1a:2b:3c
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
    no shutdown
    ip dhcp
mac-group grp01
     seq 10 match mac b2:c3:44:12:78:11
     seq 30 match mac-mask 71:14:89:f3/32


'''
```

Adding devices to the MAC group **grp03** with MAC entry sequence number and based on MAC address mask:

```
switch(config)# mac-group grp03
switch(config-mac-group)# seq 10 match mac-mask 10:14:a3:b7:55/40
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
    no shutdown
    ip dhcp
```

````
mac-group grp03
     seq 10 match mac-mask 10:14:a3:b7:55/40

```
````

Removing devices from the MAC group **grp03** based on MAC address mask:

````
switch(config)# mac-group grp03
switch(config-mac-group)# no seq 10 match mac-mask 10:14:a3:b7:55/40
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
    no shutdown
    ip dhcp
mac-group grp03

```
````

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config-mac-group`	Administrators or local user group members with execution rights for this command.

port-access cdp-group

```
port-access cdp-group <CDP-GROUP-NAME>
no port-access cdp-group <CDP-GROUP-NAME>
```

Description

Creates a CDP (Cisco Discovery Protocol) group or modifies an existing CDP group. A CDP Group is used to classify connected devices based on the CDP packet details advertised by the device. A maximum of 32 CDP groups can be configured on the switch. Each group accepts 64 match/ignore commands.

The **no** form of this command removes a CDP group.

Parameter	Description
<CDP-GROUP-NAME>	Specifies the name of the CDP group to create or modify. The maximum number of characters supported is 32. Required.

Examples

Creating a CDP group named grp01:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# match platform CISCO
switch(config-cdp-group)# match sw-version 11.2(12)P
switch(config-cdp-group)# match voice-vlan-query 512
switch(config-cdp-group)# seq 50 match platform cisco sw-version 11.2(12)P voice-
vlan-query 512
switch(config-cdp-group)# exit
switch(config)# do show running-config

Current configuration:
!
!Version AOS-CX Virtual.10.0X.000
!export-password: default
led locator on
!
!
vlan 1
port-access cdp-group grp01
     seq 10 match platform CISCO
     seq 20 match sw-version 11.2(12)P
     seq 30 match voice-vlan-query 512
     seq 50 match platform cisco sw-version 11.2(12)P voice-vlan-query 512
```

Removing a CDP group named grp01:

```
switch(config)# no port-access cdp-group grp01
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config`	Administrators or local user group members with execution rights for this command.

port-access device-profile

`port-access device-profile <DEVICE-PROFILE-NAME>`
`no port-access device-profile <DEVICE-PROFILE-NAME>`

Description

Creates a new device profile and switches to the **config-device-profile** context. A maximum of 32 device profiles can be created. This command can be issued from the interface (**config-if**) or Link Aggregation Group (**config-lag-if**) contexts.

The **no** form of this command removes a device profile.

Parameter	Description
<DEVICE-PROFILE-NAME>	Specifies the name of a device profile. Range: 1 to 32 alphanumeric characters.

Examples

Creating a device profile named **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)#
```

Removing a device profile named **profile01**:

```
switch(config)# no port-access device-profile profile01
```

Creating a device profile named **profile02** on a LAG port:

```
switch(config)#interface lag 1
switch(config-lag-if)# port-access device-profile profile01
switch(config-device-profile)#
```

Command History

Release	Modification
10.13	This command can be issued from a Link Aggregation Group (LAG) context.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority

4100i
6000
6100
	`config`
	`config-if`
	`config-lag-if`
		Administrators or local user group members with execution rights for this command.

port-access device-profile mode block-until-profile-applied

You must configure this mode in device profile only on standalone ports where there is no security configured and when you not want the port to be offline until one client is onboarded.

```
port-access device-profile mode block-until-profile-applied
no port-access device-profile mode block-until-profile-applied
```

Description

Configures the switch to block the port until a profile match occurs for a device. This configuration is required when no security feature is enabled on the port.

You must enable this mode or security on the port for local MAC match feature to operate. You must not enable both features on the same port at the same time.

You must not combine any other AAA configurations with the block-until-profile-applied mode.

This command can be issued from the interface (**config-if**) or Link Aggregation Group (**config-lag-if**) contexts. The **no** form of this command removes a rule for adding devices to a MAC group.

Example

Configuring block-until-profile applied mode on port 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# port-access device-profile
switch(config-if-deviceprofile)# mode block-until-profile-applied
switch(config-if-deviceprofile)# end
```

Command History

Release	Modification
10.13	This command can be issued from a Link Aggregation Group (LAG) context.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority

4100i
6000
6100
	`config-if`
	`config-if-deviceprofile`
	`config-lag-if`
		Administrators or local user group members with execution rights for this command.

port-access lldp-group

```
port-access lldp-group <LLDP-GROUP-NAME>
no port-access lldp-group <LLDP-GROUP-NAME>
```

Description

Creates an LLDP group or modifies an existing LLDP group. An LLDP group is used to classify connected devices based on the LLDP type-length-values (TLVs) advertised by the device. A maximum of 32 LLDP groups can be configured on the switch. Each group accepts 64 match/ignore commands.

The **no** form of this command removes an LLDP group.

Parameter	Description
<LLDP-GROUP-NAME>	Specifies the name of the LLDP group to create or modify. The maximum number of characters supported is 32. Required.

Examples

Creating an LLDP group named grp01:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)#
```

Removing an LLDP group named grp01:

```
switch(config)# no port-access lldp-group grp01
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config`	Administrators or local user group members with execution rights for this command.

show port-access device-profile

```
show port-access device-profile [[interface {all | <INTERFACE-ID>}
     [client-status <MAC-ADDR>]] | name <DEVICE-PROFILE-NAME>]
```

Description

Shows the client status for a specific MAC address or profile name.

Parameter	Description
`interface {all	<INTERFACE-ID>}`	Select **all** for all interfaces or specify the name of an interface in the format: **member/slot/port**.
`client-status <MAC-ADDR>`	Specifies a MAC address (**xx:xx:xx:xx:xx:xx**), where **x** is a hexadecimal number from 0 to F.
`name <DEVICE-PROFILE-NAME>`	Specifies the name of the device profile.

Examples

Showing the applied state of the device profiles:

```
switch# show port-access device-profile

    Profile Name     : accesspoints
    LLDP Groups      : 2920-grp
    CDP Groups       :
    MAC Groups       : 2920-mac-grp1,2920-iot-grp2
    Role             : local_role_1
    State            : Enabled

    Profile Name     : access_switches
```

```
    LLDP Groups     : 2920-grp
    CDP Groups      :
    MAC Groups      :
    Role            : local_2920_role
    State           : Enabled

    Profile Name    : iot_devices
    LLDP Groups     :
    CDP Groups      :
    MAC Groups      : iot_camera-grp1,iot_sensors-grp1
    Role            : local_2920_role
    State           : Enabled

    Profile Name    : lobbyaps
    LLDP Groups     :
    CDP Groups      : lobby_ap_cdp_grp
    MAC Groups      :
    Role            : test_ap_role
    State           : Disabled
```

Showing the applied state of the device profile on interface 1/1/3:

```
switch# show port-access device-profile interface 1/1/3 client-status
00:0c:29:9e:d1:20

Port 1/1/3, Neighbor-Mac  00:0c:29:9e:d1:20
    Profile Name    : lobbyaps
    LLDP Group      :
    CDP Group       : aruba-ap_cdp
    MAC Group       :
    Role            : test_ap_role
    Status          : Failed
    Failure Reason  : Failed to apply MAC based VLAN
```

Showing the applied state of a specific device profile:

```
switch# show port-access device-profile name lldp-group

    Profile Name            : lldp-group
    LLDP Groups             :
    CDP Groups              :
    MAC Groups              : pc-behind-phone, lldp
    Role                    : auth_role
    State                   : Enabled
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i	Manager (#)	Administrators or local user group members with execution

Platforms	Command context	Authority
6000 6100		rights for this command.

LLDP

The IEEE 802.1AB Link Layer Discovery Protocol (LLDP) provides a standards-based method for network devices to discover each other and exchange information about their capabilities. An LLDP device advertises itself to adjacent (neighbor) devices by transmitting LLDP data packets on all interfaces on which outbound LLDP is enabled, and reading LLDP advertisements from neighbor devices on ports on which inbound LLDP is enabled. Inbound packets from neighbor devices are stored in a special LLDP MIB (management information base). This information can then be queried by other devices through SNMP.

LLDP information is used by network management tools to create accurate physical network topologies by determining which devices are neighbors and through which interfaces they connect. LLDP operates at layer 2 and requires an LLDP agent to be active on each interface that sends and receives LLDP advertisements. LLDP advertisements can contain a variable number of TLV (type, length, value) information elements. Each TLV describes a single attribute of a device such as: system capabilities, management IP address, device ID, port ID.

Packet boundaries

When multiple LLDP devices are directly connected, an outbound LLDP packet travels only to the next LLDP device. An LLDP-capable device does not forward LLDP packets to any other devices, regardless of whether they are LLDP-enabled.

An intervening hub or repeater forwards the LLDP packets it receives in the same manner as any other multicast packets it receives. Therefore, two LLDP switches joined by a hub or repeater handle LLDP traffic in the same way that they would if directly connected.

Any intervening 802.1D device or Layer-3 device that is either LLDP-unaware or has disabled LLDP operation drops the packet.

LLDP-MED

LLDP-MED (ANSI/TIA-1057/D6) extends the LLDP (IEEE 802.1AB) industry standard to support advanced features on the network edge for Voice Over IP (VoIP) endpoint devices with specialized capabilities and LLDP-MED standards-based functionality. LLDP-MED in the switches uses the standard LLDP commands described earlier in this section, with some extensions, and also introduces new commands unique to LLDP-MED operation. The show commands described elsewhere in this section are applicable to both LLDP and LLDP-MED operation. LLDP-MED enables:

- Configure Voice VLAN and advertise it to connected MED endpoint devices.
- Power over Ethernet (PoE) status and troubleshooting support via SNMP.

LLDP agent

When you enable LLDP on the switch, it is automatically enabled on all data plane interfaces. You can customize this behavior by manually enabling/disabling support on each interface.

Supported standards

The LLDP agent supports the following standards: IEEE 802.1AB-2005, Station, and Media Access Control Connectivity Discovery.

Supported interfaces

LLDP is supported on interfaces mapped to a physical port. It is not supported on logical interfaces, such as loopback, tunnels, and SVIs.

Operating modes

When LLDP is enabled, the switch periodically transmits an LLDP advertisement (packet) out each active port enabled for outbound LLDP transmissions and receives LLDP advertisements on each active port enabled to receive LLDP traffic.

The LLDP agent can operate in one of the following modes:

- Transmit and receive (TxRx): This is the default setting on all ports. It enables a given port to both transmit and receive LLDP packets and to store the data from received (inbound) LLDP packets in the switch's MIB.
- Transmit only (Tx): Enables a port to transmit LLDP packets that can be read by LLDP neighbors. However, the port drops inbound LLDP packets from LLDP neighbors without reading them. This prevents the switch from learning about LLDP neighbors on that port.
- Receive only (Rx): Enables a port to receive and read LLDP packets from LLDP neighbors and to store the packet data in the switch's MIB. However, the port does not transmit outbound LLDP packets. This prevents LLDP neighbors from learning about the switch through that port.
- Disabled: Disables LLDP packet transmissions and reception on a port. In this state, the switch does not use the port for either learning about LLDP neighbors or informing LLDP neighbors of its presence.

An LLDP agent operating in TxRx mode or Tx mode sends LLDP frames to its directly connected devices both periodically and when the local configuration changes.

Sending LLDP frames

Each time the LLDP operating mode of an LLDP agent changes, its LLDP protocol state machine reinitializes. A configurable reinitialization delay prevents frequent initializations caused by frequent changes to the operating mode. If you configure the reinitialization delay, an LLDP agent must wait the specified amount of time to initialize LLDP after the LLDP operating mode changes.

Receiving LLDP frames

An LLDP agent operating in TxRx mode or Rx mode confirms the validity of TLVs carried in every received LLDP frame. If the TLVs are valid, the LLDP agent saves the information and starts an aging timer. The initial value of the aging timer is equal to the TTL value in the Time To Live TLV carried in the LLDP frame. When the LLDP agent receives a new LLDP frame, the aging timer restarts. When the aging timer decreases to zero, all saved information ages out.

TLV support

By default, the agent sends and receives the following mandatory TLVs on each interface:

- Port ID
- Chassis ID
- TTL

By default, the following ANSI/TIA-1057 TLVs for LLDP Media Endpoint Discovery (MED) are enabled on an agent. Sending them depends on the configuration and reception of any MED TLVs:

- MAC/PHY status. Includes the bit rate and auto negotiation status of the link.
- Power Via MDI: Includes Power Over Ethernet related information for supported interfaces.
- Port description
- System name
- System description
- Management address
- System capabilities
- Port VLAN ID

By default, the agent sends and receives the following ANSI/TIA-1057 TLVs for LLDP Media Endpoint Discovery (MED):

- Capabilities: Indicates MED TLV capability.
- Power Via MDI: Includes Power Over Ethernet related information.
- Network Policy: Includes the VLAN configuration for voice application.
- Location: Location identification information.
- Extended Power Via MDI: Power Over Ethernet related information

TLV advertisements

The LLDP agent transmits the following:

- Chassis-ID: Base MAC address of the switch.
- Port-ID: Port number of the physical port.
- Time-to-Live (TTL): Length of time an LLDP neighbor retains advertised data before discarding it.
- System capabilities: Identifies the primary switch capabilities (bridge, router). Identifies the primary switch functions that are enabled, such as routing.
- System description: Includes switch model name and running software version, and ROM version.
- System name: Name assigned to the switch.
- Management address: Default address selection method unless an optional address is configured.
- Port description: Physical port identifier.
- Port VLAN ID: On an L2 port, contains access or native VLAN ID. Trunk allowed VLANs information are not advertised as part of the Port VLAN ID TLV.
- Port VLAN Name: Access or native VLAN name. Enabled by default.
- Port MFS: Maximum Frame Size. Enabled by default.

LLDP MED support

LLDP-MED interoperates with directly connected IP telephony (endpoint) clients and provides the following features:

- Advertisement of the voice VLAN configured on the interface which is used by connected IP telephony (endpoint) clients.
- Advertisement of the configured location on the switch that can be used by the connected endpoint.
- Support for the fast-start capability

LLDP-MED is intended for use with VoIP endpoints and is not designed to support links between network infrastructure devices (such as switch-to-switch or switch-to-router links).

LLDP EEE

Energy Efficient Ethernet (EEE) is a mechanism defined by IEEE 802.3az, which is an extension of 802.3 IEEE standards. EEE defines support for the device to operate in the Low Power Idle (LPI) mode during low link utilization. This allows both sides of a link to disable or turn off respective transmit/receive circuitry to save power. EEE uses LLDP for exchanging optimal set of parameters for both devices.

Guidelines

- An LLDP must not contain more than one EEE TLV.
- LLDP may or may not be enabled on the interfaces supporting EEE. If LLDP is not enabled, it might not be in the optimal operational mode.
- EEE TLV should not be exchanged until it negotiates from L1 and it detects peer EEE capabilities.
- EEE TLV is disabled by default and enabled only when EEE is active.
- EEE and EEE TLV exchange can be enabled or disabled at the interface level.

Configuring the LLDP agent

Procedure

1. By default, the LLDP agent is enabled on all active interfaces. If LLDP was disabled, enable it with the command `lldp`.
2. By default, the LLDP agent transmits and receive on all interfaces. To customize LLDP behavior on a specific interface, use the commands `lldp transmit` and `lldp receive`.
3. By default, the LLDP agent sets the management address in all TLVs in the following order:
   a. LLDP management IP address.
   b. SVI.
   c. Base MAC.
4. By default, all supported TLVs are sent and received. To customize the list, use the command `lldp select-tlv`.
5. By default, support for the LLDP-MED TLV is enabled. To customize settings, use the commands `lldp med` and `lldp med-location`.
6. If required, adjust LLDP timer, holdtime, reinitialization delay, and transmit delay from their default values with the commands `lldp timer`, `lldp holdtime`, `lldp reinit`, and `lldp txdelay`.

Example

This example creates the following configuration:

- Enables LLDP support.
- Disables LLDP transmission on interface **1/1/1**.

```
switch(config)# lldp
switch(config)# interface 1/1/1
switch(config-copp)# no lldp transmit
```

LLDP commands

clear lldp neighbors

`clear lldp neighbors`

Description

Clears all LLDP neighbor details.

Examples

Clearing all LLDP neighbor details:

```
switch# clear lldp neighbors
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

clear lldp statistics

`clear lldp statistics`

Description

Clears all LLDP neighbor statistics.

Examples

Clearing all LLDP neighbor statistics:

```
switch# clear lldp statistics
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

lldp

```
lldp
no lldp
```

Description

Enables LLDP support globally on all active interfaces. By default, LLDP is enabled.

The **no** form of this command disables LLDP support globally on all active interfaces. It does not remove any LLDP configuration settings.

Examples

Enabling LLDP:

```
switch(config)# lldp
```

Disabling LLDP:

```
switch(config)# no lldp
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

lldp dot3

```
lldp dot3 {poe | macphy}
no lldp dot3 {poe | macphy}
```

Description

Sets the 802.3 TLVs to be advertised. By default, advertisement of both POE and MAC/PHY TLVs is enabled.

The **no** form of this command disables advertisement of 802.3 TLVs.

Parameter	Description
`poe`	Specifies advertisement of power over Ethernet data link classification.
`macphy`	Specifies advertisement of media access control and physical layer information.

Examples

Enabling advertisement of the POE TLV:

```
switch(config-if)# lldp dot3 poe
```

Disabling advertisement of the POE TLV:

```
switch(config-if)# no lldp dot3 poe
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution rights for this command.

lldp dot3 eee

```
lldp dot3 eee
no lldp dot3 eee
```

Description

Sets the 802.3 TLVs for Energy-Efficient Ethernet (EEE) to be advertised. By default, advertisement of EEE TLVs is enabled.

The **no** form of this command disables advertisement of 802.3 TLVs.

Parameter	Description
`eee`	Specifies advertisement of 802.3 TLVs for EEE.

Examples

Enabling advertisement of the EEE TLVs:

```
switch(config-if)# lldp dot3 eee
```

Disabling advertisement of the EEE TLVs:

```
switch(config-if)# no lldp dot3 eee
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	`config-if`	Administrators or local user group members with execution rights for this command.

lldp dot3 mfs

```
lldp dot3 mfs
no lldp dot3 mfs
```

Description

Enables the 802.3 TLV list in LLDP to advertise for maximum frame size (MFS). Enabled by default.

The **no** form of this command disables the advertisement of maximum frame size TLVs.

Examples

Enabling advertisement of maximum frame size TLVs:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp dot3 mfs
```

Disabling advertisement of maximum frame size TLVs:

```
switch(config)# interface 1/1/1
switch(config-if)# no lldp dot3 mfs
```

Command History

Release	Modification
10.11	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

lldp holdtime-multiplier

```
lldp holdtime-multiplier <multiplier>
no lldp holdtime-multiplier
```

Description

Sets the holdtime TTL multiplier value that is used to calculate the LLDP Time-to-Live value. Time-to-Live defines the length of time that neighbors consider LLDP information sent by this agent as valid. When

Time-to-Live expires, the information is deleted by the neighbor. Time-to-live is calculated by multiplying holdtime by the value of **lldp timer**.

The **no** form of this command sets the holdtime TTL multiplier to its default value of 4.

Parameter	Description
<multiplier>	Specifies the TTL multiplier in the range of 2 to 10. Default: 4.

Formula

TTL = Holdtime-multiplier x lldp timer

where:

TTL = Time-to-Live

Holdtime-multiplier = Multiplying holdtime value

lldp timer = Message transmission interval

Examples

Setting the holdtime to 8 times of the value of lldp timer:

```
switch(config)# lldp holdtime-multiplier 8
```

Setting the holdtime to the default value of 4 times of the value of lldp timer:

```
switch(config)# no lldp holdtime-multiplier
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

lldp management-address vlan

```
lldp management-address vlan <VLAN-ID>
no lldp management-address vlan <VLAN-ID>
```

Description

Sets the VLAN whose IPv4 or IPv6 address is advertised as the LLDP management authority.

The **no** form of this command removes the VLAN whose IPv4 or IPv6 address is advertised as the LLDP management authority.

The following is the precedence for the management IP address TLV in the LLDP packet (in order):

- LLDP management-IPv4-address and management-ipv6-address, if configured.
- LLDP management VLAN's Ipv4 and IPv6 address, if configured.
- Loopback IP address from the smallest configured loopback interface identifier.
- Route-only-port IP address (Layer-3 interface) or IP address of the SVI (Layer-2 interface).
- OOBM IP address.
- Base MAC address of the switch.

Parameter	Description
<VLAN-ID>	Specifies the VLAN ID.

Examples

Setting the management authority for VLAN 10:

```
switch(config)# lldp management-address vlan 10
```

Removing the management authority for VLAN 10:

```
switch(config)# no lldp management-address vlan 10
```

Command History

Release	Modification
10.12	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

lldp management-ipv4-address

```
lldp management-ipv4-address <IPV4-ADDR>
no lldp management-ipv4-address
```

Description

Defines the IPv4 management address of the switch which is sent in the management address TLV. One IPv4 and one IPv6 management address can be configured.

If you do not define an LLDP management address, then LLDP uses one of the following (in order):

- IP address of SVI [interface VLAN <*vid*>]
- Base MAC address of the switch

The **no** form of this command removes the IPv4 management address of the switch.

Parameter	Description
<IPV4-ADDR>	Specifies the management address of the switch as an IPv4 format (**x.x.x.x**), where **x** is a decimal value from 0 to 255.

Examples

Setting the management address to **10.10.10.2**:

```
switch(config)# lldp management-ipv4-address 10.10.10.2
```

Removing the management address:

```
switch(config)# no lldp management-ipv4-address
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

lldp management-ipv6-address

```
lldp management-ipv6-address <IPV6-ADDR>
no lldp management-ipv6-address
```

Description

Defines the IPv6 management address of the switch. The management address is encapsulated in the management address TLV.

If you do not define an LLDP management address, then LLDP uses one of the following (in order):

- IP address of SVI [interface VLAN <*vid*>]
- Base MAC address of the switch

The **no** form of this command removes the IPv6 management address of the switch.

Parameter	Description
<IPV6-ADDR>	Specifies an IP address in IPv6 format (**xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F.

Examples

Setting the management address to 2001:db8:85a3::8a2e:370:7334:

```
switch(config)# lldp management-ipv6-address 2001:0db8:85a3::8a2e:0370:7334
```

Removing the management address:

```
switch(config)# no lldp management-ipv6-address
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

lldp med

```
lldp med [poe [priority-override] | capability | network-policy]
no med [poe [priority-override] | capability | network-policy]
```

Description

Configures support for the LLDP-MED TLV. LLDP-MED (media endpoint devices) is an extension to LLDP developed by TIA to support interoperability between VoIP endpoint devices and other networking end-devices. The switch only sends the LLDP MED TLV after receiving a MED TLV from and connected endpoint device.

The **no** form of this command disables support for the LLDP MED TLV.

Parameter	Description
`poe [priority-override]`	Specifies advertisement of power over Ethernet data link classification. The **priority-override** option overrides user-configured port priority for Power over Ethernet. When both **lldp dot3 poe** and **lldp med poe** are enabled, the **lldp dot3 poe3** setting takes precedence. Default: enabled.
`capability`	Specifies advertisement of supported LLDP MED TLVs. The capability TLV is always sent with other MED TLVs, therefore it cannot be disabled when other MED TLVs are enabled. Default: enabled.
`network-policy`	Network policy discovery lets endpoints and network devices advertise their VLAN IDs, and IEEE 802.1p (PCP and DSCP) values for voice applications. This TLV is only sent when a voice VLAN policy is present. Default: enabled.

Examples

Enabling advertisement of the network policy TLV:

```
switch(config-if)# lldp med network-policy
```

Disabling advertisement of the network policy TLV:

```
switch(config-if)# no lldp med network-policy
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution rights for this command.

lldp med location

```
lldp med location {civic-addr   elin-addr }
no med location {civic-addr  elin-addr }
```

Description

Configures support for the LLDP-MED TLV. Supports only civic address and emergency location information number (ELIN). Coordinate-based location is not supported.

The **no** form of this command disables support for the LLDP MED TLV.

Parameter	Description
`civic-addr`	Configures the LLDP MED civic location TLV.
`elin-addr`	Configures support for the LLDP MED emergency location TLV. This feature is intended for use in ECS applications to support class 3 LLDP-MED VoIP telephones connected to a switch in an MLTS infrastructure. An ELIN is a valid NANP format telephone number assigned to MLTS operators in North America by the appropriate authority. The ELIN is used to route emergency (E911) calls to a PSAP. (Range: 1-15 numeric characters)

The **lldp med location civic-addr** command requires a minimum of one type/value pair, but typically includes multiple type/value pairs as needed to configure a complete set of data describing a given location.

CA-TYPE: This is the first entry in a type/value pair and is a number defining the type of data contained in the second entry in the type/value pair (CA-VALUE.) Some examples of CA-TYPE specifiers include: 3=city 6=street (name) 25=building name (Range: 0 - 255)

CA-VALUE: This is the second entry in a type/value pair and is an alphanumeric string containing the location information corresponding to the immediately preceding CA-TYPE entry. Strings are delimited by either blank spaces, single quotes (' ... '), or double quotes ("... ".) Each string should represent a specific data type in a set of unique type/value pairs comprising the description of a location, and each string must be preceded by a CA-TYPE number identifying the type of data in the string.

The following LLDP-MED TLV values are supported. For details on these value types, refer to RFC 4776

- 1: national subdivisions (state, canton, region, province, prefecture)
- 2: county, parish, gun (JP), district (IN)
- 3: city, township, shi (JP)
- 4: city division, borough, city district, ward, chou (JP)
- 5: neighborhood, block
- 6: group of streets below the neighborhood level
- 16: leading street direction N
- 17: trailing street suffix SW
- 18: street suffix or type
- 19: house number
- 20: house number suffix
- 21: landmark or vanity
- 22: location
- 23: name
- 24: postal/zip code
- 25: building (structure)
- 26: unit (apartment, suite)
- 27: floor
- 28: room
- 29: type of place
- 30: postal community name
- 31: post office box
- 32: additional code 13203000003
- 33: seat (desk, cubicle workstation)
- 34: primary road name 35 road section
- 36: branch road name
- 37: sub-branch road name
- 38: street name pre-modifier
- 39: street name post-modifier

Examples

Enabling support for the LLDP MED emergency location TLV:

```
switch(config-if)# lldp med location elin-addr 408-555-1212
```

Disabling support for the LLDP MED emergency location TLV:

```
switch(config-if)# no lldp med location elin-addr 408-555-1212
```

Enabling support for the LLDP MED civic address TLV:

```
switch(config-if)# lldp med location civic-addr US 1 19 123 6 Fake 18 Street
```

Disabling support for the LLDP MED civic address TLV:

```
switch(config-if)# no lldp med location civic-addr US 1 19 123 6 Fake 18 Street
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution rights for this command.

lldp receive

```
lldp receive
no lldp receive
```

Description

Enables reception of LLDP information on an interface. By default, LLDP reception is enabled on all active interfaces.

The **no** form of this command disables reception of LLDP information on an interface.

Examples

Enabling LLDP reception on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp receive
```

Disabling LLDP reception on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no lldp receive
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution rights for this command.

lldp reinit

```
lldp reinit <TIME>
no lldp reinit
```

Description

Sets the amount of time (in seconds) to wait before performing LLDP initialization on an interface.

The **no** form of this command sets the reinitialization time to its default value of 2 seconds.

Parameter	Description
<TIME>	Specifies the reinitialization time in seconds. Range: 1 to 10. Default: 2 seconds.

Examples

Setting the reinitialization time to 5 seconds:

```
switch(config)# lldp reinit 5
```

Setting the reinitialization time to the default value of 2 seconds:

```
switch(config)# no lldp reinit
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

lldp select-tlv

```
lldp select-tlv <TLV-NAME>
no lldp select-tlv <TLV-NAME>
```

Description

Selects a TLV that the LLDP agent will send and receive. By default, all supported TLVs are sent and received.

The **no** form of this command stops the LLDP agent from sending and receiving a specific TLV.

LLDP supports Organization Unique Identifiers (OUI) with the following Organization-specific TLVs:

- IEEE 802.1 (DOT1) (oui:0x00, 0x80, 0xc2)
- IEEE 802.3 (DOT3) (oui:0x00, 0x12, 0x0f)
- Aruba, a Hewlett Packard Enterprise Company (oui:0x88, 0x3a, 0x30)

Parameter	Description
`select-tlv <TLV-NAME>`	Specifies the TLV name to send. The following TLV names are supported: ■ **management-address:** Selection is based on priority in the following list (for example if first TLV name isn't selected, the next will be, progressing through this list until a selection is made): 1. IPv4 or IPV6 management address. 2. If layer 2, the IP address of the SVI. 3. Base MAC address of the switch. ■ **port-description:** Select port-description TLV. ■ **port-vlan-id:** Select port-vlan-id TLV. ■ **port-vlan-name:** Select port-vlan-name TLV. ■ **system-capabilities:** Select system-capabilities TLV. ■ **system-description:** Select system-description TLV. ■ **system-name:** Select system-name TLV.

Examples

Stopping the LLDP agent from sending the **port-description** TLV:

```
switch(config)# no lldp select-tlv port-description
```

Enabling the LLDP agent to send the **port-description** TLV:

```
switch(config)# lldp select-tlv oui
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

lldp timer

```
lldp timer <TIME>
no lldp timer
```

Description

Sets the interval (in seconds) at which local LLDP information is updated and TLVs are sent to neighboring network devices by the LLDP agent. The minimum setting for this timer must be four times the value of **lldp txdelay**.

For example, this is a valid configuration:

- **lldp timer** = 16
- **lldp txdelay** = 4

And, this is an invalid configuration:

- **lldp timer** = 5
- **lldp txdelay** = 2

When copying a saved configuration to the running configuration, the value for **lldp timer** is applied before the value of **lldp txdelay**. This can result in a configuration error if the saved configuration has a value of **lldp timer** that is not four times the value of **lldp txdelay** in the running configuration.

For example, if the saved configuration has the settings:

- **lldp timer** = 16
- **lldp txdelay** = 4

And the running configuration has the settings:

- **lldp timer** = 30
- **lldp txdelay** = 7

Then you will see an error indicating that certain configuration settings could not be applied, and you will have to manually adjust the value of **lldp txdelay** in the running configuration.

The **no** form of this command sets the update interval to its default value of 30 seconds.

Parameter	Description
<TIME>	Specifies the update interval (in seconds). Range: 5 to 32768. Default: 30.

Examples

Setting the update interval to 7 seconds:

```
switch(config)# lldp timer 7
```

Setting the update interval to the default value of 30 seconds:

```
switch(config)# no lldp timer
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

lldp transmit

```
lldp transsmit
no lldp transmit
```

Description

Enables transmission of LLDP information on specific interface. By default, LLDP transmission is enabled on all active interfaces.

The **no** form of this command disables transmission of LLDP information on an interface.

Examples

Enabling LLDP transmission on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp transsmit
```

Disabling LLDP transmission on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no lldp transsmit
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution rights for this command.

lldp txdelay

```
lldp txdelay <TIME>
no lldp txdelay
```

Description

Sets the amount of time (in seconds) to wait before sending LLDP information from any interface. The maximum value for **txdelay** is 25% of the value of **lldp tx timer**.

The **no** form of this command sets the delay time to its default value of 2 seconds.

Parameter	Description
<TIME>	Specifies the delay time in seconds. Range: 0 to 10. Default: 2.

Examples

Setting the delay time to 8 seconds:

```
switch(config)# lldp txdelay 8
```

Setting the delay time to the default value of 2 seconds:

```
switch(config)# no lldp txdelay
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

lldp trap enable

```
lldp trap enable
no lldp trap enable
```

Description

Enables sending SNMP traps for LLDP related events from a particular interface. LLDP trap generation is enabled by default on all the interfaces and has to be disabled for interfaces on which traps are not required to be generated.

The **no** form of this command disables the LLDP trap generation.

LLDP trap generation is disabled by default at the global level and must be enabled before any LLDP traps are sent.

Examples

Enabling LLDP trap generation on global level:

```
switch(config)# lldp trap enable
```

Enabling LLDP trap generation on interface level:

```
switch(config-if)# lldp trap enable
```

Disabling LLDP trap generation on global level:

```
switch(config)# no lldp trap enable
```

Disabling LLDP trap generation on interface level:

```
switch(config-if)# no lldp trap enable
```

Displaying LLDP global configuration:

```
switch# show lldp configuration


LLDP Global Configuration
=========================
LLDP Enabled                : No
LLDP Transmit Interval      : 30
LLDP Hold Time Multiplier   : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval  : 2
LLDP Trap Enabled           : No


TLVs Advertised
===============
Management Address
Port Description
Port VLAN-ID
System Description
System Name

LLDP Port Configuration
=======================
PORT           TX-ENABLED          RX-ENABLED          INTF-TRAP-ENABLED
--------------------------------------------------------------------------
1/1/1          Yes                 Yes                 Yes
1/1/2          Yes                 Yes                 Yes
1/1/3          Yes                 Yes                 Yes
1/1/4          Yes                 Yes                 Yes
1/1/5          Yes                 Yes                 Yes
1/1/6          Yes                 Yes                 Yes
...........
...........
mgmt           Yes                 Yes                 Yes
```

Displaying LLDP Configuration for the interface:

```
switch# show lldp configuration 1/1/1

LLDP Global Configuration
=========================
LLDP Enabled                 : Yes
LLDP Transmit Interval       : 30
LLDP Hold Time Multiplier    : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval   : 2
LLDP Trap Enabled            : No


LLDP Port Configuration
=======================
PORT           TX-ENABLED          RX-ENABLED          INTF-TRAP-ENABLED
---------------------------------------------------------------------------
1/1/1          Yes                 Yes                 Yes
```

Displaying LLDP Configuration for the management interface:

```
switch# show lldp configuration mgmt

LLDP Global Configuration
=========================
LLDP Enabled                 : Yes
LLDP Transmit Interval       : 30
LLDP Hold Time Multiplier    : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval   : 2
LLDP Trap Enabled            : Yes


LLDP Port Configuration
=======================
PORT           TX-ENABLED          RX-ENABLED          INTF-TRAP-ENABLED
---------------------------------------------------------------------------
mgmt           Yes                 Yes                 Yes
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config` and `config-if`	Administrators or local user group members with execution rights for this command.

show lldp configuration

`show lldp configuration [<INTERFACE-ID>]`

Description

Shows LLDP configuration settings for all interfaces or a specific interface.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: **member/slot/port**.

Example

Showing configuration settings for all interfaces:

This example shows configuration settings for interface **1/1/1**.

```
switch# show lldp configuration 1/1/1

LLDP Global Configuration
=========================
LLDP Enabled                 : Yes
LLDP Transmit Interval       : 30
LLDP Hold Time Multiplier    : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval   : 2
LLDP Trap Enabled            : No


LLDP Port Configuration
=======================
PORT            TX-ENABLED          RX-ENABLED          INTF-TRAP-ENABLED
--------------------------------------------------------------------------
1/1/1           Yes                 Yes                 Yes
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp local-device

`show lldp local-device`

Description

Shows global LLDP information advertised by the switch, as well as port-based data. If VLANs are configured on any active interfaces, the VLAN ID is only shown for trunk native or untagged VLAN IDs on access interfaces.

Example

Showing global LLDP information only:

```
switch# show lldp local-device

Global Data
===========

Chassis-ID             : 88:3a:30:47:c1:c0
System Name            : 6100
System Description     : Aruba JL679A  PL.10.06.0001-346-g56a12a8f4cf15
Management Address     : 88:3a:30:47:c1:c0
Capabilities Available : Bridge, Router
Capabilities Enabled   : Bridge, Router
TTL                    : 120
```

Showing all ports except **1/1/11** as administratively down:

```
switch# show lldp local-device

Global Data
===========

Chassis-ID             : 88:3a:30:47:c1:c0
System Name            : 6100
System Description     : Aruba JL679A  PL.10.06.0001-346-g56a12a8f4cf15
Management Address     : 11.11.11.11
Capabilities Available : Bridge, Router
Capabilities Enabled   : Bridge, Router
TTL                    : 120

Port Based Data
===============

Port-ID           : 1/1/11
Port-Desc         : "1/1/11"
Port Mgmt-Address : 11.11.11.11
Port VLAN ID      : 1
Parent Interface  : interface 1/1/11

switch#
```

```
switch# show lldp local-device

Global Data
===========

Chassis-ID             : 88:3a:30:47:c1:c0
System Name            : 4100i
System Description     : Aruba JL817A  RL.10.08.0001
Management Address     : 11.11.11.11
Capabilities Available : Bridge, Router
Capabilities Enabled   : Bridge, Router
TTL                    : 120

Port Based Data
===============

Port-ID           : 1/1/11
Port-Desc         : "1/1/11"
Port Mgmt-Address : 11.11.11.11
```

```
Port VLAN ID      : 1
Parent Interface  : interface 1/1/11

switch#
```

In this example, all the ports except **1/1/11** are administratively down, and VLAN ID 100 is configured on this access interface.

```
switch# show lldp local-device

Global Data
===========

Chassis-ID              : 1c:98:ec:e3:45:00
System Name             : switch
System Description      : Aruba
Management Address      : 192.168.10.1
Capabilities Available : Bridge, Router
Capabilities Enabled    : Bridge, Router
TTL                     : 120

Port Based Data
===============

Port-ID           : 1/1/11
Port-Desc         : "1/1/11"
Port VLAN ID      : 100
Parent Interface  : interface 1/1/11
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp neighbor-info

`show lldp neighbor-info [<INTERFACE-NAME>]`

Description

Displays information about neighboring devices for all interfaces or for a specific interface. The information displayed varies depending on the type of neighbor connected and the type of TLVs sent by the neighbor.

Parameter	Description
<INTERFACE-NAME>	Specifies the interface for which to show information for neighboring devices. Use the format **member/slot/port** (for example, **1/3/1**).

Examples

Showing LLDP information for all interfaces:

```
switch# show lldp neighbor-info

LLDP Neighbor Information
=========================

Total Neighbor Entries          : 1
Total Neighbor Entries Deleted  : 5
Total Neighbor Entries Dropped  : 0
Total Neighbor Entries Aged-Out : 3

LOCAL-PORT  CHASSIS-ID         PORT-ID       PORT-DESC     TTL      SYS-NAME
--------------------------------------------------------------------------------
1/1/2       38:21:c7:5c:df:40  1/1/2         1/1/2         120      switch
1/1/3       f8:60:f0:c9:e0:a0  1/1/3         1/1/3         120      switch
```

Showing information for interface **1/1/3** when it has only one switch connected as a neighbor:

```
Aruba-6100-Switch2# show lldp neighbor-info 1/1/3

Port                             : 1/1/3
Neighbor Entries                 : 1
Neighbor Entries Deleted         : 1
Neighbor Entries Dropped         : 0
Neighbor Entries Aged-Out        : 1
Neighbor Chassis-Name            : 6100
Neighbor Chassis-Description     : Aruba JL679A PL.10.06.0001
Neighbor Chassis-ID              : 88:3a:30:47:d1:c0
Neighbor Management-Address      : 88:3a:30:47:d1:c0
Chassis Capabilities Available   : Bridge, Router
Chassis Capabilities Enabled     : Bridge, Router
Neighbor Port-ID                 : 1/1/3
Neighbor Port-Desc               : 1/1/3
Neighbor Port VLAN ID            : 1
Neighbor Port VLAN Name          : DEFAULT_VLAN_1
Neighbor Port MFS                : 1500
TTL                              : 120
Neighbor Mac-Phy details
Neighbor Auto-neg Supported      : True
Neighbor Auto-neg Enabled        : True
Neighbor Auto-ned Advertised     : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighnor MAU Type                : 1000 BASETFD
```

Showing neighbor information for interface 1/3/2 when it has EEE enabled and successfully auto-negotiated:

```
switch# show lldp neighbor-info 1/3/2

Port                              : 1/3/2
Neighbor Entries                  : 1
Neighbor Entries Deleted          : 1
Neighbor Entries Dropped          : 0
Neighbor Entries Aged-Out         : 1
Neighbor Chassis-Name             : BLDG01-F1-6300
Neighbor Chassis-Description      : Aruba JL668A  FL.10.07.0001BN
Neighbor Chassis-ID               : 88:3a:30:92:a5:c0
Neighbor Management-Address       : 10.6.9.15
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled      : Bridge, Router
Neighbor Port-ID                  : 1/1/1
Neighbor Port-Desc                : 1/1/1
Neighbor Port VLAN ID             : 1
Neighbor Port VLAN Name           : DEFAULT_VLAN_1
Neighbor Port MFS                 : 1500
TTL                               : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported       : true
Neighbor Auto-Neg Enabled         : true
Neighbor Auto-Neg Advertised      : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type                 : 1000 BASETFD

Neighbor EEE information          : DOT3
Neighbor TX Wake time             : 17 us
Neighbor RX Wake time             : 17 us
Neighbor Fallback time            : 17 us
Neighbor TX Echo time             : 17 us
Neighbor RX Echo time             : 17 us
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp neighbor-info detail

`show lldp neighbor-info detail`

Description

Shows detailed LLDP neighbor information for all LLDP neighbor connected interfaces.

Examples

Showing detailed LLDP information for all interfaces:

```
switch# show lldp neighbor-info detail

LLDP Neighbor Information
=========================

Total Neighbor Entries          : 6
Total Neighbor Entries Deleted  : 2
Total Neighbor Entries Dropped  : 0
Total Neighbor Entries Aged-Out : 2

--------------------------------------------------------------------------------

Port                            : 1/1/1
Neighbor Entries                : 1
Neighbor Entries Deleted        : 0
Neighbor Entries Dropped        : 0
Neighbor Entries Aged-Out       : 0
Neighbor Chassis-Name           : 6300
Neighbor Chassis-Description    : Aruba ...
Neighbor Chassis-ID             : 38:11:17:1a:d5:00
Neighbor Management-Address     : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled    : Bridge, Router
Neighbor Port-ID                : 1/1/4
Neighbor Port-Desc              : 1/1/4
Neighbor Port VLAN ID           : 1
Neighbor Port VLAN Name         : DEFAULT_VLAN_1
Neighbor Port MFS               : 1500
TTL                             : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported     : true
Neighbor Auto-Neg Enabled       : true
Neighbor Auto-Neg Advertised    : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type               : 1000 BASETFD

--------------------------------------------------------------------------------

Port                            : 1/1/2
Neighbor Entries                : 1
Neighbor Entries Deleted        : 0
Neighbor Entries Dropped        : 0
Neighbor Entries Aged-Out       : 0
Neighbor Chassis-Name           : 6300
Neighbor Chassis-Description    : Aruba ...
Neighbor Chassis-ID             : 38:11:17:1a:d5:00
Neighbor Management-Address     : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled    : Bridge, Router
Neighbor Port-ID                : 1/1/5
Neighbor Port-Desc              : 1/1/5
Neighbor Port VLAN ID           : 1
Neighbor Port VLAN Name         : DEFAULT_VLAN_1
Neighbor Port MFS               : 1500
TTL                             : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported     : true
Neighbor Auto-Neg Enabled       : true
Neighbor Auto-Neg Advertised    : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type               : 1000 BASETFD
```

```
--------------------------------------------------------------------------------

Port                           : 1/1/3
Neighbor Entries               : 1
Neighbor Entries Deleted       : 0
Neighbor Entries Dropped       : 0
Neighbor Entries Aged-Out      : 0
Neighbor Chassis-Name          : 6300
Neighbor Chassis-Description   : Aruba ...
Neighbor Chassis-ID            : 38:11:17:1a:d5:00
Neighbor Management-Address    : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID               : 1/1/6
Neighbor Port-Desc             : 1/1/6
Neighbor Port VLAN ID          : 1
Neighbor Port VLAN Name        : DEFAULT_VLAN_1
Neighbor Port MFS              : 1500
TTL                            : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported    : true
Neighbor Auto-Neg Enabled      : true
Neighbor Auto-Neg Advertised   : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type              : 1000 BASETFD

--------------------------------------------------------------------------------

Port                           : 1/1/46
Neighbor Entries               : 1
Neighbor Entries Deleted       : 0
Neighbor Entries Dropped       : 0
Neighbor Entries Aged-Out      : 0
Neighbor Chassis-Name          : 6300
Neighbor Chassis-Description   : Aruba ...
Neighbor Chassis-ID            : 38:11:17:1a:d5:00
Neighbor Management-Address    : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID               : 1/1/19
Neighbor Port-Desc             : 1/1/19
Neighbor Port VLAN ID          : 1
Neighbor Port VLAN Name        : DEFAULT_VLAN_1
Neighbor Port MFS              : 1500
TTL                            : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported    : true
Neighbor Auto-Neg Enabled      : true
Neighbor Auto-Neg Advertised   : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type              : 1000 BASETFD

--------------------------------------------------------------------------------

Port                           : 1/1/47
Neighbor Entries               : 1
Neighbor Entries Deleted       : 0
Neighbor Entries Dropped       : 0
Neighbor Entries Aged-Out      : 0
Neighbor Chassis-Name          : 6300
Neighbor Chassis-Description   : Aruba ...
Neighbor Chassis-ID            : 38:11:17:1a:d5:00
```

```
Neighbor Management-Address    : 38:11:17:1a:d5:00
Chassis Cap
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp statistics

`show lldp statistics [<INTERFACE-ID>]`

Description

Shows global LLDP statistics or statistics for a specific interface.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: **member/slot/port**.

Example

Showing global statistics for all interfaces:

```
switch# show copp-policy statistics
switch# show lldp statistics
LLDP Global Statistics
==========================================================================

Total Packets Transmitted            : 25
Total Packets Received               : 20
Total Packets Received And Discarded : 0
Total TLVs Unrecognized              : 0

LLDP Port Statistics
==========================================================================

PORT-ID         TX-PACKETS     RX-PACKETS     RX-DISCARDED   TLVS-UNKNOWN
--------------------------------------------------------------------------
1/1/1           25             20             0              0
1/1/2           0              0              0              0
1/1/3           0              0              0              0
1/1/4           0              0              0              0
1/1/5           0              0              0              0
1/1/6           0              0              0              0
1/1/7           0              0              0              0
```

```
1/1/8          0           0           0           0
1/1/9          0           0           0           0
1/1/10         0           0           0           0
1/1/11         0           0           0           0
1/1/12         0           0           0           0
1/1/13         0           0           0           0
1/1/14         0           0           0           0
1/1/15         0           0           0           0
1/1/16         0           0           0           0
1/1/17         0           0           0           0
1/1/18         0           0           0           0
1/1/19         0           0           0           0
1/1/20         0           0           0           0
1/1/21         0           0           0           0
1/1/22         0           0           0           0
1/1/23         0           0           0           0
1/1/24         0           0           0           0
1/1/25         0           0           0           0
1/1/26         0           0           0           0
1/1/27         0           0           0           0
1/1/28         0           0           0           0
```

Showing statistics for interface **1/1/1**:

```
switch# show lldp statistics 1/1/1

LLDP Statistics
===============

Port Name                          : 1/1/1
Packets Transmitted                : 159
Packets Received                   : 163
Packets Received And Discarded     : 0
Packets Received And Unrecognized : 0
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp tlv

`show lldp tlv`

Description

Shows the LLDP TLVs that are configured for send and receive.

Example

```
switch# show lldp tlv

TLVs Advertised
===============

Management Address
Port Description
Port VLAN-ID
System Capabilities
System Description
System Name
VLAN Name
MFS
OUI
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Cisco Discovery Protocol (CDP)

Cisco Discovery Protocol (CDP) is a proprietary layer 2 protocol supported by most Cisco devices. It is used to exchange information, such as software version, device capabilities, and voice VLAN information, between directly connected devices, such as a VoIP phone and a switch.

CDP support

By default, CDP is enabled on each active switch port. This is a read-only capability, which means the switch can receive and store information about adjacent CDP devices, but does not generate CDP packets (except when communicating with Cisco IP phones.)

The switch supports CDPv2 only and does not support SNMP MIB traps.

When a CDP-enabled port receives a CDP packet from another CDP device, it enters data for that device into the CDP Neighbors table, along with the port number on which the data was received. It does not forward the packet. The switch also periodically purges the table of any entries that have expired. (The holdtime for any data entry in the switch CDP Neighbors table is configured in the device transmitting the CDP packet and cannot be controlled in the switch receiving the packet.) A switch reviews the list of CDP neighbor entries every three seconds and purges any expired entries.

Support for legacy Cisco IP phones

Autoconfiguration of legacy Cisco IP phones for tagged voice VLAN support requires CDPv2.

On initial boot-up, and sometimes periodically, a Cisco phone queries the switch and advertises information about itself using CDPv2. When the switch receives the VoIP VLAN Query TLV (type 0x0f) from the phone, the switch immediately responds with the voice VLAN ID in a reply packet using the

VoIP VLAN Reply TLV (type 0x0e). This enables the Cisco phone to boot properly and send traffic on the advertised voice VLAN ID.

The switch CDP packet includes these TLVs:

- CDP Version: 2
- CDP TTL: 180 seconds
- Checksum
- Capabilities (type 0x04): 0x0008 (is a switch)
- Native VLAN: The PVID of the port
- VoIP VLAN Reply (type 0xe): voice VLAN ID (same as advertised by LLDP-MED)
- Trust Bitmap (type 0x12): 0x00
- Untrusted port CoS (type 0x13): 0x00

CDP commands

cdp

`cdp`

Description

Configures CDP support globally on all active interfaces or on a specific interface. By default, CDP is enabled on all active interfaces.

When CDP is enabled, the switch adds entries to its CDP Neighbors table for any CDP packets it receives from neighboring CDP devices.

When CDP is disabled, the CDP Neighbors table is cleared and the switch drops all inbound CDP packets without entering the data in the CDP Neighbors table.

The **no** form of this command disables CDP support globally on all active interfaces or on a specific interface.

Examples

Enabling CDP globally:

```
switch(config)# cdp
```

Disabling CDP globally:

```
switch(config)# no cdp
```

Enabling CDP on interface **1/1/1**:

```
switch(config)# interface 1/1/1s
switch(config-if)# cdp
```

Disabling CDP on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no cdp
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config` `config-if`	Administrators or local user group members with execution rights for this command.

clear cdp counters

`clear cdp counters`

Description

Clears CDP counters.

Examples

Clearing CDP counters:

```
switch(config) clear cdp counters
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

clear cdp neighbor-info

`clear cdp neighbor-info`

Description

Clears CDP neighbor information.

Examples

Clearing CDP neighbor information:

```
switch(config) clear neighbor-info
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

show cdp

`show cdp`

Description

Shows CDP information for all interfaces.

Examples

Showing CDP information:

```
switch(config)# show cdp
CDP Global Information
======================
CDP           : Enabled
CDP Mode      : Rx only
CDP Hold Time : 180 seconds

Port         CDP
--------  --------------
1/1/1          Enabled
1/1/2          Enabled
1/1/3          Enabled
1/1/4          Enabled
1/1/5          Enabled
1/1/6          Enabled
1/1/7          Enabled
1/1/8          Enabled
1/1/9          Enabled
1/1/10         Enabled
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

show cdp neighbor-info

```
show cdp neighbor-info <INTERFACE-ID>
```

Description

Shows CDP information for all neighbors or for CDP information on a specific interface.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: **member/slot/port**.

Examples

Showing all CDP neighbor information:

```
switch(config)# show cdp neighbor-info
Port        Device ID                Platform                 Capability
--------------------------------------------------------------------------------
1/1/1       myswitch                 cisco WS-C2950-12        SI
```

Showing CDP information for interface **1/1/1**:

```
switch(config)# show cdp neighbor-info 1/1/1
Local Port : 1/1/1
MAC                 : 3c:a8:2a:7b:6b:2b
Device ID           : SEPd4adbd2a30d6
Address             : 2.71.0.230
Platform            : Cisco IP Phone 3905
Duplex              : full
Capability          : host
Voice VLAN Support : Yes
Neighbor Port-ID    : Port 1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

show cdp traffic

```
show cdp neighbor-info
```

Description

Shows CDP statistics for each interface.

Examples

Showing CDP traffic statistics:

```
switch(config)# show cdp traffic
CDP Statistics
====================
Port        Transmitted Frames       Received Frames          Discarded Frames
--------------------------------------------------------------------------------
1/1/1          0                        4                        0
1/1/2          0                        0                        0
1/1/3          0                        2                        0
1/1/4          0                        0                        0
1/1/5          0                        0                        0
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

Chapter 14

Zero Touch Provisioning

Zero Touch Provisioning (ZTP) enables the auto-configuration of factory default switches without a network administrator onsite.

When a switch is booted from its factory default configuration, ZTP autoprovisions the switch by automatically downloading and installing a firmware file, a configuration file, or both. With ZTP, even a nontechnical user (for example: a store manager in a retail chain or a teacher in a school) can deploy devices at a site.

ZTP support

The switch supports standards-based Zero Touch Provisioning (ZTP) operations as follows:

- The switch must be running the factory default configuration.
- ZTP operations are supported over IPv4 connections only. IPv6 connections are not supported for ZTP operations.
- You must configure the DHCP server to provide a standards-based ZTP server solution. Options and features that are specific to Network Management Solution (NMS) tools, such as AirWave, are not supported.
  - Supported DHCP options are:

DHCP option	Description
43	Vendor Specific Information
43 suboption 144	Name of the configuration file
43 suboption 145	Name of the firmware image file
43 suboption 148	FQDN or IPv4 address of the HTTP Proxy
60	Vendor Class Identifier (VCI)
66	IPv4 address of the TFTP server (Specifying a host name instead of an IP address is not supported.)
67	Name of the configuration file (Option 43 suboption 144 takes precedence over this option.)

- The configuration file is a text file or JSON file that becomes the startup and running configuration on the switch after the ZTP operation is complete. The configuration can be in CLI or in JSON format.
- When the switch is started using the factory default configuration, the ZTP operation is started automatically and is active until any running configuration of the switch is modified. There is no CLI command required to start the operation.

The switch supports the following standards:

- RFC 2131, *Dynamic Host Configuration Protocol*.
- RFC 2132, *DHCP Options and BOOTP Vendor Extensions*. Support is limited to the options listed in the table "Supported DHCP options for ZTP on AOS-CX."

Hewlett Packard Enterprise recommends that you implement ZTP in a secure and private environment. Any public access can compromise the security of the switch, as follows:

- ZTP is enabled only in the factory default configuration of the switch, DHCP snooping is not enabled. The Rogue DHCP server must be manually managed.
- The DHCP offer is in plain data without encryption.

Setting up ZTP on a trusted network

The following procedure is an overview of setting up a Zero Touch Provisioning (ZTP) environment to provision newly installed switches automatically. The procedure is intended for network administrators who are familiar with automatically provisioning switches in a network, and does not provide detailed information about configuring or managing switches.

Procedure

1. For each switch model to be provisioned using ZTP, do the following:
    a. Obtain the switch firmware image file.
    b. Prepare the switch configuration file. The configuration file becomes the running configuration and the startup configuration on the switch.
2. Set up a TFTP server and record its IP address. The address is required when you set up the DHCP server. The switch must be able to reach the TFTP server and DHCP server on the same subnet.

    Switches support provisioning through a network connected to a data port or through a network connected to the in-band management interface VLAN 1.
3. Publish the configuration files and image files to the TFTP server. You need to know the locations of the files and the IP address of the TFTP server when you set up the vendor class options on the DHCP server.
4. On the DHCP server, set up vendor classes for each switch model you plan to provision. To do this you need the following information:
    - The IP address of the TFTP server. Using a host name is not supported.
    - The path to the switch configuration and firmware image files on the TFTP server.
    - The vendor class identifier (VCI) for each switch model.

    You can obtain the VCI by entering the `show dhcp client vendor-class-identifier` command from a switch CLI command prompt in the manager context. The VCI is the text string in the response that starts with `Aruba`.

    For example:

    ```
    switch# show dhcp client vendor-class-identifier
    Vendor Class Identifier:  Aruba xxxxx xxxx
    ```

    Where x indicates the switch model number.
5. At the installation site, provide the switch installer with a Cat6 network cable connected to the network that includes the DHCP and TFTP servers, and information about the switch port to use. The switch installer plugs the cable into the data port you specify.

The ZTP operation begins when power is applied to the switch after the network cable is installed.

6. Assuming the downloaded configuration includes a way to access the CLI of the switch, you can enter the following command to show the options offered by the DHCP server and the status of the ZTP operation:
   `show ztp information`

ZTP process during switch boot

1. The switch boots up with the factory default configuration.

   If the ZTP operation detects that the switch configuration is different from the factory default configuration, the ZTP operation ends. The switch must be configured at the installation site.
2. The switch sends out a DHCP discovery from the in-band management interface.

   The switch waits to receive DHCP options indefinitely or until the running configuration is modified. If a DHCP IP address is received but no DHCP options are received, the switch waits an additional minute before ending the ZTP operation.

   After the ZTP operation ends, there is no automatic retry. You can either attempt to boot the switch with the factory default configuration again, configure the switch at the installation site, or use the ZTP force-provision CLI to trigger the ZTP process, ignoring the present running configuration of the switch.
   - Once force-provision is enabled, new DHCP requests are sent from the switch. Disabling force-provision does not stop the DHCP already in progress, but only changes the switch configuration status of force-provision.
   - If ZTP fails while force-provision is enabled, there is no automatic retry. To retry, `ztp force-provision` should be disabled and re-enabled to clear the current ZTP state and send a new DHCP request. When `ztp force-provision` is already enabled on the switch, re-enabling it results in no operation.
   - If the DHCP server is configured to provide both ZTP image and configuration options and there is a non-default startup configuration present on the switch, clearing the non-default startup configuration before triggering `ztp force-provision` is recommended. If an image is downloaded via ZTP, the switch reboots once the image download is complete and ZTP force-provision configuration is lost, causing ZTP to enter into a failed state. ZTP force-provision will need to be enabled again to continue the process.
3. The DHCP server responds with an offer containing the following:
   - The IPv4 address of the TFTP server
   - One or both of the following:
     - The name of the firmware image file
     - The name of the configuration file
   - Aruba Central Location (optional) including the shared token value for the on-premise server
   - HTTP Proxy Location (optional)
4. If a firmware image file is offered, the ZTP operation downloads the image file from the TFTP server to the switch. If the current switch image and downloaded firmware image version do not match, then the switch boots with the downloaded image:
   - If the image upgrade fails, the switch retains its original firmware image and the ZTP operation ends with a failed status.

- If the image upgrade succeeds, the ZTP operation is started again after the switch reboots. Because the downloaded image file matches the image file installed on the switch, the ZTP operation continues, and checks if a configuration file is offered.

5. If a configuration file is offered, the ZTP operation downloads the configuration file copies the file to the running-config and then to the startup-config of the switch:
   - If the startup configuration update fails, the switch retains its factory-default running configuration and the ZTP operation ends with a failed status.
     - If the copy operation fails, the ZTP operation ends with a failed status.
     - If the copy operation succeeds, the ZTP operation ends successfully.

ZTP commands

show ztp information

`show ztp information`

Description

Shows information about Zero Touch Provisioning (ZTP) operations performed on the switch.

Usage

When a switch configured to use ZTP is booted from a factory default configuration, the switch contacts a DHCP server, which offers options for obtaining files used to provision the switch:

- The IP address of the TFTP server
- The name of the image file
- The name of the configuration file
- The Aruba Central FDQN or IPv4 address
- The HTTP proxy FDQN or IPv4 address

The **show ztp information** command shows the options offered by the DHCP server and the status of the ZTP operation.

The status of the ZTP operation is one of the following:

Success

The ZTP operation succeeded.

One of the following is true:

- Both the running configuration and the startup configuration were updated.
- The IP address of the TFTP server was received, but the offer did not include a configuration file or a firmware image file.
- Any combination of vendor encapsulated DHCP options are received as configured, along with the firmware image and switch configuration file.
- Only vendor encapsulated DHCP options are configured and are received accordingly.

Failed - Custom startup configuration detected

The switch was booted from a configuration that is not the factory default configuration. For example, the administrator password has been set.

Failed - Timed out while waiting to receive ZTP options

Either the switch received the DHCP IPv4 address but no ZTP options were received within 1 minute or ZTP force-provision is triggered and no ZTP options are received within 3 minutes.

Failed - Detected change in running configuration

The running configuration was modified by a user while the ZTP operation was in progress.

Failed - TFTP server unreachable

The TFTP server is not reachable at the specified IP address.

Failed - TFTP server information unavailable

The image file name or config file name is provided without the TFTP server location to fetch the files from and ZTP enters failed state.

Failed - Invalid configuration file received

Either the file transfer of the configuration file failed, or the configuration file is invalid (an error occurred while attempting to apply the configuration).

Failed - Invalid image file received

Either the file transfer of the firmware image file failed, or the firmware image file is invalid (an error occurred while verifying the image).

In the case of reconnection, connect with a main or alternative location to the COP instance as a user. The current connection is shown in the **Central location** field.

Scenario 1: If the location the device is currently connected on is updated, the system reconnects in order to connect with the new location.

Scenario 2: If the location in which the device is not currently connected on is updated, the DUT does not go through the reconnection process.

Examples

Showing switch image download in progress after receiving ZTP options:

```
switch# show ztp information
TFTP Server                        : 10.0.0.2
Image File                         : TL_10_02_0001.swi
Configuration File                 : config_file
ZTP Status                         : In-progress - Image download and verification
Aruba Central Location             : secure.arubanetworks.com
Alternative Aruba Central Location: NA
Aruba Central Shared Token         : aruba123
Force-Provision                    : Disabled
HTTP Proxy Location                : http.proxy.arubanetworks.com
```

Showing switch image download failure after receiving ZTP options:

```
switch# show ztp information
TFTP Server                        : 10.0.0.2
Image File                         : TL_10_02_0001.swi
Configuration File                 : config_file
ZTP Status                         : Failed - Unable to download image
Aruba Central Location             : secure.arubanetworks.com
Alternative Aruba Central Location: NA
Aruba Central Shared Token         : aruba123
Force-Provision                    : Disabled
HTTP Proxy Location                : http.proxy.arubanetworks.com
```

Showing switch configuration download in progress after receiving ZTP options:

```
switch# show ztp information
TFTP Server                          : 10.0.0.2
Image File                           : TL_10_02_0001.swi
Configuration File                   : config_file
ZTP Status                           : In-progress - Configuration download
Aruba Central Location               : secure.arubanetworks.com
Alternative Aruba Central Location : NA
Aruba Central Shared Token           : aruba123
Force-Provision                      : Disabled
HTTP Proxy Location                  : http.proxy.arubanetworks.com
```

Showing switch configuration download failure after receiving ZTP options:

```
switch# show ztp information
TFTP Server                          : 10.0.0.2
Image File                           : TL_10_02_0001.swi
Configuration File                   : config_file
ZTP Status                           : Failed - Unable to download configuration
Aruba Central Location               : secure.arubanetworks.com
Alternative Aruba Central Location : NA
Aruba Central Shared Token           : aruba123
Force-Provision                      : Disabled
HTTP Proxy Location                  : http.proxy.arubanetworks.com
```

Showing switch failure to update start-up configuration after downloading configuration received from ZTP options:

```
switch# show ztp information
TFTP Server                         : 10.0.0.2
Image File                          : TL_10_02_0001.swi
Configuration File                  : config_file
ZTP Status                          : Failed - Could not copy to start-up configuration
Aruba Central Location              : secure.arubanetworks.com
Alternative Aruba Central Location: NA
Aruba Central Shared Token          : aruba123
Force-Provision                     : Disabled
HTTP Proxy Location                 : http.proxy.arubanetworks.com
```

In the following example, the ZTP operation succeeded, and both an image file and a configuration file were provided.

```
switch# show ztp information
TFTP Server            : 20.1.1.4
Image File             : PL_10_06_0001BT.swi
Configuration File     : bristol_maxlimit
Status                 : Success
Force-Provision        : Disabled
switch#
```

In the following example, the ZTP option succeeded. A configuration file was not provided, but an image file was provided.

```
switch# show ztp information
TFTP Server             : 20.1.1.4
Image File              : NA
Configuration File      : bristol_maxlimit
Status                  : Success
Force-Provision         : Disabled
switch#
```

In the following example, the ZTP operation failed because the TFTP server was unreachable.

```
switch# show ztp information
TFTP Server             : 20.1.1.4
Image File              : PL_10_06_0001BT.swi
Configuration File      : bristol_maxlimit
Status                  : Failed - TFTP server unreachable
Force-Provision         : Disabled
switch#
```

In the following example, the ZTP operation was stopped because the switch did not receive any options from the DHCP server for ZTP within 1 minute of receiving the IP address from the server.

```
switch## show ztp information
TFTP Server             : NA
Image File              : NA
Configuration File      : NA
Status                  : Failed - Timed out while waiting to receive ZTP options
Force-Provision         : Disabled
switch#
```

In the following example, the ZTP operation was stopped because the switch was booted from a configuration that was not the factory default configuration.

```
switch# show ztp information
TFTP Server             : 20.1.1.4
Image File              : PL_10_06_0001BT.swi
Configuration File      : bristol_maxlimit
Status                  : Failed - Custom startup configuration detected
Force-Provision         : Disabled
```

In the following example, the switch received the image file and the TFTP-sever and config file from the DHCP server for ZTP was successful:

```
switch# show ztp information
TFTP Server                        : 10.0.0.2
Image File                         : TL_10_02_0001.swi
Configuration File                 : ztp.cfg
ZTP Status                         : Success
Aruba Central Location             : NA
Alternative Aruba Central Location: NA
Aruba Central Shared Token         : NA
Force-Provision                    : Disabled
HTTP Proxy Location                : NA
```

In the following example, the switch received the image file and the TFTP-sever and config file from the DHCP server entered the failed state as teh TFTP server was not reachable:

```
switch# show ztp information
TFTP Server                          : 10.0.0.2
Image File                           : TL_10_02_0001.swi
Configuration File                   : ztp.cfg
ZTP Status                           : Failed - TFTP server unreachable
Aruba Central Location               : NA
Alternative Aruba Central Location: NA
Aruba Central Shared Token           : NA
Force-Provision                      : Disabled
HTTP Proxy Location                  : NA
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

ztp force provision

```
ztp force-provision
no ztp force-provision
```

Description

Starts on-demand ZTP.

Usage

DHCP options received are processed independent of he current state of configuration on the switch. Previous ZTP TFTP Server, Image File, Configuration File, Aruba Central Location, and HTTP Proxy location options are cleared and the switch sends a DHCP request.

Examples

In the following example, force-provision is enabled.

```
switch# configure terminal
switch(config)# ztp force-provision
```

In the following example, force-provision status is checked while enabled.

```
switch# show ztp information
TFTP Server                       : 10.0.0.2
Image File                        : TL_10_02_0001.swi
Configuration File                : ztp.cfg
Status                            : Success
Aruba Central Location            : NA
```

```
Aruba Central Shared Token      : NA
Force-Provision                 : Enabled
HTTP Proxy Location             : NA
```

In the following example, force-provision is disabled.

```
switch# configure terminal
switch(config)# no ztp force-provision
```

In the following example, force-provision status is checked while disabled.

```
switch# show ztp information
TFTP Server                     : 10.0.0.2
Image File                      : TL_10_02_0001.swi
Configuration File              : ztp.cfg
Status                          : Success
Aruba Central Location          : NA
Aruba Central Shared Token      : NA
Force-Provision                 : Disabled
HTTP Proxy Location             : NA
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

Chapter 15
vSphere agent

The HPE Aruba Networking CX vSphere agent allows administrators to trace how Virtual Machines are connected to the physical switches. The ability to visualize how a Virtual Machine connects to a CX port helps to resolve connectivity issues. This functionality helps operators in several troubleshooting scenarios, such as diagnosing a connectivity issue, or ensuring adequate distribution of VMs across the physical infrastructure.

You can install the vSphere agent on one or many of the switches in the fabric. You can perform the following tasks, from any switch where the agent is installed:

1. Run the show VMs command to list all the VMs managed by the vCenter Appliance integrated with the agent. This functionality also helps to view VMs connections to the ports on other switches on the fabric.
2. Search for a specific VM by name. To find the connection between the VM and a switch port, attach the switch to at least one host managed by the integrated vCenter.

For more information about cloud capabilities of vSphere agent, refer to *New Central Online Help*.

> The **show neighbors** command examines the LLDP data that is advertised from any switch to a distributed virtual switch and provides connection details for third-party switches.

Installation of vSphere agent

To install a container, perform the following steps:

1. Download the vSphere agent from ASP and select the arm vs. x86_64 architecture for the switch platform. Refer to the list of supported switches mentioned earlier to select the corresponding architecture.
2. Host the agent on a local network that can be accessed using the HTTP protocol. Usage of the https protocol is a limitation of the container framework. The directory where the images of the agents are placed will be set in step 3.d. The HTTP server must be reachable on the mgmt vrf - or otherwise configured vrf. Any simple HTTP server helps to accomplish this step.

Configure the CLI using the following commands:

1. To enter the config mode, run the container vsphere command.
2. To set the management vrf as the communication path for the container, run the vrf mgmt command. The vrf is mgmt by default, or you can specify any other non-default vrf configured by the user. mgmt is the only supported vrf currently.
3. To limit memory usage of the container, run the restrict memory command. Set 1GB for vSphere instance.
4. To set the installation path to the path accessible by the HTTP protocol noted in Step 2, run the image location command.

5. To pass attributes to be configured for the container from CX CLI, run the env command. Use the following format to pass the attributes to establish the configuration instance:
   a. VSPHERE_VCENTER_N, where N takes value in the range 1–4.
   b. The encrypted plaintext command with arguments in the form of vSphere credentials, for example: user@<domain>:password@<fqdn of vcenter>

      The following example displays a valid running config:

```
container vsphere
image-location http://X.X.X.X:8000//vsphere-agent.img vrf mgmt restrict
cpu 20
restrict memory 512 vrf mgmt
env VSPHERE_VCENTER_1 encrypted plaintext
USER@vsphere.local:PASSWORD@X.X.X.X
```

Show commands

There are four top-level show commands.

- vms - shows virtual machine connection information.
- connection-status - shows information about configured vSphere vCenters.
- neighbors - shows learned neighbor information (Note: This information is learned from LLDP on vCenter distributed vswitches, not the switch).
- help - shows agent version, description, and configurable environment variables.

show vms

The following show command describes the status of the connection to vSphere, and show the VMs that are attached to switches:

At least one command parameter must be used if the parameters are available.

For example, with the show vms command, we force use of the brief option so the user is aware of all available options.

If not specified, all commands default to brief.

```
igor-sw-02# container vsphere exec show vms brief Hypervisor VM Name Physical
Switch Connections Interface

laser-02-esxi.lab.plexxi.com vm4 laser-sw-01 (b8:d4:e7:d9:af:00) 1/1/3 vm2 laser-
sw-02 (b8:d4:e7:da:40:00) 1/1/4
laser-sw-02 (b8:d4:e7:da:40:00) 1/1/3 vm5 laser-sw-02 (b8:d4:e7:da:40:00) 1/1/4
laser-sw-02 (b8:d4:e7:da:40:00) 1/1/3 laser-sw-01 (b8:d4:e7:d9:af:00) 1/1/3
laser-01-esxi.lab.plexxi.com vm1 laser-sw-01 (b8:d4:e7:d9:af:00) 1/1/1 vm3 laser-
sw-01 (b8:d4:e7:d9:af:00) 1/1/2
laser-sw-02 (b8:d4:e7:da:40:00) 1/1/1
igor-sw-02# container vsphere exec show vms vm-name vm4 Hypervisor VM Name
Physical Switch Connections Interface

laser-02-esxi.lab.plexxi.com vm4 laser-sw-01 (b8:d4:e7:d9:af:00) 1/1/3 igor-sw-02#
container vsphere exec show vms vm-name vm4 detailed VM Name : vm4
Hypervisor : laser-02-esxi.lab.plexxi.com Vendor : vmware
Hostname : vm4
OS : CentOS 4/5 or later (64-bit) VM status : on
Virtual NIC : Network adapter 2 IP Address : 172.20.12.204
```

```
MAC Address : 00:50:56:89:3c:bf Virtual Switch : dvs4
Port Group : dpg4 Vlan : 0
Physical Switch Connections Interface : 1/1/3
Chassis ID : b8:d4:e7:d9:af:00 Switch Name : laser-sw-01
Switch Description : Aruba JL635A GL.10.12.0001G Management Address :
172.20.11.253

Container exec commands are top-level, and can be run in any CLI context.

igor-sw-02# container vsphere exec show connection-status brief Hypervisor Last
Sync Connection State

laser-vcsa.lab.plexxi.com 2023-03-17 23:30:03 connected igor-sw-02# config
igor-sw-02(config)# container vsphere exec show connection-status brief Hypervisor
Last Sync Connection State

laser-vcsa.lab.plexxi.com 2023-03-17 23:30:07 connected igor-sw-02(config)#
interface 1/1/1
igor-sw-02(config-if)# container vsphere exec show connection-status brief
Hypervisor Last Sync Connection State

laser-vcsa.lab.plexxi.com 2023-03-17 23:30:15 connected
```

show vms

This command displays the connectivity status for the vSphere agent.

The connection timestamp is updated whenever a successful connection is made with the vSphere agent. The sync timestamp is updated whenever a full sync, event poll, or event process occurs.

```
igor-sw-02(config-container-vsphere)# container vsphere exec show connection-
status brief Hypervisor Last Sync Connection State


laser-vcsa.lab.plexxi.com 2023-03-18 02:36:06 connected
igor-sw-02(config-container-vsphere)# container vsphere exec show connection-
status detailed Hypervisor : laser-vcsa.lab.plexxi.com
Connection State : connected Last Sync : 2023-03-18 02:35:08
Last Connection : 2023-03-18 02:35:09
igor-sw-02(config-container-vsphere)# container vsphere exec show connection-
status detailed Hypervisor : laser-vcsa.lab.plexxi.com
Connection State : disconnected Last Sync : 2023-03-18 02:04:05
Last Connection : 2023-03-18 02:09:30
Connection Failure Reason : Could Not Resolve Hostname
```

show neighbors

show neighbors command displays the virtual network adapter neighbor information.

Neighbors are pulled from distributed virtual switches of the vSphere and cached locally. Neighbors are used to map VMs to physical switch ports.

If you do not see a neighbor on the port, you will not see any VM connectivity on that port either. detailed and brief options are available. Filtering can be done on switch-name and/or interface.

```
igor-sw-02(config-container-vsphere)# container vsphere exec show neighbors brief
Chassis ID Switch Name Interface

b8:d4:e7:da:40:00 laser-sw-02 1/1/1 1/1/2
b8:d4:e7:d9:af:00 laser-sw-01 1/1/1 1/1/2
```

```
igor-sw-02(config-container-vsphere)# container vsphere exec show neighbors
detailed Chassis ID : b8:d4:e7:da:40:00
Switch Name : laser-sw-02
Switch Description : Aruba JL635A GL.10.10.1000 Management Address : 192.168.200.1
Interface : 1/1/1 Advertisement Type : lldp TTL : 102
Interface : 1/1/2 Advertisement Type : lldp TTL : 102
Chassis ID : b8:d4:e7:d9:af:00 Switch Name : laser-sw-01
Switch Description : Aruba JL635A GL.10.10.1000 Management Address : 192.168.200.0
Interface : 1/1/1 Advertisement Type : lldp TTL : 102
Interface : 1/1/2 Advertisement Type : lldp TTL : 102
```

Supported switches

The following switch platforms are validated for data center. The agent itself has no context about the switch that it is installed on, except that the chipset architecture must be either x86_64 or arm:

- 8100 (x86_64)
- 8320 (x86_64)
- 8325 (x86_64)
- 8360 (arm)
- 9300 (x86_64)
- 10000 (x86_64)

The HPE Aruba Networking CX vSphere agent is a containerized agent developed for data center applications. The plugin consumes memory and CPU that would otherwise be allocated to the switch. Thus, the targeted platforms have higher system resources. Those resources are capped to prevent its interference with performance of the switch.

The solution was tested and qualified on the platforms listed here. Other platforms may work at low scale, but they are not recommended outside of the labor test environments due to the resource requirements for vSphere.

Troubleshooting

Overview

In data center networking, the most advanced step in troubleshooting often involves directly accessing the closest physical device related to the issue at hand. This trouble shooting mechanism allows the diagnosis of problems with connected endpoints. While administrators can use management tools to avoid accessing the physical device, there are scenarios where the tools are not utilized. The lack of utilization is either due to their absence, do not have of familiarity, or some of the administrators prefer device interaction. To address this, it is important to embrace these variations and aim for better integration between device-based approaches and management tools, rather than viewing them as conflicting options.

The complexity of troubleshooting the devices is multiplied by the dynamically varying endpoints within the data center. To troubleshoot, admins might extensively research endpoint metadata before beginning the process. Customers use different methods to store inventory data about their endpoints. This method can result in searching for incorrect or outdated information, causing confusion.

Additionally, admins may struggle to identify the correct network device to start with or make mistakes in the process.

Understanding the reasons behind entering the "Troubleshooting of last resort" scenario is essential for developing more effective solutions. Often, this scenario arises either due to reported connectivity problems or when admins attempt to establish connectivity for a new device or service. Interestingly, the first step in this scenario is to resort to the last option is asking a series of questions about MAC addresses, IP addresses, switches, ports, and more. Users frequently do not have this information and must look it up from spreadsheets, paper records, or outdated inventory systems.

Troubleshooting Tools

Admins can use MAC tables, ARP tables, and ping and trace tools to troubleshoot the issues. You can run the show mac-address-table command to troubleshoot:

```
cedar-sw-01# show mac-address-table MAC age-time        : 300 seconds Number of
MAC addresses : 15

MAC Address    VLAN    Type    Port

f4:03:43:d3:be:b8      1       dynamic 1/1/4
f4:03:43:c0:e1:38      1       dynamic lag256
f4:03:43:c0:e1:30      1       dynamic lag2
00:50:56:5e:7a:69      1       dynamic lag1
00:50:56:5b:a0:40      1       dynamic lag1
04:90:81:00:08:a2      10      dynamic lag256
```

The MAC address table helps you to check if you have seen the MAC address of the endpoint in the last 5 minutes. If the user does not report a problem or share the MAC address, find out the MAC address used by the endpoint. However, if the device has many virtual network parts such as switches, it can be unclear which MAC it is using.

You can also run the show arp command to troubleshoot the issue.

```
show arp
cedar-sw-01# show arp all-vrfs

IPv4 Address   MAC     Port    Physical Port   State   VRF


10.10.3.23     00:50:56:ac:3d:c6 vlan30        lag1    reachable default
1.1.1.12       04:90:81:00:08:a2 vlan99        lag256  reachable default
1.1.1.1        04:90:81:00:08:a2 1/1/48        1/1/48  reachable default
```

For the following MAC address 04:90:81:00:08:a2, the output shows that there was an ARP request for this MAC address. The device with IPV4 address 1.1.1.1 is the VSX keepalive interface on the peer switch.

Endpoint metadata

Solving the issue involves getting the metadata right down to the switch. When users use specific commands, we can add extra details to the responses using metadata. For instance, if we know that the IP address 1.1.1.1 corresponds to a user VM named "Simon," we must include that information in the output.

```
show arp (decoration)
cedar-sw-01# show arp all-vrfs
```

```
IPv4 Address   MAC     Port    Physical Port   State   VRF


10.10.3.23     00:50:56:ac:3d:c6 vlan30        lag1    reachable default
1.1.1.12       04:90:81:00:08:a2 vlan99        lag256  reachable default
1.1.1.1 (simons-vm)    04:90:81:00:08:a2 1/1/48        1/1/48  reachable default
```

The previous example is generic, to show the value of extra meta-data on standard switch screens, but in the vSphere context we have to research further to get more useful results.

VMware Use Cases

If you focus only on physical devices, you must figure out the specific switch to which your endpoint is connected. Other switches in the network might not have complete data about the endpoint, so they might not provide any information. In this scenario, more advanced data and perspectives become valuable for our administrators. By using VMware vSphere data as a foundation for your metadata, you gain a broader range of information beyond individual switches. You acquire a comprehensive set of data about topology of the fabric and endpoints. Operators use this information without requiring them to know the precise location of the endpoints.

Following queries that users might perform to gather more information about the endpoint:

Where can I find this endpoint (whether it is a virtual machine, VMkernel, or host)?

You might have a differing set of data on the endpoint, depending on their situation, and they want to know where it resides based on the limited information they have.

Create this query with a flexible set of data:

- VMs name, DNS name (can be different from VM name), VMs IP (more than one IP is possible), VMs NIC name (VMs version, not ESXi), VMs MAC

  VMkernel looks like a VM in some ways, just a virtual interface used for some important VMware control functions such as vMotion
- ESXi name, DNS name (often the same as ESXi name), IP, vmnic, vmnic MAC

What does the user want in response to their query?

- Switch, LAG , or Interface endpoint last seen
- When it was last seen
- Previous location (can be useful for detecting loops), will move if vMotion or depending on vmware load balancing

What additional data is available for this endpoint?

There is additional metadata for vSphere on the endpoint that can be useful for the customer. For a complete integration and for full visualization and automation, this data is used and is applied to Switch CLI.

For VMs

- The basic inventory data on this VM
  - VMs name, DNS name (often different from VM name), VMs IP (they can have more than one), VMs NIC name (VMs version, not ESXi), VMs MAC
  - The VMs association with the Virtual network in ESXi/vsphere
    - Portgroup (VLAN, VLAN type), vSwitch, ESXi Host, ESXi Host NIC (name, IP, and MAC)

- For VMkernel
  - Basic inventory data on this VMkernel
    - Name, IP, MAC
  - The VMKs association with the Virtual network in ESXi or vsphere
    - Portgroup (VLAN, VLAN type), vSwitch, ESXi Host, ESXi Host NIC (name, IP, and MAC)
- For ESXi
  - The basic inventory data on this
    - ESXi name, DNS name, ESXi IP, ESXi Nics, and MACs
    - For each ESXi NIC, the association with vSwitch

What devices are connected to this port?

You can run show mac-table-address interface x/t/z command, to display active MACs.

The advantage of using VM metadata is that you can find out the location of the VM without active MACs. You can find out the portgroup or vSwitch location through LLDP from the host, and from the VM association you can find out the location of the switch port.

To this end, the show VMs interface x/t/z could respond with a mix of active and inactive VMs.

clear events

`clear events`

Description

Clears up event logs. Using the `show events` command will only display the logs generated after the `clear events` command.

Examples

Clearing all generated event logs:

```
switch# show events
---------------------------------------------------
show event logs
---------------------------------------------------
2018-10-14:06:57:53.534384|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource
utilization poll interval is changed to 27
2018-10-14:06:58:30.805504|lldpd|103|LOG_INFO|MSTR|1|Configured LLDP tx-timer to
36
2018-10-14:07:01:01.577564|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource
utilization poll interval is changed to 49

switch# clear events

switch# show events
---------------------------------------------------
show event logs
---------------------------------------------------
2018-10-14:07:03:05.637544|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource
utilization poll interval is changed to 34
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

console baud-rate

```
console baud-rate <SPEED>
no console baud-rate <SPEED>
```

Description

Sets the console serial port speed.

The no form of this command resets the console port speed to its default of 115200 bps.

Parameter	Description
<SPEED>	Selects the console port speed in bps, either `9600` or `115200`.

Usage

The speed change occurs immediately for the active console session. The console will be inaccessible until the client terminal settings are updated to match the console port speed that you set. After the command is executed you will be prompted to log in again.

Examples

Setting the console port speed to 9600 bps:

```
switch(config)# console baud-rate 9600
This command will configure the baud rate immediately for the active serial
console session. After the command is executed the user will be prompted to
re-login. The serial console will be inaccessible until the terminal client
settings are updated to match the baud rate of the switch.
Continue (y/n)? y
```

Resetting the console port to its default speed 115200 bps:

```
switch(config)# no console baud-rate
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

domain-name

```
domain-name <NAME>
no domain-name [<NAME>]
```

Description

Specifies the domain name of the switch.

The **no** form of this command sets the domain name to the default, which is no domain name.

Parameter	Description
<NAME>	Specifies the domain name to be assigned to the switch. The first character of the name must be a letter or a number. Length: 1 to 32 characters.

Examples

Setting and showing the domain name:

```
switch# show domain-name

switch# config
switch(config)# domain-name example.com
switch(config)# show domain-name
example.com
switch(config)#
```

Setting the domain name to the default value:

```
switch(config)# no domain-name
switch(config)# show domain-name

switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

front-panel-security factory-reset

```
front-panel-security factory-reset
no front-panel-security factory-reset
```

Description

Enable the **front-panel factory-reset** feature to allow a factory reset via the reset button on the front panel of the switch, without console access. Once the feature is enabled, this setting will be saved across reboots, until a factory-reset is performed. The **no** form of this command disables the front-panel factory-reset feature after it has been enabled. This feature is disabled by default.

Usage

If you enable front-panel factory-reset feature:

- pressing the reset button and releasing it before 5 seconds will reboot the system.
- pressing the reset button and releasing it between 5 to 20 seconds will start a hard reset of the system
- pressing the reset button and releasing it between 20 to 25 seconds will start a hard reset of the system, and the system will undergo the factory reset process.
- pressing the reset button for more than 25 seconds will not trigger any action.

If the front-panel factory-reset feature is left at its default disabled setting in the running configuration :

- pressing the reset button and releasing it before 5 seconds will reboot the system.
- pressing the reset button and releasing it between 5 to 20 seconds mark will start a hard reset of the system.
- pressing the reset button for more than 20 seconds will not trigger any action.

For more information about the LED indicators on this switch, refer to the Installation Guide for your switch.

Example

```
switch(config)# front-panel-security factory-reset
This command will enable front-panel factory reset capability, where user can
trigger factory-reset via reset button. This feature will remain enabled until
it is disabled, or a factory-reset is performed.

Continue (y/n)?
```

Command History

Release	Modification
10.13.1000	Command introduced

Command Information

Platforms	Command context	Authority
6000 6100	`config`	Administrators or local user group members with execution rights for this command.

hostname

```
hostname <HOSTNAME>
no hostname [<HOSTNAME>]
```

Description

Sets the host name of the switch.

The **no** form of this command sets the host name to the default value, which is `switch`.

Parameter	Description
<HOSTNAME>	Specifies the host name. The first character of the host name must be a letter or a number. Length: 1 to 32 characters. Default: `switch`

Examples

Setting and showing the host name:

```
switch# show hostname
switch
switch# config
switch(config)# hostname myswitch
myswitch(config)# show hostname
myswitch
```

Setting the host name to the default value:

```
myswitch(config)# no hostname
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

power consumption-average-period

`power consumption-average-period <PERIOD-IN-SECONDS>`

Description

Configures a time period for average power consumption in seconds.

Parameter	Description
<PERIOD-IN-SECONDS>	Specifies the period in seconds for average power consumed. Range: 60-3600. Default: 600

Example

Configuring a time period of 60 seconds for average power consumption:

```
switch(config)# power consumption-average-period 60
```

Command History

Release	Modification
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
4100i	`config`	Administrators or local user group members with execution rights for this command.

show boot-history

`show boot-history [all|{vsf member <1-10>}]`

Description

Shows boot history information. When no parameters are specified, shows the most recent information about the current boot operation, and the three previous boot operations for the switch. When the **all** parameter is specified, the output of this command shows the boot information for the active management module.

.

To view boot-history on a standby, the command must be sent on the conductor console.

Parameter	Description
`all`	Optional. Shows boot information for the active management module.
`vsf member <1-10>`	Optional. Display boot history for the specified VSF member

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

The output of this command includes the following information:

Parameter	Description
`Index`	The position of the boot in the history file. Range: 0 to 3.
`Boot ID`	A unique ID for the boot . A system-generated 128-bit string.

Parameter	Description
`Current Boot, up for <time>`	For the current boot, the **show boot-history** command shows the number of seconds the module has been running on the current software.
`<Timestamp>: boot reason`	For previous boot operations, the **show boot-history** command shows the time at which the operation occurred and the reason for the boot. The reason for the boot is one of the following values: ■ ***<DAEMON-NAME>* crash**: The daemon identified by *<DAEMON-NAME>* caused the module to boot. ■ **Kernel crash**: The operating system software associated with the module caused the module to boot. ■ **Uncontrolled reboot**: The reason for the reboot is not known. ■ **Reboot requested through database**: The reboot occurred because of a request made through the CLI or other API.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=================

Index : 2
Boot ID : c34a2c2499004a02bbeeff4992e1fdbd
Current Boot, up for 1 days 13 hrs 13 mins 27 secs

Index : 1
Boot ID : bfba9bc486304e57904ac717a0ccbdcd
02 Sep 23 02:55:33 : CPU request reset with 0x20201, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:55:33 : Switch boot count is 2

Index : 0
Boot ID : a88a71b7ca9a4574af7e3b811ddfdc7e
02 Sep 23 02:49:26 : Reboot requested by user, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:50:02 : Switch boot count is 1

Index : 3
Boot ID : f00ba10c8c44457f83fee303d014a89a
25 Aug 23 10:27:42 :  Power on reset with 0x1, Version: FL.10.14.0000-1465-
g9df95249d06b0~dirty
25 Aug 23 10:28:18 :  Switch boot count is 3
25 Aug 23 10:29:02 :  Primary overtemperature fault detected with 0x2 in PSU 1/1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show capacities

`show capacities <FEATURE>`

Description

Shows system capacities and their values for all features or a specific feature.

Parameter	Description
<FEATURE>	Specifies a feature. For example, **aaa**.

Usage

Capacities are expressed in user-understandable terms. Thus they may not map to a specific hardware or software resource or component. They are not intended to define a feature exhaustively.

Examples

Showing classifier-related capacities on the 6100:

```
switch# show capacities classifier

System Capacities: Filter Classifier
Capacities Name                                                            Value
---------------------------------------------------------------------------------
--
Maximum number of Access Control Entries configurable in a system
4096
Maximum number of Access Control Lists configurable in a system
512
Maximum number of class entries configurable in a system
4096
Maximum number of classes configurable in a system
512
Maximum number of entries in an Access Control List
1024
Maximum number of entries in a class
1024
Maximum number of entries in a policy
64
Maximum number of classifier policies configurable in a system
512
Maximum number of policy entries configurable in a system
4096
```

Showing all available capacities on the 6100:

```
switch# show capacities

System Capacities:
Capacities Name
Value
----------------------------------------------------------------------------------
--------
Maximum number of Access Control Entries configurable in a system
 4096
Maximum number of Access Control Lists configurable in a system
  512
Maximum number of class entries configurable in a system
 4096
Maximum number of classes configurable in a system
  512
Maximum number of entries in an Access Control List
 1024
Maximum number of entries in a class
 1024
Maximum number of entries in a policy
   64
Maximum number of classifier policies configurable in a system
  512
Maximum number of policy entries configurable in a system
 4096
Maximum number of clients supported for tracking the IP address in the system
  128
Maximum number of dynamic VLANs that can be allowed using MVRP
  256
Maximum number of nexthops per IP ECMP group
    1
Maximum number of IP neighbors (IPv4+IPv6) supported in the system
 1024
Maximum number of IPv4 neighbors(# of ARP entries) supported in the system
 1024
Maximum number of IPv6 neighbors(# of ND entries) supported in the system
  512
Maximum number of L2 MAC addresses supported in the system
 8192
Maximum number of L3 Groups for IP Tunnels and ECMP Groups
    1
Maximum number of L3 Destinations for Routes, Nexthops in Tunnels and ECMP groups
 1024
Maximum number of configurable LAG ports
    8
Maximum number of members supported by a LAG port
    8
Maximum number of VLANs across ports allowed in loop-protect
 3328
Maximum number of IGMP/MLD groups supported
  512
Maximum number of IGMP/MLD snooping groups supported
  512
Maximum number of Mirror Sessions configurable in a system
    4
Maximum number of enabled Mirror Sessions in a system
    4
Maximum number of mstp instances configurable in a system
   16
Maximum number of Clients that can be authenticated on a port
   32
Maximum number of Device Profiles allowed to be created on the system
```

```
    8
Maximum number of Port Access Roles allowed to be created on the system
   32
Maximum number of MAC Address that can be authorized on a port
   32
Maximum number of Port Access Role VLAN IDs allowed to be created on the system
   50
Maximum number of Port Access Role VLAN names allowed to be created on the system
   50
Maximum number of RBAC rules per user group
 1024
Maximum number of RPVST VLANs configurable on the system
   16
Maximum number of RPVST VPORTs supported in a system
  512
Maximum number of SVIs supported in the system
   16
Maximum number of routes (IPv4+IPv6) on the system
  512
Maximum number of IPv4 routes on the system
  512
Maximum number of IPv6 routes on the system
  512
Maximum number of VLANs supported in the system
  512

System Capacities:
Capacities Name
Value
----------------------------------------------------------------------------------
--------
Maximum number of Access Control Entries configurable in a system
 4096
Maximum number of Access Control Lists configurable in a system
  512
Maximum number of class entries configurable in a system
 4096
Maximum number of classes configurable in a system
  512
Maximum number of entries in an Access Control List
 1024
Maximum number of entries in a class
 1024
Maximum number of entries in a policy
   64
Maximum number of classifier policies configurable in a system
  512
Maximum number of policy entries configurable in a system
 4096
Maximum number of clients supported for tracking the IP address in the system
  128
Maximum number of dynamic VLANs that can be allowed using MVRP
  256
Maximum number of nexthops per IP ECMP group
    1
Maximum number of IP neighbors (IPv4+IPv6) supported in the system
 1024
Maximum number of IPv4 neighbors(# of ARP entries) supported in the system
 1024
Maximum number of IPv6 neighbors(# of ND entries) supported in the system
  512
Maximum number of L2 MAC addresses supported in the system
```

```
 8192
Maximum number of L3 Groups for IP Tunnels and ECMP Groups
    1
Maximum number of L3 Destinations for Routes, Nexthops in Tunnels and ECMP groups
 1024
Maximum number of configurable LAG ports
    8
Maximum number of members supported by a LAG port
    8
Maximum number of VLANs across ports allowed in loop-protect
 3328
Maximum number of IGMP/MLD groups supported
  512
Maximum number of IGMP/MLD snooping groups supported
  512
Maximum number of Mirror Sessions configurable in a system
    4
Maximum number of enabled Mirror Sessions in a system
    4
Maximum number of mstp instances configurable in a system
   16
Maximum number of Clients that can be authenticated on a port
   32
Maximum number of Device Profiles allowed to be created on the system
    8
Maximum number of Port Access Roles allowed to be created on the system
   32
Maximum number of MAC Address that can be authorized on a port
   32
Maximum number of Port Access Role VLAN IDs allowed to be created on the system
   50
Maximum number of Port Access Role VLAN names allowed to be created on the system
   50
Maximum number of RBAC rules per user group
 1024
Maximum number of RPVST VLANs configurable on the system
   16
Maximum number of RPVST VPORTs supported in a system
  512
Maximum number of SVIs supported in the system
   16
Maximum number of routes (IPv4+IPv6) on the system
  512
Maximum number of IPv4 routes on the system
  512
Maximum number of IPv6 routes on the system
  512
Maximum number of VLANs supported in the system
  512
```

```
switch# show capacities

System Capacities:
Capacities Name
Value
-----------------------------------------------------------------------------------
--------
Maximum number of Access Control Entries configurable in a system
 4096
Maximum number of Access Control Lists configurable in a system
  512
```

```
Maximum number of class entries configurable in a system
 4096
Maximum number of classes configurable in a system
  512
Maximum number of entries in an Access Control List
 1024
Maximum number of entries in a class
 1024
Maximum number of entries in a policy
   64
Maximum number of classifier policies configurable in a system
  512
Maximum number of policy entries configurable in a system
 4096
Maximum number of clients supported for tracking the IP address in the system
  128
Maximum number of dynamic VLANs that can be allowed using MVRP
  256
Maximum number of nexthops per IP ECMP group
    1
Maximum number of IP neighbors (IPv4+IPv6) supported in the system
 1024
Maximum number of IPv4 neighbors(# of ARP entries) supported in the system
 1024
Maximum number of IPv6 neighbors(# of ND entries) supported in the system
  512
Maximum number of L2 MAC addresses supported in the system
 8192
Maximum number of L3 Groups for IP Tunnels and ECMP Groups
    1
Maximum number of L3 Destinations for Routes, Nexthops in Tunnels and ECMP groups
 1024
Maximum number of configurable LAG ports
    8
Maximum number of members supported by a LAG port
    8
Maximum number of VLANs across ports allowed in loop-protect
 3328
Maximum number of IGMP/MLD groups supported
  512
Maximum number of IGMP/MLD snooping groups supported
  512
Maximum number of Mirror Sessions configurable in a system
    4
Maximum number of enabled Mirror Sessions in a system
    4
Maximum number of mstp instances configurable in a system
   16
Maximum number of Clients that can be authenticated on a port
   32
Maximum number of Device Profiles allowed to be created on the system
    8
Maximum number of Port Access Roles allowed to be created on the system
   32
Maximum number of MAC Address that can be authorized on a port
   32
Maximum number of Port Access Role VLAN IDs allowed to be created on the system
   50
Maximum number of Port Access Role VLAN names allowed to be created on the system
   50
Maximum number of RBAC rules per user group
 1024
```

```
Maximum number of RPVST VLANs configurable on the system
   16
Maximum number of RPVST VPORTs supported in a system
  512
Maximum number of SVIs supported in the system
   16
Maximum number of routes (IPv4+IPv6) on the system
  512
Maximum number of IPv4 routes on the system
  512
Maximum number of IPv6 routes on the system
  512
Maximum number of VLANs supported in the system
  512

System Capacities:
Capacities Name
Value
----------------------------------------------------------------------------------
--------
Maximum number of Access Control Entries configurable in a system
 4096
Maximum number of Access Control Lists configurable in a system
  512
Maximum number of class entries configurable in a system
 4096
Maximum number of classes configurable in a system
  512
Maximum number of entries in an Access Control List
 1024
Maximum number of entries in a class
 1024
Maximum number of entries in a policy
   64
Maximum number of classifier policies configurable in a system
  512
Maximum number of policy entries configurable in a system
 4096
Maximum number of clients supported for tracking the IP address in the system
  128
Maximum number of dynamic VLANs that can be allowed using MVRP
  256
Maximum number of nexthops per IP ECMP group
    1
Maximum number of IP neighbors (IPv4+IPv6) supported in the system
 1024
Maximum number of IPv4 neighbors(# of ARP entries) supported in the system
 1024
Maximum number of IPv6 neighbors(# of ND entries) supported in the system
  512
Maximum number of L2 MAC addresses supported in the system
 8192
Maximum number of L3 Groups for IP Tunnels and ECMP Groups
    1
Maximum number of L3 Destinations for Routes, Nexthops in Tunnels and ECMP groups
 1024
Maximum number of configurable LAG ports
    8
Maximum number of members supported by a LAG port
    8
Maximum number of VLANs across ports allowed in loop-protect
 3328
```

```
Maximum number of IGMP/MLD groups supported
  512
Maximum number of IGMP/MLD snooping groups supported
  512
Maximum number of Mirror Sessions configurable in a system
    4
Maximum number of enabled Mirror Sessions in a system
    4
Maximum number of mstp instances configurable in a system
   16
Maximum number of Clients that can be authenticated on a port
   32
Maximum number of Device Profiles allowed to be created on the system
    8
Maximum number of Port Access Roles allowed to be created on the system
   32
Maximum number of MAC Address that can be authorized on a port
   32
Maximum number of Port Access Role VLAN IDs allowed to be created on the system
   50
Maximum number of Port Access Role VLAN names allowed to be created on the system
   50
Maximum number of RBAC rules per user group
 1024
Maximum number of RPVST VLANs configurable on the system
   16
Maximum number of RPVST VPORTs supported in a system
  512
Maximum number of SVIs supported in the system
   16
Maximum number of routes (IPv4+IPv6) on the system
  512
Maximum number of IPv4 routes on the system
  512
Maximum number of IPv6 routes on the system
  512
Maximum number of VLANs supported in the system
  512
```

Showing all available capacities for mirroring:

```
switch# show capacities mirroring

System Capacities: Filter Mirroring
Capacities Name                                                             Value
---------------------------------------------------------------------------------
-
Maximum number of Mirror Sessions configurable in a system
4
Maximum number of enabled Mirror Sessions in a system
4
```

Showing all available capacities for MSTP:

```
switch# show capacities mstp

System Capacities: Filter MSTP
Capacities Name                                                             Value
---------------------------------------------------------------------------------
```

```
-
Maximum number of mstp instances configurable in a system
64
```

Showing all available capacities for VLAN count:

```
switch# show capacities vlan-count

System Capacities: Filter VLAN Count
Capacities Name                                                              Value
----------------------------------------------------------------------------------
-
Maximum number of VLANs supported in the system
4094
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show capacities-status

`show capacities-status <FEATURE>`

Description

Shows system capacities status and their values for all features or a specific feature.

Parameter	Description
<FEATURE>	Specifies the feature, for example **aaa** or **vrrp** for which to display capacities, values, and status. Required.

Examples

Showing the system capacities status for all features:

```
switch# show capacities-status

System Capacities Status
Capacities Status Name                                                   Value
Maximum
----------------------------------------------------------------------------------
----
Number of Access Control Entries currently configured                     0     4096
```

```
Number of Access Control Lists currently configured                    0     512
Number of class entries currently configured                           0    4096
Number of classes currently configured                                 0     512
Number of policies currently configured                                0     512
Number of policy entries currently configured                          0    4096
Number of dynamic VLANs currently learnt using MVRP                     0     256
Number of IP neighbor (IPv4+IPv6) entries                              1    1024
Number of IPv4 neighbor(ARP) entries                                   1    1024
Number of IPv6 neighbor(ND) entries                                    0     512
Number of L3 Groups for IP Tunnels and ECMP Groups currently configured  0       1
Number of L3 Destinations for Routes, Nexthops in ECMP groups and
   Tunnels currently configured                                        0    1024
Number of Mirror Sessions currently configured                         0       4
Number of Mirror Sessions currently enabled                            0       4
Number of mstp instances currently configured                          0      16
Number of RPVST VLANs currently configured                             0      16
Number of routes (IPv4+IPv6) currently configured                      1     512
Number of IPv4 routes currently configured                             1     512
Number of IPv6 routes currently configured                             0     512
Number of VLANs currently configured                                   1     512
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show console

`show console`

Description

Shows the serial console port current speed.

Examples

Showing the console port current speed:

```
switch# show console
Baud Rate: 9600
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show core-dump

`show core-dump all`

Description

Shows core dump information about the specified module. When no parameters are specified, shows only the core dumps generated in the current boot of the management module. When the **all** parameter is specified, shows all available core dumps.

Parameter	Description
`all`	Shows all available core dumps.

Usage

When no parameters are specified, the `show core-dump` command shows only the core dumps generated in the current boot of the management module. You can use this command to determine when any crashes are occurring in the current boot.

If no core dumps have occurred, the following message is displayed: **No core dumps are present**

To show core dump information for the standby management module, you must use the **standby** command to switch to the standby management module and then execute the **show core-dump** command.

In the output, the meaning of the information is the following:

`Daemon Name`
Identifies name of the daemon for which there is dump information.

`Instance ID`
Identifies the specific instance of the daemon shown in the **Daemon Name** column.

`Present`
Indicates the status of the core dump:

`Yes`
The core dump has completed and available for copying.

`In Progress`
Core dump generation is in progress. Do not attempt to copy this core dump.

`Timestamp`
Indicates the time the daemon crash occurred. The time is the local time using the time zone configured on the switch.

`Build ID`
Identifies additional information about the software image associated with the daemon.

Examples

Showing core dump information for the current boot of the active management module only:

```
switch# show core-dump
===================================================================================
Daemon Name      | Instance ID | Present     | Timestamp                | Build ID
===================================================================================
```

```
hpe-fand          1399          Yes         2017-08-04 19:05:34      1246d2a
hpe-sysmond       957           Yes         2017-08-04 19:05:29      1246d2a
==================================================================================
Total number of core dumps : 2
==================================================================================

==================================================================================
Daemon Name      | Instance ID | Present     | Timestamp               | Build ID
==================================================================================
hpe-fand          1399          Yes         2017-08-04 19:05:34      1246d2a
hpe-sysmond       957           Yes         2017-08-04 19:05:29      1246d2a
==================================================================================
Total number of core dumps : 2
==================================================================================
```

Showing all core dumps:

```
switch# show core-dump all
=============================================================================
Management Module core-dumps
=============================================================================
Daemon Name      | Instance ID | Present     | Timestamp             | Build ID
=============================================================================
hpe-sysmond       513           Yes         2017-07-31 13:58:05    e70f101
hpe-tempd         1048          Yes         2017-08-13 13:31:53    e70f101
hpe-tempd         1052          Yes         2017-08-13 13:41:44    e70f101

Line Module core-dumps
=============================================================================
Line Module : 1/1
=============================================================================
dune_agent_0      18958         Yes         2017-08-12 11:50:17    e70f101
dune_agent_0      18842         Yes         2017-08-12 11:50:09    e70f101
=============================================================================
Total number of core dumps : 5
=============================================================================
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show domain-name

`show domain-name`

Description

Shows the current domain name.

Usage

If there is no domain name configured, the CLI displays a blank line.

Example

Setting and showing the domain name:

```
switch# show domain-name

switch# config
switch(config)# domain-name example.com
switch(config)# show domain-name
example.com
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show environment fan

`show environment fan` is not available for JL679A.

```
show environment fan
```

Description

Shows the status information for all fans and fan trays (if present) in the system.

Usage

For fan trays, **Status** is one of the following values:

- ready:The fan tray is operating normally.
- fault:The fan tray is in a fault event. The status of the fan tray does not indicate the status of fans.
- empty:The fan tray is not installed in the system.

For fans:

```
 Speed: Indicates the relative speed of the fan based on the nominal speed range of the
fan.
   N/A:This value is not applicable to the 6000 or 6100.
Direction: The direction of airflow through the fan. Values are:
   left-to-right:Air flows from the left of the system to the right of the system.
   N/A:The fan is not installed.
Status: Fan status. Values are:
   uninitialized:The fan has not completed initialization.
```

```
ok: The fan is operating normally.
fault: The fan is in a fault state.
empty: The fan is not installed.
```

Examples

Showing output for systems with fan trays for 6100 switch series:

```
switch# show environment fan
Fan information
---------------------------------------------------------------------------
Mbr/Fan       Product  Serial Number  Speed   Direction      Status  RPM
              Name
---------------------------------------------------------------------------
1/1           N/A      N/A            N/A     left-to-right  ok      N/A
```

```
switch# show environment fan

Fan information
---------------------------------------------------------------
Fan   Serial Number  Speed   Direction      Status        RPM
---------------------------------------------------------------
1     SGXXXXXXXXXX   slow    front-to-back  ok            6000
2     SGXXXXXXXXXX   normal  front-to-back  ok            8000
3     SGXXXXXXXXXX   medium  front-to-back  ok            11000
4     SGXXXXXXXXXX   fast    front-to-back  ok            14000
5     SGXXXXXXXXXX   max     front-to-back  fault         16500
6     N/A            N/A     N/A            empty
...
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show environment led

`show environment led`

Description

Shows state and status information for all the configurable LEDs in the system.

Example

Showing state and status for LED:

```
switch# show environment led
Mbr/Name        State     Status
-------------------------------
1/locator       off       ok
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show environment power-consumption

`show environment power-consumption <DETAIL>`

Description

Displays power consumption information.

Parameter	Description
<DETAIL>	Displays detailed power consumption information.
<MEMBER-ID>	For VSF supported platforms only. Displays the power consumption information for the specified VSF member. Range: 1-10.

Usage

Power consumed values are updated every minute. The total power consumed is the total power used in a chasis. The power consumed average is calculated from the total power consumed as a running average over a period of time. The average period has a default of 10 minutes. The period can be configured using **power-consumption-average-period.**

The following information is provided for a summary of power consumption:

- line module: power used by line module
- management module: power used by management module
- fabric module: power used by fabric module
- chassis module: power used by chassis module
- fan module: power used by fan module
- power total: total power consumption
- average power: average for total power consumption that is calculated over a given period
- average period: time to calculate power average

Example

Showing the power consumption for a switch with a single line card:

```
switch# show environment power-consumption

Power Consumption Averaging Period: 60 seconds

Name     Description          Instantaneous Power (W)      Average Power (W)
-----------------------------------------------------------------------------
1        8360-32YAC Switch                    300.00                 311.50
```

Showing the power consumption for a VSF stack:

```
switch# show environment power-consumption

Power Consumption Averaging Period: 60 seconds

Name        Description              Instantaneous Power (W)    Average Power (W)
----------------------------------------------------------------------------------
1           6200M 24G 24G 4SPF+ SW                    300.00               311.50
2           6200M 24G 24G 4SPF+ SW                    280.00               275.50
```

Showing the power consumption for a switch with multiple line cards, in brief:

```
switch# show environment power-consumption

Power Consumption Averaging Period: 60 seconds

Name     Description        Instantaneous Power (W)       Average Power (W)
----------------------------------------------------------------------------
1        6410 v2 Chassis                    1300.00                 1311.50
```

Showing the power consumption for a switch with multiple line cards, in detail:

```
switch# show environment power-consumption detail

Power Consumption Averaging Period: 60 seconds

Name  Module Type      Instantaneous Power (W)       Average Power (W)
------------------------------------------------------------------------
1     chassis total                    1000.00                 1002.50
1/1     fabric                           40.00
1/1     line                            120.00
1/2     line                            350.00
1/1     management                       20.00
        other                           470.00
```

Showing the power consumption for 4 member stack, in detail:

```
switch# show environment power-consumption detail

Power Consumption Averaging Period: 60 seconds

Name  Module Type      Instantaneous Power (W)       Average Power (W)
------------------------------------------------------------------------
```

```
1     chassis total                      300.00                   302.50
2     chassis total                      320.00                   319.50
3     chassis total                      280.00                   282.50
4     chassis total                      310.00                   311.50

Total power consumption 1210.00
```

Showing the power consumption for VSF member 2:

```
switch# show environment power-consumption member 2

Power Consumption Averaging Period: 60 seconds

Name  Module Type        Instantaneous Power (W)       Average Power (W)
--------------------------------------------------------------------------
2     6200M 24G 4SFP+ SW                  280.00                   275.00
```

Showing the power consumption for VSF invalid member:

```
switch# show environment power-consumption member 5
Member 5 is not present
```

Command History

Release	Modification
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
4100i	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show environment power-supply

`show environment power-supply`

Description

Shows status information about all power supplies in the switch.

Usage

The following information is provided for each power supply:

Parameter	Description
`Mbr/PSU`	Shows the member and slot number of the power supply.

Parameter	Description
`Product Number`	Shows the product number of the power supply.
`Serial Number`	Shows the serial number of the power supply, which uniquely identifies the power supply.
`PSU Status`	The status of the power supply. Values are: ■ **OK**:Power supply is operating normally. ■ **OK***: Power supply is operating normally, but it is the only power supply in the chassis. One power supply is not sufficient to supply full power to the switch. When this value is shown, the output of the command also shows a message at the end of the displayed data. ■ **Absent**: No power supply is installed in the specified slot. ■ **Input fault**: The power supply has a fault condition on its input. ■ **Output fault**: The power supply has a fault condition on its output. ■ **Warning**: The power supply is not operating normally. ■ **Wattage Maximum**: Shows the maximum amount of wattage that the power supply can provide.

Example

Showing the output when only one power supply is installed in the switch:

```
switch# show environment power-supply
         Product  Serial                PSU           Wattage
Mbr/PSU  Number   Number                Status        Maximum
--------------------------------------------------------------
1/1      N/A      N/A                   OK            500
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show environment temperature

`show environment temperature [detail]`

Description

Shows the temperature information from sensors in the switch that affect fan control.

Parameter	Description
`detail`	Shows detailed information from each temperature sensor.

Usage

Temperatures are shown in Celsius.

Valid values for status are the following: \

Parameter	Description
`normal`	Sensor is within nominal temperature range.
`max`	Highest temperature from this sensor.
`low_critical`	Lowest threshold temperature for this sensor.
`critical`	Highest threshold temperature for this sensor.
`fault`	Fault event for this sensor.
`emergency`	Over temperature event for Over temperature event for this sensor.

Examples

Showing current temperature information for a 6100 switch:

```
switch# show environment temperature
Temperature information
-----------------------------------------------------------------------------
                                                     Current
Mbr/Slot-Sensor                 Module Type        temperature  Status
-----------------------------------------------------------------------------
1/1-COMe-Daughter-Boar          line-card-module     66.45 C    normal
1/1-PCIE-Switch                 line-card-module     95.82 C    normal
1/1-Processor                   line-card-module     00.00 C    fault
1/1-Switch-ASIC                 line-card-module     116.36 C   emergency
1/1-Switch-ASIC-Internal        line-card-module     108.25 C   critical
1/2-COMe-Daughter-Boar          line-card-module     67.29 C    normal
1/2-PCIE-Switch                 line-card-module     95.82 C    normal
1/2-Processor-1                 line-card-module     72.92 C    normal
1/2-Processor-2                 line-card-module     73.05 C    normal
1/2-Switch-ASIC                 line-card-module     97.41 C    normal
1/2-Switch-ASIC-Internal        line-card-module     97.62 C    normal
...
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show events

```
show events [ -e <EVENT-ID> |
    -s {emergency | alert | critical | error | warning | notice | info | debug} |
    -r |
    -a |
    -n <COUNT> |
    -i  <MEMBER-SLOT> |
    -m {active | standby} |
    -c {lldp | ospf | ...} |
    -d {lldpd | bgpd | fand | ...}]
```

Description

Shows event logs generated by the switch modules since the last reboot.

Parameter	Description
`-e <EVENT-ID>`	Shows the event logs for the specified event ID. Event ID range: 101 through 99999.
`-s {emergency | alert | critical | error | warning | notice | info | debug}`	Shows the event logs for the specified severity. Select the severity from the following list: ■ `emergency`: Displays event logs with severity emergency only. ■ `alert`: Displays event logs with severity alert and above. ■ `critical`: Displays event logs with severity critical and above. ■ `error`: Displays event logs with severity error and above. ■ `warning`: Displays event logs with severity warning and above. ■ `notice`: Displays event logs with severity notice and above. ■ `info`: Displays event logs with severity info and above. ■ `debug`: Displays event logs with all severities.
`-r`	Shows the most recent event logs first.
`-a`	Shows all event logs, including those events from previous boots.
`-n <COUNT>`	Displays the specified number of event logs.
`-c {lldp | ospf | ...}`	Shows the event logs for the specified event category. Enter `show event -c` for a full listing of supported categories with descriptions.
`-d {lldpd | bgpd | fand | ...}`	Shows the event logs for the specified process. Enter `show event -d` for a full listing of supported daemons with descriptions.

Examples

Showing event logs:

```
switch# show events
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
   70:72:cf:51:50:7c
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to
   up for bridge_normal interface
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1
   in Hardware
```

Showing the most recent event logs first:

```
switch# show events -r
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1
   in Hardware
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to
   up for bridge_normal interface
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
   70:72:cf:51:50:7c
```

Showing all event logs:

```
switch# show events -a
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
   70:72:cf:51:50:7c
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to
   up for bridge_normal interface
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1
   in Hardware
```

Showing event logs related to LACP:

```
switch# show events -c lacp
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
   70:72:cf:51:50:7c
```

Showing event logs as per the specified member/slot ID:

```
switch# show events -i 1/1
---------------------------------------------------
show event logs
---------------------------------------------------
```

```
2017-08-17:22:32:25.743991|hpe-sysmond|6301|LOG_INFO|LC|1/1|System resource
   utilization poll interval is changed to 313
2017-08-17:22:33:01.692860|hpe-sysmond|6301|LOG_INFO|LC|1/1|System resource
   utilization poll interval is changed to 23
2017-08-17:22:33:06.181436|hpe-sysmond|6301|LOG_INFO|LC|1/1|System resource
   utilization poll interval is changed to 512
2017-08-17:22:33:06.181436|systemd-coredump|1201|LOG_CRIT|LC|1/1|hpe-sysmond
   crashed due to signal:11
```

Showing event logs as per the specified process:

```
switch# show events -d lacpd
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
   70:72:cf:51:50:7c
```

Displaying the specified number of event logs:

```
switch# show events -n 5
---------------------------------------------------
show event logs
---------------------------------------------------
2018-03-21:06:12:15.500603|arpmgrd|6101|LOG_INFO|AMM|-|ARPMGRD daemon has started
2018-03-21:06:12:17.734405|lldpd|109|LOG_INFO|AMM|-|Configured LLDP tx-delay to 2
2018-03-21:06:12:17.740517|lacpd|1307|LOG_INFO|AMM|-|LACP system ID set to
   70:72:cf:d4:34:42
2018-03-21:06:12:17.743491|vrfmgrd|5401|LOG_INFO|AMM|-|Created a vrf entity
   42cc3df7-1113-412f-b5cb-e8227b8c22f2
2018-03-21:06:12:17.904008|vrfmgrd|5401|LOG_INFO|AMM|-|Created a vrf entity
   4409133e-2071-4ab8-adfe-f9662c06b889
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

show front-panel-security status

`show front-panel-security status`

Description

Displays the configuration status of the front panel factory reset feature and the time stamp of the first occurrence of factory reset using the reset button.

Example

Displays the status of the front panel factory reset feature:

```
switch# show front-panel-security status
Front panel factory reset                    : enabled
First occurrence of front-panel factory reset : Wed 2023-09-06 02:57:50 +08
```

Displays the status of the front panel factory reset feature:

```
switch# show front-panel-security status
Front panel factory reset                    : enabled but unsupported
First occurrence of front-panel factory reset : NA
```

If the status of the feature is displayed as **enabled but unsupported**, then users should issue the **allow-unsafe-updates** command and reboot the switch.

```
switch# config
switch(config)# allow-unsafe-updates 30

This command will enable non-failsafe updates of programmable devices for
the next 30 minutes.  You will first need to wait for all line and fabric
modules to reach the ready state, and then reboot the switch to begin
applying any needed updates.  Ensure that the switch will not lose power,
be rebooted again, or have any modules removed until all updates have
finished and all line and fabric modules have returned to the ready state.

WARNING: Interrupting these updates may make the product unusable!

Continue (y/n)? y

    Unsafe updates      : allowed (less than 30 minute(s) remaining)
```

Command History

Release	Modification
10.13.1000	Command introduced

Command Information

Platforms	Command context	Authority
6000 6100	Manager (#)	Administrators or local user group members with execution rights for this command.

show hostname

`show hostname`

Description

Shows the current host name.

Example

Setting and showing the host name:

```
switch# show hostname
switch
switch# config
switch(config)# hostname myswitch
myswitch(config)# show hostname
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show images

`show images`

Description

Shows information about the software in the primary and secondary images.

Example

Showing the primary and secondary images on a 6100 switch:

```
switch(config)# show images
---------------------------------------------------------------------------
AOS-CX Primary Image
---------------------------------------------------------------------------
Version : PL.10.06.0002E
Size    : 243 MB
Date    : 2020-11-25 22:00:47 PST
SHA-256 : 61fe9233b2c842e8ac1731ad46949bd63e269c5c72d69290932ef19c1ebb0730

---------------------------------------------------------------------------
AOS-CX Secondary Image
---------------------------------------------------------------------------
Version : PL.10.07.0000E-201-gba0c336
Size    : 271 MB
Date    : 2020-11-25 21:09:08 UTC
SHA-256 : 2fdfc646a8013efcca75729584bbb0fa54604086bad3f46e1c5d4e706b8b30ee

Default Image : primary
Boot Profile Timeout : 2 seconds

---------------------------------------------------------------------------
Management Module 1/1 (Active)
---------------------------------------------------------------------------
```

```
Active Image       : primary
Service OS Version : PL.01.07.0003-internal
BIOS Version       : PL.01.0001
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show module

`show module`

Description

Shows information about installed line modules and management modules.

Although this switch does not have removable modules, this command will still return information about the switch, referring to management modules and line modules.

Usage

Identifies and shows status information about the line modules and management modules that are installed in the switch.

To show the configuration information—if any—associated with that line module slot, use the **show running-configuration** command.

Status is one of the following values:

`Active`
This is the active management module.
`Deinitializing`
The switch is being deinitialized.
`Diagnostic`
The switch is in a state used for troubleshooting.
`Down`
The switch is physically present but is powered down.
`Failed`
The switch has experienced an error and failed.
`Initializing`
The switch is being initialized.
`Present`
The switch hardware is installed in the chassis.
`Ready`
The switch is available for use.
`Updating`
A firmware update is being applied to the switch.

Examples

Showing all installed modules on a 6100 switch:

```
switch# show module

Management Modules
==================

     Product                                              Serial
Name Number  Description                                  Number     Status
---- ------- -------------------------------------- ---------- ----------------
1/1  JL678A  6100F 24G 4SFP+ Swch                         SG0ZKW600P Active (local)

Line Modules
============

     Product                                              Serial
Name Number  Description                                  Number     Status
---- ------- -------------------------------------- ---------- ----------------
1/1  JL678A  6100F 24G 4SFP+ Swch                         SG0ZKW600P Ready

Management Modules
==================

     Product                                              Serial
Name Number  Description                                  Number     Status
---- ------- -------------------------------------- ---------- ----------------
1/1  JL678A  6100F 24G 4SFP+ Swch                         SG0ZKW600P Active (local)

Line Modules
============

     Product                                              Serial
Name Number  Description                                  Number     Status
---- ------- -------------------------------------- ---------- ----------------
1/1  JL678A  6100F 24G 4SFP+ Swch                         SG0ZKW600P Ready
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show running-config

`show running-config [<FEATURE>] [all]`

Description

Shows the current nondefault configuration running on the switch. No user information is displayed.

Parameter	Description
<FEATURE>	Specifies the name of a feature. For a list of feature names, enter the `show running-config` command, followed by a space, followed by a question mark (?).
all	Shows all default values for the current running configuration.

Examples

Showing the current running configuration:

```
switch> show running-config
Current configuration:
!
!Version AOS-CX PL.10.06.0001-346-g56a12a8f4cf15
!export-password: default
ntp enable
!
!
!
!
ssh server vrf default
vlan 1
spanning-tree
spanning-tree instance 1 vlan 1,2,4-10
interface 1/1/1
    no shutdown
    vlan access 1
    portfilter 1/1/2-1/1/3

interface 1/1/2
    no shutdown
    vlan access 1
interface 1/1/3
    no shutdown
    vlan access 1
interface 1/1/4
    no shutdown
    vlan access 1
interface 1/1/5
    no shutdown
    vlan access 1
interface 1/1/6
    no shutdown
    vlan access 1
interface 1/1/7
    no shutdown
    vlan access 1
interface 1/1/8
    no shutdown
    vlan access 1
interface 1/1/9
    no shutdown
    vlan access 1
interface 1/1/10
    no shutdown
    vlan access 1
interface 1/1/11
    no shutdown
    vlan access 1
```

```
interface 1/1/12
    no shutdown
    vlan access 1
interface 1/1/13
    no shutdown
    vlan access 1
interface 1/1/14
    no shutdown
    vlan access 1
interface 1/1/15
    no shutdown
    vlan access 1
interface 1/1/16
    no shutdown
    vlan access 1
interface vlan 1
    ip dhcp
snmp-server vrf default
snmp-server community public
snmp-server host 1.1.1.1 inform version v2c
snmp-server host 1.1.1.1 trap version v2c
snmpv3 context A vrf default
!
!
!
!
!
https-server vrf default
```

Showing the current running configuration in json format:

```
switch> show running-config json
Running-configuration in JSON:
{
    "Monitoring_Policy_Script": {
        "system_resource_monitor_mm1.1.0": {
            "Monitoring_Policy_Instance": {
                "system_resource_monitor_mm1.1.0/system_resource_monitor_
mm1.1.0.default": {
                    "name": "system_resource_monitor_mm1.1.0.default",
                    "origin": "system",
                    "parameters_values": {
                        "long_term_high_threshold": "70",
                        "long_term_normal_threshold": "60",
                        "long_term_time_period": "480",
                        "medium_term_high_threshold": "80",
                        "medium_term_normal_threshold": "60",
                        "medium_term_time_period": "120",
                        "short_term_high_threshold": "90",
                        "short_term_normal_threshold": "80",
                        "short_term_time_period": "5"
                    }
                }
            },
...
...
...
...
```

Show the current running configuration without default values:

```
switch(config)# show running-config
Current configuration:
!
!Version AOS-CX PL.10.06.0001-346-g56a12a8f4cf15
!export-password: default
!
!
!
!
ssh server vrf default
vlan 1
spanning-tree
interface 1/1/1
    no shutdown
    vlan access 1
interface 1/1/2
    no shutdown
    vlan access 1
interface 1/1/3
    no shutdown
    vlan access 1
interface 1/1/4
    no shutdown
    vlan access 1
interface 1/1/5
    no shutdown
    vlan access 1
interface 1/1/6
    no shutdown
    vlan access 1
interface 1/1/7
    no shutdown
    vlan access 1
interface 1/1/8
    no shutdown
    vlan access 1
interface 1/1/9
    no shutdown
    vlan access 1
interface 1/1/10
    no shutdown
    vlan access 1
interface 1/1/11
    no shutdown
    vlan access 1
interface 1/1/12
    no shutdown
    vlan access 1
interface 1/1/13
    no shutdown
    vlan access 1
interface 1/1/14
    no shutdown
    vlan access 1
interface 1/1/15
    no shutdown
    vlan access 1
interface 1/1/16
    no shutdown
    vlan access 1
interface vlan 1
    ip dhcp
```

```
!
!
!
!
!
https-server vrf default
switch#
switch#                                             PL.10.06.0001-346-
g56a12a8f4cf15
!export-password: default
!
!
!
!
ssh server vrf default
vlan 1
spanning-tree
interface 1/1/1
    no shutdown
    vlan access 1
interface 1/1/2
    no shutdown
    vlan access 1
interface 1/1/3
    no shutdown
    vlan access 1
interface 1/1/4
    no shutdown
    vlan access 1
interface 1/1/5
    no shutdown
    vlan access 1
interface 1/1/6
    no shutdown
    vlan access 1
interface 1/1/7
    no shutdown
    vlan access 1
interface 1/1/8
    no shutdown
    vlan access 1
interface 1/1/9
    no shutdown
    vlan access 1
interface 1/1/10
    no shutdown
    vlan access 1
interface 1/1/11
    no shutdown
    vlan access 1
interface 1/1/12
    no shutdown
    vlan access 1
interface 1/1/13
    no shutdown
    vlan access 1
interface 1/1/14
    no shutdown
    vlan access 1
interface 1/1/15
    no shutdown
    vlan access 1
```

```
interface 1/1/16
    no shutdown
    vlan access 1
interface vlan 1
    ip dhcp
!
!
!
!
!
https-server vrf default
switch#
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show running-config current-context

`show running-config current-context`

Description

Shows the current non-default configuration running on the switch in the current command context.

Usage

You can enter this command from the following configuration contexts:

- Any child of the global configuration (`config`) context. If the child context has instances—such as interfaces—you can enter the command in the context of a specific instance. Support for this command is provided for one level below the `config` context. For example, entering this command for a child of a child of the `config` context not supported. If you enter the command on a child of the `config` context, the current configuration of that context and the children of that context are displayed.
- The global configuration (`config`) context. If you enter this command in the global configuration (`config`) context, it shows the running configuration of the entire switch. Use the `show running-configuration` command instead.

Examples

Showing the running configuration for the current interface:

```
switch(config-if)# show running-config current-context
```

```
interface 1/1/1
    no shutdown
    description Example interface
vlan access 1
    exit
```

Showing the current running configuration for the in-band management interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)#description IN-BAND Management Interface
switch(config-if-vlan)#ip dhcp
switch(config-if-vlan)#no shutdown
switch(config-if-vlan)#end
switch#
```

Showing the current running configuration for the in-band management interface without DHCP:

```
switch(config)# interface vlan 1
switch(config-if-vlan)#description IN-BAND Management Interface
switch(config-if-vlan)#no ip dhcp
switch(config-if-vlan)#ip address 192.168.10.1/24
switch(config-if-vlan)#no shutdown
switch(config-if-vlan)#end
switch#
```

Showing the running configuration for the external storage share named `nasfiles`:

```
switch(config-external-storage-nasfiles)# show running-config current-context
external-storage nasfiles
   address 192.168.0.1
   vrf default
   username nasuser
   password ciphertext
AQBapalKj+XMsZumHEwIc9OR6YcOw5Z6Bh9rV+9ZtKDKzvbaBAAAAB1CTrM=
   type scp
   directory /home/nas
   enable
switch(config-external-storage-nasfiles)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config` or a child of `config`. See Usage.	Administrators or local user group members with execution rights for this command.

show startup-config

show startup-config [json]

Description

Shows the contents of the startup configuration.

Switches in the `factory-default` configuration do not have a startup configuration to display.

Parameter	Description
json	Display output in JSON format.

Examples

Showing the startup-configuration in JSON format:

```
switch# show startup-config json
Startup configuration:
{
    "AAA_Server_Group": {
        "local": {
            "group_name": "local"
        },
        "none": {
            "group_name": "none"
        }
    },
...
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show system

show system [serviceos password-prompt]

Description

Shows general status information about the system.

Parameter	Description
serviceos password-prompt	Shows the Service OS password prompt status.

Usage

CPU utilization represents the average utilization across all the CPU cores.

System Description, System Contact, and System Location can be set with the `snmp-server` command.

Examples

Showing system information:

```
switch# show system
Hostname                : switch
System Description      : switch description
System Contact          : contact
System Location         : location

Vendor                  : Aruba
Product Name            : Xxxxxx ...
Chassis Serial Nbr      : XXXXXXXXXX
Base MAC Address        : xxxxxx-xxxxxx
AOS-CX Version          : XX.99.99.9999

Time Zone               : UTC

Up Time                 : 1 week, 5 hours, 28 minutes
CPU Util (%)            : 5
CPU Util (% avg 1 min)  : 11
CPU Util (% avg 5 min)  : 10
Memory Usage (%)        : 35
```

Showing the Service OS password prompt status:

```
switch# show system serviceos password-prompt
password-prompt: disabled
```

Command History

Release	Modification
10.12	Added `CPU Util (% avg 1 min)` and `CPU Util (% avg 5 min)`.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show system resource-utilization

`show system resource-utilization [all]`

Description

Shows the system resource utilization data.

Parameter	Description
`all`	Shows the resource utilization data for the entire switch.

Examples

Showing system resource utilization data:

```
switch# show system resource-utilization
System Resources:
Processes                          :  144
CPU usage(%)                       :   10
CPU usage(% average over 1 minute) :   11
CPU usage(% average over 5 minute) :   15
Memory usage(%)                    :   22
Open FD's                          : 1358
Storage 1: Endurance utilization = 10-20% (mmc-type-a), 0-10% (mmc-type-b),
   Health = normal

Data written to various partitions since boot
Nos      :   5 MB
Log      :   1 MB
Coredump :  23 MB
Security :   2 MB
Selftest : 405 KB
Swap     :  14 MB

Storage partition usage(%)
Nos      :   5
Log      :  60
Coredump :  23
Security :   2
Selftest :   1
Swap     :   0
```

Attempting to show resource utilization data when system resource utilization polling is disabled:

```
switch# show system resource-utilization
System resource utilization data poll is currently disabled
```

Showing system resource utilization data for all:

```
switch# show system resource-utilization all
---------------------------------------------------------------------------
Resource utilization data for Management Module
---------------------------------------------------------------------------

System Resources:
Processes                          :  244
CPU usage(%)                       :   10
CPU usage(% average over 1 minute) :   11
CPU usage(% average over 5 minute) :   15
```

```
Memory usage(%)                         :   11
Open FD's                               : 1854
Storage 1: Endurance utilization = 0-10% (mmc-type-a), 0-10% (mmc-type-b),
   Health = normal

Data written to various partitions since boot
Nos       :  15 MB
Log       :   1 MB
Coredump :  23 MB
Security :   2 MB
Selftest : 405 KB
Swap      :   0 KB

Storage partition usage(%)
Nos       :   5
Log       :  60
Coredump :  23
Security :   2
Selftest :   1
Swap      :   0
```

Command History

Release	Modification
10.12	The output of this command includes **CPU usage(% average over 1 minute)** and **CPU usage(% average over 5 minute)**.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show tech

`show tech [basic | <FEATURE>] [local-file]`

Description

Shows detailed information about switch features by automatically running the `show` commands associated with the feature. If no parameters are specified, the `show tech` command shows information about all switch features. Technical support personnel use the output from this command for troubleshooting.

Parameter	Description
`basic`	Specifies showing a basic set of information.
<FEATURE>	Specifies the name of a feature. For a list of feature names, enter the `show tech` command, followed by a space, followed by a

Parameter	Description
	question mark (?).
`local-file`	Shows the output of the `show tech` command to a local text file.

Usage

To terminate the output of the `show tech` command, enter **Ctrl+C**.

If the command was not terminated with **Ctrl+C**, at the end of the output, the `show tech` command shows the following:

- The time consumed to execute the command.
- The list of failed `show` commands, if any.

To get a copy of the local text file content created with the show tech command that is used with the local-file parameter, use the `copy show-tech local-file` command.

Example

Showing the basic set of system information:

```
switch# show tech basic
==============================================================
Show Tech executed on Wed Sep  6 16:50:37 2017
==============================================================
==============================================================
[Begin] Feature basic
==============================================================

*******************************
Command : show core-dump all
*******************************
no core dumps are present

...
==============================================================
[End] Feature basic
==============================================================

==============================================================
1 show tech command failed
==============================================================
Failed command:
  1. show boot-history
==============================================================
Show tech took 3.000000 seconds for execution
```

Directing the output of the **show tech basic** command to the local text file:

```
switch# show tech basic local-file
Show Tech output stored in local-file. Please use 'copy show-tech local-file'
to copy-out this file.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show usb

`show usb`

Description

Shows the USB port configuration and mount settings.

Examples

If USB has not been enabled:

```
switch> show usb
Enabled: No
Mounted: No
```

If USB has been enabled, but no device has been mounted:

```
switch> show usb
Enabled: Yes
Mounted: No
```

If USB has been enabled and a device mounted:

```
switch> show usb
Enabled: Yes
Mounted: Yes
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show usb file-system

`show usb file-system [<PATH>]`

Description

Shows directory listings for a mounted USB device. When entered without the <PATH> parameter the top level directory tree is shown.

Parameter	Description
<PATH>	Specifies the file path to show. A leading "/" in the path is optional.

Usage

Adding a leading "/" as the first character of the *<PATH>* parameter is optional.

Attempting to enter '..' as any part of the *<PATH>* will generate an invalid path argument error. Only fully-qualified path names are supported.

Examples

Showing the top level directory tree:

```
switch# show usb file-system
/mnt/usb:
'System Volume Information'  dir1'

/mnt/usb/System Volume Information':
IndexerVolumeGuid  WPSettings.dat

/mnt/usb/dir1:
dir2  test1

/mnt/usb/dir1/dir2:
test2
```

Showing available path options from the top level:

```
switch# show usb file-system /
total 64
drwxrwxrwx 2 32768 Jan 22 16:27 'System Volume Information'
drwxrwxrwx 3 32768 Mar  5 15:26 dir1
```

Showing the contents of a specific folder:

```
switch# show usb file-system /dir1
total 32
drwxrwxrwx 2 32768 Mar  5 15:26 dir2
-rwxrwxrwx 1     0 Feb  5 18:08 test1

switch# show usb file-system dir1/dir2
total 0
-rwxrwxrwx 1 0 Feb  6 05:35 test2
```

Attempting to enter an invalid character in the path:

```
switch# show usb file-system dir1/../..
Invalid path argument
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show version

`show version`

Description

Shows version information about the network operating system software, service operating system software, and BIOS.

Example

Showing version information for a 4100i switch:

```
DocTeam-4100i# show version
-----------------------------------------------------------------------------
ArubaOS-CX
(c) Copyright 2017-2023 Hewlett Packard Enterprise Development LP
-----------------------------------------------------------------------------
Version      : RL.10.13.1000S
Build Date   : 2023-11-15 15:29:10 UTC
Build ID     : ArubaOS-CX:RL.10.13.1000S:ac4c3941a38c:202311151508
Build SHA    : ac4c3941a38c5d66f8f0212706af8f35fa01adfd
Hot Patches  :
Active Image : secondary
Service OS Version : RL.01.09.0003
BIOS Version       : RL.01.0003
```

Showing version information for a 6100 switch:

```
switch(config)# show version
-----------------------------------------------------------------------------
AOS-CX
(c) Copyright 2017-2022 Hewlett Packard Enterprise Development LP
-----------------------------------------------------------------------------
Version      : PL.10.xx.xxxxx
Build Date   : 2022-05-27 17:00:46 PDT
Build ID     : AOS-CX:PL©10.xx.xxxxx:9e8bf51170a6:202012062115
```

```
Build SHA    : 9e8bf51170a602370f12e0bde814e8d8f49bf706
Active Image : secondary

Service OS Version : PL.10.xx.xxxxx
BIOS Version       : PL.01.0002
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

system resource-utilization poll-interval

`system resource-utilization poll-interval <SECONDS>`

Description

Configures the polling interval for system resource information collection and recording such as CPU and memory usage.

Parameter	Description
<SECONDS>	Specifies the poll interval in seconds. Range: 10-3600. Default: 10.

Example

Configuring the system resource utilization poll interval:

```
switch(config)# system resource-utilization poll-interval 20
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

top cpu

`top cpu`

Description

Shows CPU utilization information.

Example

Showing top CPU information:

```
switch# top cpu
top - 09:42:55 up 3 min, 3 users, load average: 3.44, 3.78, 1.70
Tasks:  76 total, 2 running, 74 sleeping,  0 stopped,  0 zombie
%Cpu(s): 31.4 us, 32.7 sy, 0.5 ni, 34.4 id, 04. wa, 0.0 hi, 0.6 si, 0.0 st
KiB Mem : 4046496 total,  2487508 free,  897040 used,   661948 buff/cache
KiB Swap:       0 total,        0 free,       0 used,  2859196 avail Mem

  PID USER     PRI  NI   VIRT    RES   SHR S  %CPU %MEM      TIME+ COMMAND
...
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

top memory

`top memory`

Description

Shows memory utilization information.

Example

Showing top memory:

```
switch> top memory
top - 09:42:55 up 3 min, 3 users, load average: 3.44, 3.78, 1.70
Tasks:  76 total, 2 running, 74 sleeping,  0 stopped,  0 zombie
%Cpu(s): 31.4 us, 32.7 sy, 0.5 ni, 34.4 id, 04. wa, 0.0 hi, 0.6 si, 0.0 st
KiB Mem : 4046496 total,  2487508 free,  897040 used,   661948 buff/cache
KiB Swap:       0 total,        0 free,       0 used,  2859196 avail Mem

  PID USER     PRI  NI   VIRT    RES   SHR S  %CPU %MEM      TIME+ COMMAND
```

```
...
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

usb

```
usb
no usb
```

Description

Enables the USB ports on the switch. This setting is persistent across switch reboots and management module failovers. Both active and standby management modules are affected by this setting.

The **no** form of this command disables the USB ports.

Example

Enabling USB ports:

```
switch(config)# usb
```

Disabling USB ports when a USB drive is mounted:

```
switch(config)# no usb
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

usb mount | unmount

```
usb {mount | unmount}
```

Description

Enables or disables the inserted USB drive.

Parameter	Description
`mount`	Enables the inserted USB drive.
`unmount`	Disables the inserted USB drive in preparation for removal.

Usage

If USB has been enabled in the configuration, the USB port on the active management module is available for mounting a USB drive. The USB port on the standby management module is not available.

An inserted USB drive must be mounted each time the switch boots or fails over to a different management module.

A USB drive must be unmounted before removal.

The supported USB file systems are FAT16 and FAT32.

Examples

Mounting a USB drive in the USB port:

```
switch# usb mount
```

Unmounting a USB drive:

```
switch# usb unmount
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Chapter 17

Support and Other Resources

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	https://www.arubanetworks.com/support-services/
AOS-CX Switch Software Documentation Portal	https://www.arubanetworks.com/techdocs/AOS-CX/help_portal/Content/home.htm
HPE Aruba Networking Support Portal	https://networkingsupport.hpe.com/home
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-408-754-1200 (Primary - Toll Number) +1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working)
International telephone	https://www.arubanetworks.com/support-services/contact-support/

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	https://developer.arubanetworks.com/hpe-aruba-networking-aoscx/docs/about
Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
AOS-CX Software Technical Update channel on	Videos on new features introduced in this release: https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP-Q_UL3CskS

YouTube.	
HPE Aruba Networking Hardware Documentation and Translations Portal	https://www.arubanetworks.com/techdocs/hardware/DocumentationPortal/Content/home.htm
HPE Aruba Networking software	https://networkingsupport.hpe.com/downloads
Software licensing and Feature Packs	https://licensemanagement.hpe.com/
End-of-Life information	https://www.arubanetworks.com/support-services/end-of-life/

Accessing Updates

You can access updates from the Aruba Support Portal or the HPE My Networking Website.

Aruba Support Portal

https://networkingsupport.hpe.com/downloads

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

My Networking

https://www.hpe.com/networking/support

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

https://support.hpe.com/portal/site/hpsc/aae/home/

Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HP Passport set up with relevant entitlements.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

https://networkingsupport.hpe.com/notifications/subscriptions (requires an active HPE Aruba Networking support account to manage subscriptions). Security notices are viewable without a networking support account.

Warranty Information

To view warranty information for your product, go to https://www.arubanetworks.com/support-services/product-warranties/.

Regulatory Information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts

Additional regulatory information

Aruba is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see https://www.arubanetworks.com/company/about-us/environmental-citizenship/.

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